Multi-Omics Analysis of DNA Methylation and Gene Expression in Esophageal Squamous Cell Carcinoma (ESCC) cell and Normal Tissue

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1. Introduction

Esophageal squamous cell carcinoma (ESCC) is the sixth leading cause of cancer-related deaths worldwide, accounting for 90% of esophageal cancer cases in Asia, Eastern Europe, and Africa (Tarazi et al., 2021). Despite its prevalence, ESCC has a low survival rate of 10–30% (Conway et al., 2023). Understanding its etiology is crucial for developing prevention and treatment strategies.

DNA methylation plays a significant role in cancer development. Comparing methylation patterns between normal and tumor tissues reveals epigenetic changes in key genes, while high-throughput sequencing uncovers genomic alterations and their impact on tumorigenesis.

This study investigates differentially expressed genes between ESCC and normal tissues, integrating RNA-seq analysis to explore potential effects on gene expression.

1.1. Data

This report is based on data obtained from ESCC tissue and adjacent normal tissues collected from different patients.

	Infinium Data	RNA-seq Data
Accession ID	GSE157807	GSE194116
Number of Samples	10	12

1.1.1. Infinium Data (GSE157807)

The methylation profile data was obtained from the Illumina Infinium HumanMethylation850 BeadChip platform. It includes DNA methylation data from five ESCC samples and five paired adjacent normal tissue samples, resulting in a total of 10 samples. The dataset utilizes a paired design, where each ESCC sample has a corresponding normal control sample. This can be obtained from https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE157807.

Sample ID	PatientID	TissueType
GSM4775294	ca-973	ESCC
GSM4775295	ca-973	NORMAL
GSM4775296	ca-962	ESCC
GSM4775298	ca-944	ESCC
GSM4775299	ca-944	NORMAL
GSM4775300	ca-942	ESCC
GSM4775301	ca-942	NORMAL
GSM4775320	ca-933	ESCC
GSM4775303	ca-933	NORMAL

1.1.2. RNA-seq Data (GSE194116)

The RNA-seq data obtained by using the Illumina HiSeq 2500. It includes raw gene count data from six paired ESCC and normal tissue samples. Quality control, trimming, and mapping of the data were performed as described in the GEO metadata. This can be obtained from https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE194116.

	Sample Type
GSM5828359	ESC-1
GSM5828360	Normal-1
GSM5828361	ESC-2
GSM5828362	Normal-2
GSM5828363	ESC-2
GSM5828364	Normal-3
GSM5828365	ESC-2
GSM5828366	Normal-4
GSM5828367	ESC-2
GSM5828368	Normal-5
GSM5828359	ESC-2
GSM5828359	Normal-6

2. Method

2.1. Methylation profiling analysis

Data preprocessing, quality control, and quantile normalization were conducted using the *minfi* and *ENmix* packages. Low-quality probes, such as those with unreliable signal or high missingness, were filtered out during preprocessing. Beta values and M-values were calculated for downstream analysis.

A paired design was applied to account for patient-specific effects. Tissue type (tumor vs. normal) was modeled in the design matrix, and contrasts were defined to test differential methylation, with patient IDs included as a blocking factor in the analysis. The *limma* package was used to fit linear models, The intra-patient correlation was estimated using *duplicateCorrelation*, and the *limma* package was used to fit linear models. Empirical Bayes smoothing was applied to improve statistical power and identify significant CpG sites.

Checking methylation for top results was done with getM() function, interesting annotations were also added to the output. Region-level analysis was conducted using *bumphunter*, identifying clusters of CpG sites with consistent methylation changes. Significant CpGs and regions were annotated with genomic features, such as gene association and CpG island relation, using the *IlluminaHumanMethylationEPICanno.ilm10b4.hg19* package.

Gene set enrichment analysis was performed with *goana* to identify enriched Gene Ontology (GO) terms, and clusterProfiler ackage was used to visualize differentially methylated Biological Process Terms.

Visualization of results included volcano and MA plots for differential methylation, as well as histograms to assess the distribution of p-values.

The results of CpG GSA, DMRs GSA, CpG Entrez IDs, and DMRs Entrez IDs were saved as "CpG_GSA_results.txt", "bmp_GSA_results.txt", "EntrezIDs_CpG_results.txt", and "EntrezIDs_bmp_results.txt", respectively (see Appendix A).

2.2. Expression profiling analysis

Quality control was performed on an HPC environment, while differential expression analysis was conducted on raw count data using the *edgeR* package. Lowly expressed genes were filtered out, and data normalization was applied using method in *edgeR* to account for library size differences. A design matrix was constructed to incorporate both the paired nature of the data and experimental conditions.

Differential expression analysis employed generalized linear models with a likelihood ratio test (LRT) to identify significant genes.

Gene Ontology (GO) enrichment analysis was performed using the *goana* and *clusterProfiler* package to identify differentially expressed Biological Process terms.

Results were visualized through MDS plots, Volcano plots, and MA plots.

Result of differentially expressed genes are saved as “RNAseq_results.txt”, and result of gene set analysis was saved as “RNAseq_GSA_results” (see Appendix B).

2.3. Integration of Data Sources

The results of the RNA-seq and CpG methylation analyses were compared, providing insight into the robustness of the findings. This comparison included both gene-level and gene set-level overlap. RNA-seq results were filtered based on an FDR threshold of 0.05, ensuring only statistically significant results were considered. Duplicated Entrez IDs and missing values were removed to ensure a unique set of genes for analysis.

A Venn diagram was created to visualize the overlap between the significant RNA-seq results and both differentially methylated regions (bumps) and CpG analysis. (see Appendix C).

3. Results

3.1. Methylation Data Analysis

The MDS plot (Figure 1) shows distinct clustering of Tumor and ESCC samples, indicating a strong tissue type effect on methylation. Blocking for patient effect ensures that the differences observed are due to tissue type, not individual patient variability. Therefore, in the matrix design, individual variability is controlled by treating the patient effect as a blocking factor, rather than explicitly including individual as a variable. This allows the model to focus on the tissue type differences while adjusting for patient-specific variability.

A total of 35,994 CpG probes (4.18%) were significantly differentially methylated when filtered based on adj.P.Val threshold of 0.05.Figure 2 is a MA-plot of log fold changes between M-values in ESCC and normal adjacent tissue. Among 6,885 DMRs, only 3 regions

were significant (FWER < 0.1). Gene set analysis on genes within or near these DMRs revealed key GO terms such as response to growth factors and regulation of cell differentiation.

For validation, individual CpG sites were analyzed. The nearest genes were identified, excluding duplicates. Top GO terms for CpG analysis included cell morphogenesis and neuron projection development (see figure 3). 6 of the top 50 gene sets overlapped between DMRs and CpG analysis. This includes regulation of cell projection organization.

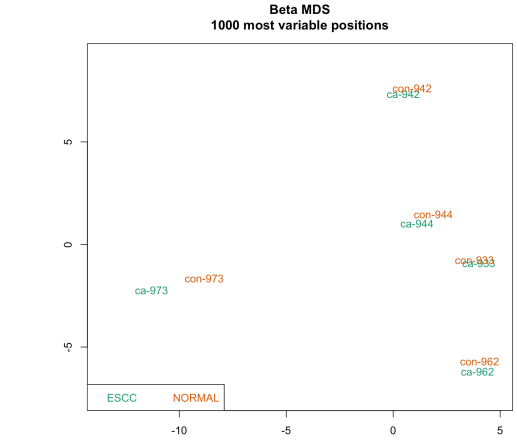


Figure 1. MDS plot of beta-values.

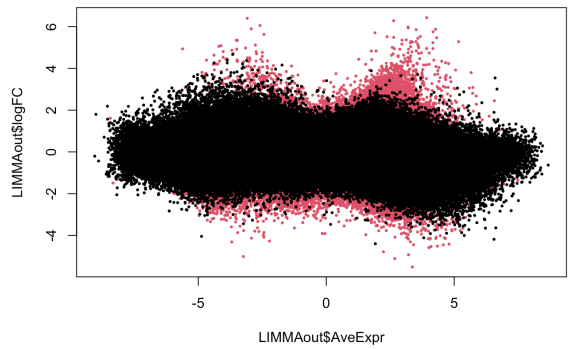


Figure 2. MA plot of log fold changes between M-values in ESCC and normal tissue.

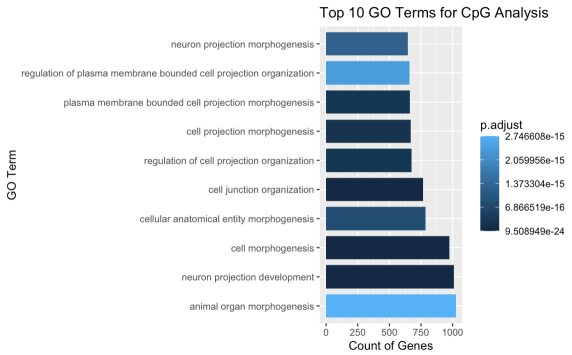


Figure 3. Top 10 GO Terms for CpG analysis.

3.2. RNA-seq Data Analysis

The MDS plot (Figure 4) highlights clear separation between Normal and ECA samples, suggesting a significant group effect. To account for individual variability, a patient effect was incorporated in matrix design as a factor factor in the statistical model.

Using the likelihood ratio test with the glmQLFTest() function, a total of 3,780 genes (upregulated, logFC > 0) and 3,525 genes (downregulated, logFC < 0) were identified as significantly differentially expressed among 24,315 genes, based on a FDR threshold of 0.05. The results are visualized in the MA-plot in Figure 5, and the top significant genes are listed in Table 3. Gene set analysis was performed on the significant genes using over-representation analysis of GO terms. After applying Benjamini-Hochberg FDR correction, 2,066 GO Biological Process terms were identified as significantly differentially regulated. Figure 6 shows the top 10 GO terms for CpG analysis, which include protein localization to organelle, small GTPase-mediated signal transduction, and RNA splicing.

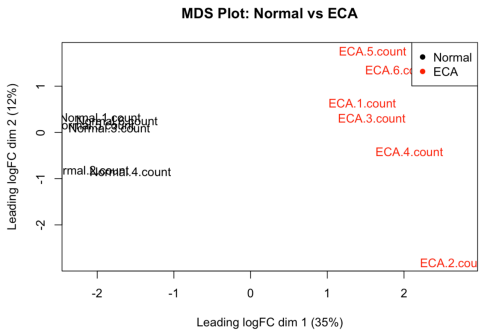


Figure 4. MDS Plot: Clustering of Normal vs ECA Samples

Gene_ name	logFC	P_value	FDR
CST1	-9.96	2.97e-12	8.14e-08
KRT75	-7.88	1.12e-11	1.53e-07
MMP11	-6.49	4.3e-10	3.96e-06
P2RX2	6.82	7.18e-10	4.92e-06
ADIPOQ	8.19	1.91e-09	1.05e-05
UNC13C	4.56	2.31e-09	1.06e-05
ADAMTS20	-5.09	3.76e-09	1.46e-05
COL10A1	-7.56	4.38e-09	1.46e-05
MISP	-4.64	4.79e-09	1.46e-05
UPK1A	4.60	6.68e-09	1.83e-05

Table 3. Key genes with differential expression between tumor and healthy tissues identified in the RNA-seq analysis.

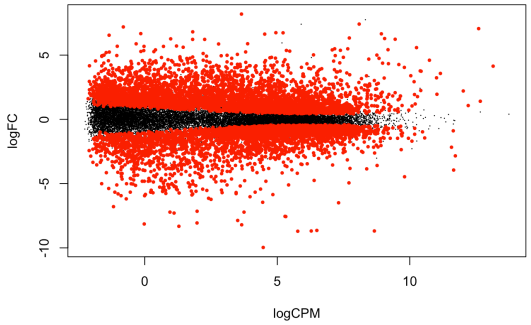


Figure 5. MA-plot of log fold changes between expression values in ESC and normal tissue.

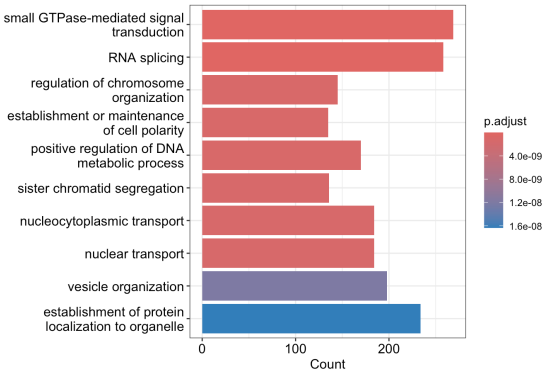


Figure 6. Top 10 GO Terms for CpG analysis.

3.3. Comparison of Different Results

The overlap between gene expression and methylation analyses was further investigated. This included assessing the overlap of differentially methylated CpG sites and differentially methylated regions (DMRs).

A Venn diagram comparing RNA-seq and DMRs (see Figure 7) showed no overlap between the two. However, when comparing RNA-seq data with CpG methylation results, 1000 genes out of 6305 and 1194 genes exhibited overlapping results (see figure 8).

Additionally, a comparison of the top 10 GO Biological Process terms between the two analyses revealed 26 overlapping terms (see Figure 9). Some of the most significant overlapping terms include: "positive regulation of cellular component organization," "regulation of organelle organization," and "regulation of cellular processes."

Lastly, genes of interest (GOI) were selected based on common genes identified in both the RNA-seq and CpG analyses. Upregulated genes include CFLAR, KDM1A, and RBM6, while downregulated genes such as STPG1, SEMA3F, ANKIB1, HECW1, MAD1L1, and ICA1 were found to have reduced expression.

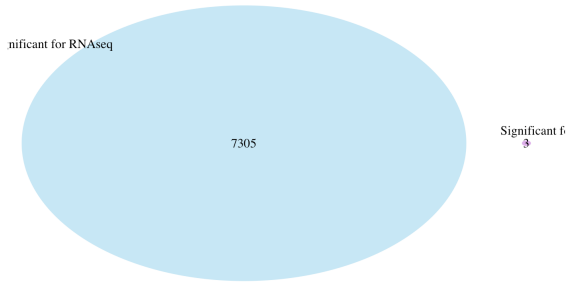


Figure 7. overlap between RNAseq results and DMRs

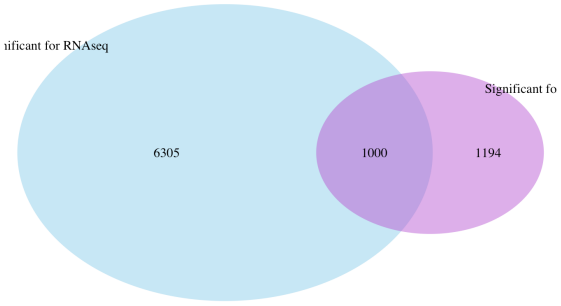


Figure 8. overlap between RNAseq results and CpG methylation results

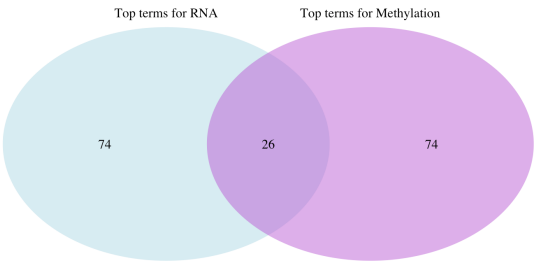


Figure 9. overlap between RNAseq results and CpG methylation results

4. Discussion

Alterations in DNA methylation patterns are commonly observed in tumors, making it crucial to understand how these changes contribute to disease. The most widely used method for analyzing methylation differences between samples is the sliding window approach to identify differentially methylated regions (DMRs) (VanderKraats et al., 2013). Ideally, we would expect more overlap with DMRs than with individual CpG sites. However, in my analysis, no overlap was detected between significant RNA-

seq genes and DMRs, while 1000 CpGs showed overlap. This indicates that changes in gene expression may reflect methylation changes at specific CpG sites, which are likely to play a significant role in regulating gene expression.

The possible reasons for this could be:

CpG Sensitivity: CpG sites are more responsive to methylation changes and often directly affect gene expression, particularly in promoter or regulatory regions. In contrast, DMRs require significant methylation changes across multiple CpGs, making them less sensitive to subtle changes in gene expression. Additionally, the stricter statistical criteria for identifying DMRs may exclude smaller methylation changes at individual CpGs, indicating that the lack of DMRs does not rule out their association with gene expression.

DMR Analysis Settings: The bumphunter method applied in my study used stringent parameters (cutoff = 0.25, B=500), which might limit the detection of DMRs with smaller methylation differences in CpGs.

Sample Size and Variability: With a sample size of only 10-12, there may not be enough statistical power to detect significant DMRs, even though changes in individual CpGs are more noticeable. A larger sample size would likely enhance the ability to identify subtle methylation differences.

Lastly, interesting genes, such as *STPG1, SEMA3F, HECW1, MAD1L1, ICA1, CFLAR, MTMR7, KDM1A, and RBM6,* which overlapped in both differential expression and CpG analyses, were found to be downregulated. Although these genes are not directly associated with esophageal squamous cell carcinoma (ESCC) or esophageal cancer, they have been linked to the progression of other cancers, including prostate cancer, lymphoma, and gastric cancer (Chen et al., 2022), (Doçi et al., 2015).

Additionally, the primary biological processes that showed overlap between differential expression and methylation analyses included GO terms related to cellular component organization and biogenesis. Structural changes in cells are a hallmark of cancer. These alterations can significantly influence processes like tumor formation, metastasis, and evasion of apoptosis. As a result, the changes in gene expression and methylation associated with these processes are essential in cancer research (Xu et al., 2021).

5. Reflection and Challenges

Difference from Published Results

The most significant discrepancy was the absence of *SPOCD1* as a key gene in my RNA-seq analysis, despite a related study showing it as highly differentially expressed between normal and ESCC samples. This warrants further investigation into data or methodological differences (Lin et al., 2022).

Goodness-of-Fit (GoF) Analysis:

The qq-plot of residual deviance showed deviations from normality, suggesting model misfit (see Figure 9). This may indicate issues with outliers, sample size, or model sensitivity, which could be addressed with adjustments or alternative techniques

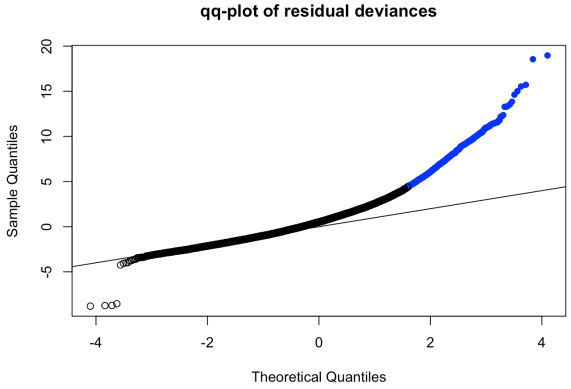


Figure 9. QQ-plot for model residuals

6. Conclusion

The integration of methylation and gene expression data provides valuable insights into the molecular mechanisms of ESCC. The overlap of key genes and biological processes associated with cancer underscores the importance of considering both epigenetic and transcriptional changes in understanding cancer pathogenesis. Further research with larger sample sizes and refined analysis techniques could enhance the robustness of these findings.

References

Chen, Y., Jiang, L., Xia, L., Zhang, G., & Chen, L. (2022). Thpok inhibits the immune escape of gastric cancer cells by inducing STPG1 to inactivate the Erk pathway. *BMC Immunology*, *23*(1). https://doi.org/10.1186/s12865-022-00485-5

Conway, E., Wu, H., & Tian, L. (2023). Overview of risk factors for esophageal squamous cell carcinoma in China. *Cancers*, *15*(23), 5604. https://doi.org/10.3390/cancers15235604

Doçi, C. L., Mikelis, C. M., Lionakis, M. S., Molinolo, A. A., & Gutkind, J. S. (2015). Genetic identification of *sema3f* as an antilymphangiogenic metastasis suppressor gene in head and neck squamous carcinoma. *Cancer Research*, *75*(14), 2937–2948. https://doi.org/10.1158/0008-5472.can-14-3121

Lin, Z., Chen, L., Wu, T., Zhang, Y., Huang, X., Chen, Y., Chen, J., & Xu, Y. (2022). Prognostic value of SPOCD1 in esophageal squamous cell carcinoma: A comprehensive study based on bioinformatics and validation. *Frontiers in Genetics*, *13*. https://doi.org/10.3389/fgene.2022.872026

Tarazi, M., Chidambaram, S., & Markar, S. R. (2021). Risk factors of esophageal squamous cell carcinoma beyond alcohol and smoking. *Cancers*, *13*(5), 1009. https://doi.org/10.3390/cancers13051009

VanderKraats, N. D., Hiken, J. F., Decker, K. F., & Edwards, J. R. (2013). Discovering high-resolution patterns of differential DNA methylation that correlate with gene expression changes. *Nucleic Acids Research*, *41*(14), 6816–6827. https://doi.org/10.1093/nar/gkt482

Xu, G. B., Wang, H., Chen, S. A., & Chen, H. (2021). Epigenetic regulation in cancer metastasis. *Medical Epigenetics*, 471–489. https://doi.org/10.1016/b978-0-12-823928-5.00032-3

Appendix A:

Differential Methylation Analysis of Esophageal Squamous Cell Carcinoma (ESCC) and Paired Normal Tissue Samples

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2024-12-13

This R script performs differential methylation analysis on Illumina Infinium HumanMethylation850 BeadChip data from ESCC and paired normal tissue samples. It begins with data preprocessing and quality control using minfi and ENmix, followed by normalization of raw methylation intensities. The script then conducts differential methylation analysis using the limma package and visualizes the results with volcano and MA plots. It also annotates the differentially methylated CpGs with genomic features and performs gene-level analysis, identifying the most significant genes with methylation changes.

General info

Species: Homo Sapiens Microarray type: Illumina Infinium HumanMethylation850 BeadChip Samples: The dataset includes five ESCC samples and five matched adjacent normal tissue samples

load required packages

```
##Load packages
library(affy)
library(arrayQualityMetrics)
library(ArrayExpress)
library(limma)
library(siggenes)
library(GEOquery)
library(ArrayExpress)
library(htmltools)
library(biomaRt)
library(tximport)
library(tximeta)
library(minfi)
library(ENmix)
library(IlluminaHumanMethylationEPICmanifest)
library(readr)
library(dplyr)
library(tidyr)
library(minfi)
library(FDb.InfiniumMethylation.hg19)
library(minfiData)
library(IlluminaHumanMethylationEPICanno.ilm10b4.hg19)
library(GenomicRanges)
library(ggplot2)
```

set the working directory

```
setwd("/Users/kimchaeyoung/Desktop/Applied_HT_analysis/Project")
dir <- "/Users/kimchaeyoung/Desktop/Applied_HT_analysis/Project/GSE157807_RAW"
```

1. Load the Infinium data (.idat files)

```
fibromethyl_RG <-read.metharray.exp(dir, extended = TRUE, force = TRUE)
pheno <- read.table("phenotype_data.txt", header = T, sep = " ")
head(pheno)

##     SampleID PatientID TissueType
## 1 GSM4775294    ca-973      ESCC
## 2 GSM4775295   con-973     NORMAL
## 3 GSM4775296    ca-962      ESCC
## 4 GSM4775297   con-962     NORMAL
## 5 GSM4775298    ca-944      ESCC
## 6 GSM4775299   con-944     NORMAL
```

Match the samplenames to the phenotypic sample IDs

```
sampleNames(fibromethyl_RG) <- substr(sampleNames(fibromethyl_RG), 1, 10)
all(sampleNames(fibromethyl_RG) == pheno$SampleID)

## [1] TRUE
```

Add the phenotypic annotation

```
rownames(pheno) <- pheno$SampleID
pData(fibromethyl_RG) <- DataFrame(pheno)
```

Take a look at your data

```
fibromethyl_RG
```

```
## class: RGChannelSetExtended
## dim: 1051539 10
## metadata(0):
## assays(5): Green Red GreenSD RedSD NBeads
## rownames(1051539): 1600101 1600111 ... 99810990 99810992
## rowData names(0):
## colnames(10): GSM4775294 GSM4775295 ... GSM4775302 GSM4775303
## colData names(3): SampleID PatientID TissueType
## Annotation
##   array: IlluminaHumanMethylationEPIC
##   annotation: ilm10b4.hg19
```

```
head(pData(fibromethyl_RG)) # phenotypic data
```

```
## DataFrame with 6 rows and 3 columns
##              SampleID   PatientID  TissueType
##           <character> <character> <character>
## GSM4775294  GSM4775294      ca-973       ESCC
## GSM4775295  GSM4775295     con-973     NORMAL
## GSM4775296  GSM4775296      ca-962       ESCC
## GSM4775297  GSM4775297     con-962     NORMAL
## GSM4775298  GSM4775298      ca-944       ESCC
## GSM4775299  GSM4775299     con-944     NORMAL
```

```
head(getAnnotation(fibromethyl_RG)) # probe/gene annotation data
```

```
## DataFrame with 6 rows and 46 columns
##                   chr       pos      strand       Name    AddressA
##           <character> <integer> <character> <character> <character>
## cg18478105       chr20  61847650          -  cg18478105    46761277
## cg09835024        chrX  24072640          -  cg09835024    16745152
## cg14361672        chr9 131463936          +  cg14361672    51800947
## cg01763666       chr17  80159506          +  cg01763666    37768834
## cg12950382       chr14 105176736          +  cg12950382     8726444
## cg02115394       chr13 115000168          +  cg02115394    68602543
##               AddressB            ProbeSeqA            ProbeSeqB
##           <character>          <character>          <character>
## cg18478105    86644198 AAATAAATTTCACTCTCAAA.. AAATAAATTTCGCTCTCAAA..
## cg09835024    81631976 AATAAACACCAACCCCAAAC.. AATAAACGCCGACCCCGAAC..
## cg14361672     7743487 ATCACTACCTAATCTATAAC.. ATCACTACCTAATCTATAAC..
## cg01763666    23754592 AAACAAAAATAAACAAACTC.. GAACAAAAATAAACGAACTC..
## cg12950382    76660327 ATACCAAAAAATAACAATAT.. GTACCGAAAAATAACAATAT..
## cg02115394    59659581 AAAATCACTACAACACCTCT.. AAAATCGCTACGACGCCTCT..
##                  Type    NextBase       Color    Probe_rs Probe_maf
##           <character> <character> <character> <character> <numeric>
```

```
## cg18478105          I          C        Grn         NA       NA
## cg09835024          I          A        Red         NA       NA
## cg14361672          I          T        Red         NA       NA
## cg01763666          I          C        Grn         NA       NA
## cg12950382          I          A        Red  rs12882277  0.378464
## cg02115394          I          A        Red         NA       NA
##               CpG_rs   CpG_maf      SBE_rs   SBE_maf          Islands_Name
##          <character> <numeric> <character> <numeric>           <character>
## cg18478105         NA        NA          NA        NA chr20:61846843-61848..
## cg09835024         NA        NA          NA        NA chrX:24072558-24073135
## cg14361672         NA        NA          NA        NA chr9:131464843-13146..
## cg01763666         NA        NA          NA        NA
## cg12950382         NA        NA          NA        NA
## cg02115394         NA        NA          NA        NA chr13:115000148-1150..
##           Relation_to_Island      Forward_Sequence             SourceSeq
##                  <character>           <character>           <character>
## cg18478105            Island TCCCGTCTTACGGGATGGAT.. CGGTCCCCGCCCCGCCGCAC..
## cg09835024            Island AGCCCCGTCATAGGTGGGCG.. GGTGGGCGCCGACCCCGAGC..
## cg14361672           N_Shore TCACCTTCCCACCTCCTGGA.. CGCAGGATGCCAGGACGGGC..
## cg01763666           OpenSea CTGGAATGCCAGCTGCTGCT.. CGCCTGCCTCAGCTTAGCTG..
## cg12950382           OpenSea CCCTGCTGCCACCACCTCGG.. CGCCACTCGGGCACGTGCAC..
## cg02115394            Island TTCTGGGGAAAGAAGGCTCA.. GGAATCGCTGCGGCGCCTCT..
##           UCSC_RefGene_Name UCSC_RefGene_Accession UCSC_RefGene_Group
##                 <character>           <character>       <character>
## cg18478105           YTHDF1             NM_017798            TSS200
## cg09835024           EIF2S3             NM_001415           TSS1500
## cg14361672             PKN3             NM_013355           TSS1500
## cg01763666           CCDC57             NM_198082              Body
## cg12950382        INF2;INF2 NM_022489;NM_001031714         Body;Body
## cg02115394      CDC16;CDC16 NM_003903;NM_001078645     TSS200;TSS200
##           Phantom4_Enhancers Phantom5_Enhancers        DMR X450k_Enhancer
##                  <character>       <character> <character>    <character>
## cg18478105
## cg09835024
## cg14361672
## cg01763666
## cg12950382
## cg02115394
##                      HMM_Island Regulatory_Feature_Name
##                     <character>           <character>
## cg18478105   20:61317142-61318498    20:61846284-61847956
## cg09835024                          X:24071907-24073667
## cg14361672
## cg01763666   17:77752688-77752973
## cg12950382 14:104247518-104247873
## cg02115394 13:114018251-114018976  13:115000009-115001429
##           Regulatory_Feature_Group   GencodeBasicV12_NAME
##                         <character>           <character>
## cg18478105     Promoter_Associated         YTHDF1;YTHDF1
## cg09835024     Promoter_Associated               EIF2S3
## cg14361672                                       PKN3
## cg01763666
## cg12950382
## cg02115394     Promoter_Associated CDC16;CDC16;CDC16;CD..
```

```
##           GencodeBasicV12_Accession  GencodeBasicV12_Group
##                         <character>           <character>
## cg18478105    ENST00000370334.4;EN..         TSS200;TSS200
## cg09835024        ENST00000253039.4               TSS200
## cg14361672        ENST00000291906.4              TSS1500
## cg01763666
## cg12950382
## cg02115394    ENST00000375312.3;EN.. TSS200;TSS1500;TSS15..
##              GencodeCompV12_NAME GencodeCompV12_Accession
##                      <character>             <character>
## cg18478105         YTHDF1;YTHDF1   ENST00000370334.4;EN..
## cg09835024   EIF2S3;EIF2S3;EIF2S3   ENST00000487075.1;EN..
## cg14361672                 PKN3       ENST00000291906.4
## cg01763666
## cg12950382            INF2;INF2   ENST00000474229.1;EN..
## cg02115394 CDC16;CDC16;CDC16;CD..   ENST00000360383.3;EN..
##             GencodeCompV12_Group DNase_Hypersensitivity_NAME
##                      <character>               <character>
## cg18478105         TSS200;TSS200     chr20:61847520-61847..
## cg09835024 TSS1500;TSS1500;TSS200     chrX:24072600-24073395
## cg14361672              TSS1500
## cg01763666                           chr17:80159145-80159..
## cg12950382         5'UTR;TSS1500
## cg02115394 TSS200;TSS1500;TSS15..
##           DNase_Hypersensitivity_Evidence_Count    OpenChromatin_NAME
##                                      <character>          <character>
## cg18478105                                    3
## cg09835024                                    3
## cg14361672                                      chr9:131463740-13146..
## cg01763666                                    3
## cg12950382                                      chr14:105171651-1051..
## cg02115394                                      chr13:114999804-1150..
##           OpenChromatin_Evidence_Count   TFBS_NAME TFBS_Evidence_Count
##                                <character> <character>        <character>
## cg18478105
## cg09835024
## cg14361672                           3
## cg01763666
## cg12950382                           6
## cg02115394                           6
##           Methyl27_Loci Methyl450_Loci Random_Loci
##             <character>    <character> <character>
## cg18478105                       TRUE
## cg09835024                       TRUE
## cg14361672                       TRUE
## cg01763666                       TRUE
## cg12950382                       TRUE
## cg02115394                       TRUE
```

```
head(getRed(fibromethyl_RG)) # red intensities
```

```
##        GSM4775294 GSM4775295 GSM4775296 GSM4775297 GSM4775298 GSM4775299
## 1600101       8583      12827       3830       9367       4954      10840
## 1600111       3453       2761       2645       2980       2439       2774
```

```
## 1600115      17883      22055      16548      18121      15096      19506
## 1600123       8856       3802       6330       4031       8918       3456
## 1600131       8768       9344       9284       9278       8703       9871
## 1600135       6165       4901       1791       2787       6313       3551
##         GSM4775300 GSM4775301 GSM4775302 GSM4775303
## 1600101       5862      11990       9392      12172
## 1600111       2854       3314       5044       3472
## 1600115      18868      20872      20537      21858
## 1600123      10723       4022       4632       4070
## 1600131       9592      10419      10582      10919
## 1600135       6684       3320       3028       3217
```

```
head(getGreen(fibromethyl_RG)) # green intensities
```

```
##         GSM4775294 GSM4775295 GSM4775296 GSM4775297 GSM4775298 GSM4775299
## 1600101      10460       6324      11666       9540      12510       8582
## 1600111       8063       8561       8776       7726       8740       9129
## 1600115       2273       2644       1756       2398       2162       2650
## 1600123       3899       7585       4028       6637       4704       8566
## 1600131        452        701        378       1095        725       1101
## 1600135       8546       9023       8894       9067       8439      10444
##         GSM4775300 GSM4775301 GSM4775302 GSM4775303
## 1600101      12631      10981      11332      10098
## 1600111       9813      10796       8260       9675
## 1600115       2296       2969       2813       3319
## 1600123       5983       8557       9007       7526
## 1600131        838       1308        822       2712
## 1600135       9149      12027      11627      12507
```

```
head(getNBeads(fibromethyl_RG)) # number of beads on the array
```

```
##         GSM4775294 GSM4775295 GSM4775296 GSM4775297 GSM4775298 GSM4775299
## 1600101         17         12         10          3         14         14
## 1600111         16         13         11         15         10         20
## 1600115         14         13          6         21          9          9
## 1600123          6          6         13         12          9          9
## 1600131         16         10         14         15         17         12
## 1600135         13         12          8         19         20         10
##         GSM4775300 GSM4775301 GSM4775302 GSM4775303
## 1600101         18         15         15         12
## 1600111         11         15         11         16
## 1600115         15         14         19         11
## 1600123         11         10         12         10
## 1600131         18         12         15         14
## 1600135          6         18         16         22
```

Subset the data

```
fibromethyl_RG <- fibromethyl_RG[, pData(fibromethyl_RG)$TissueType %in% c("ESCC",
"NORMAL")]
fibromethyl_RG
```

```
## class: RGChannelSetExtended
## dim: 1051539 10
```

```
## metadata(0):
## assays(5): Green Red GreenSD RedSD NBeads
## rownames(1051539): 1600101 1600111 ... 99810990 99810992
## rowData names(0):
## colnames(10): GSM4775294 GSM4775295 ... GSM4775302 GSM4775303
## colData names(3): SampleID PatientID TissueType
## Annotation
##   array: IlluminaHumanMethylationEPIC
##   annotation: ilm10b4.hg19
```

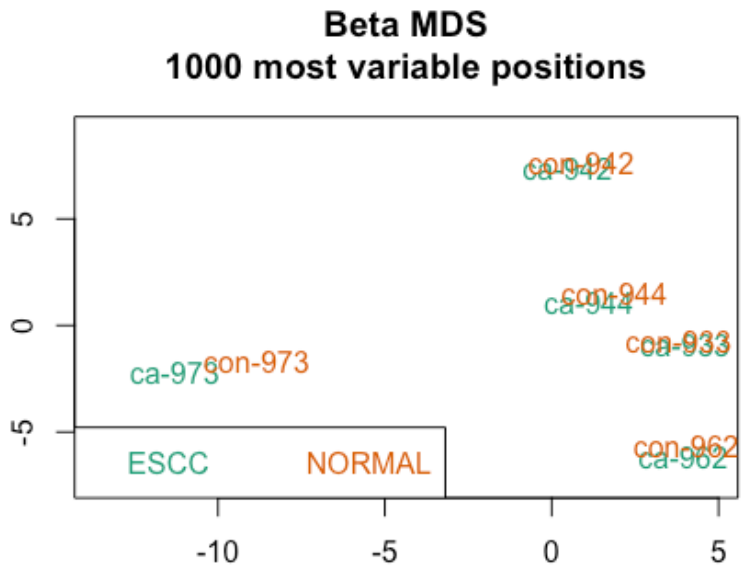
Change sample names to something more meaningful

```
sampleNames(fibromethyl_RG) <- paste(sampleNames(fibromethyl_RG),
pData(fibromethyl_RG)$PatientID, sep = "_")
```

MDSplot

The MDS plot shows distinct clustering of Tumor and ESCC samples, indicating a strong tissue type effect on methylation.

```
mdsPlot(fibromethyl_RG, sampNames = pheno$PatientID, sampGroups = pheno$TissueType)
```



2. Convert RGChannelSetExtended to MethylSet (nopreprocessing)

```
fibromethyl_raw <- preprocessRaw(fibromethyl_RG)
fibromethyl_raw

## class: MethylSet
## dim: 865859 10
```

```
## metadata(0):
## assays(2): Meth Unmeth
## rownames(865859): cg18478105 cg09835024 ... cg10633746 cg12623625
## rowData names(0):
## colnames(10): GSM4775294_ca-973 GSM4775295_con-973 ...
##   GSM4775302_ca-933 GSM4775303_con-933
## colData names(3): SampleID PatientID TissueType
## Annotation
##   array: IlluminaHumanMethylationEPIC
##   annotation: ilm10b4.hg19
## Preprocessing
##   Method: Raw (no normalization or bg correction)
##   minfi version: 1.50.0
##   Manifest version: 0.3.0
```

Function to calculate M-values

```
getM <- function(Methylset, offset = 1){
  M <- log2((getMeth(Methylset)+offset)/(getUnmeth(Methylset)+offset))
  return(M)
}
```

Take a look at the raw M- and beta-values

```
head(getB(fibromethyl_raw, offset = 100)) # offset is 100 by default
```

```
##           GSM4775294_ca-973 GSM4775295_con-973 GSM4775296_ca-962
## cg18478105        0.02037731         0.03109891        0.02054859
## cg09835024        0.06172306         0.06293996        0.02378288
## cg14361672        0.61400311         0.56718493        0.65933080
## cg01763666        0.82792407         0.74343134        0.83734793
## cg12950382        0.95796548         0.95489454        0.97870947
## cg02115394        0.07590685         0.09547401        0.09380235
##           GSM4775297_con-962 GSM4775298_ca-944 GSM4775299_con-944
## cg18478105         0.03490071         0.02270157         0.02921835
## cg09835024         0.03853107         0.03552134         0.04718218
## cg14361672         0.60851766         0.61657085         0.62235043
## cg01763666         0.84669986         0.79726187         0.80676104
## cg12950382         0.97014384         0.96592052         0.95276292
## cg02115394         0.11122933         0.08732973         0.09858528
##           GSM4775300_ca-942 GSM4775301_con-942 GSM4775302_ca-933
## cg18478105        0.04628510         0.03395232        0.02584976
## cg09835024        0.06220547         0.02824983        0.03910984
## cg14361672        0.67525773         0.65724069        0.44645227
## cg01763666        0.78959714         0.79593858        0.84665083
## cg12950382        0.97379239         0.96207481        0.95842527
## cg02115394        0.05235188         0.10389376        0.11233870
##           GSM4775303_con-933
## cg18478105         0.03023432
## cg09835024         0.03624792
## cg14361672         0.64133140
## cg01763666         0.77531340
## cg12950382         0.97697621
## cg02115394         0.08184721
```

```r
head(getM(fibromethyl_raw, offset = 1)) # offset is 1 by default
```

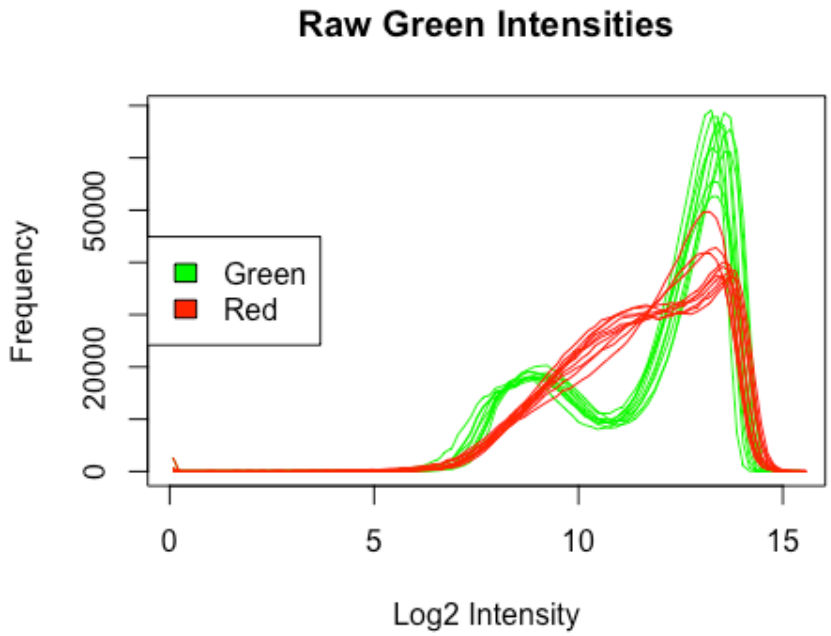
```
##           GSM4775294_ca-973 GSM4775295_con-973 GSM4775296_ca-962
## cg18478105         -5.572929         -4.9498211        -5.5544487
## cg09835024         -3.907059         -3.8815180        -5.3301306
## cg14361672          0.695017          0.4072892         0.9819998
## cg01763666          2.353933          1.5770407         2.4752358
## cg12950382          5.046578          4.7951514         6.5451254
## cg02115394         -3.586201         -3.2293933        -3.2539252
##           GSM4775297_con-962 GSM4775298_ca-944 GSM4775299_con-944
## cg18478105         -4.7730217        -5.4109550         -5.039479
## cg09835024         -4.6200376        -4.7373754         -4.316086
## cg14361672          0.6589631         0.7108818          0.744739
## cg01763666          2.5679166         2.0456904          2.132450
## cg12950382          5.7611918         5.4810015          4.717919
## cg02115394         -2.9813925        -3.3693331         -3.177278
##           GSM4775300_ca-942 GSM4775301_con-942 GSM4775302_ca-933
## cg18478105         -4.352037         -4.8188211        -5.2214340
## cg09835024         -3.890386         -5.0823355        -4.5982407
## cg14361672          1.084901          0.9641675        -0.2968053
## cg01763666          1.979098          2.0227699         2.5564314
## cg12950382          6.173927          5.2650289         4.9940694
## cg02115394         -4.157782         -3.0928352        -2.9667276
##           GSM4775303_con-933
## cg18478105         -4.991068
## cg09835024         -4.715291
## cg14361672          0.859330
## cg01763666          1.839024
## cg12950382          6.327118
## cg02115394         -3.474192
```

3. Quality control on raw data

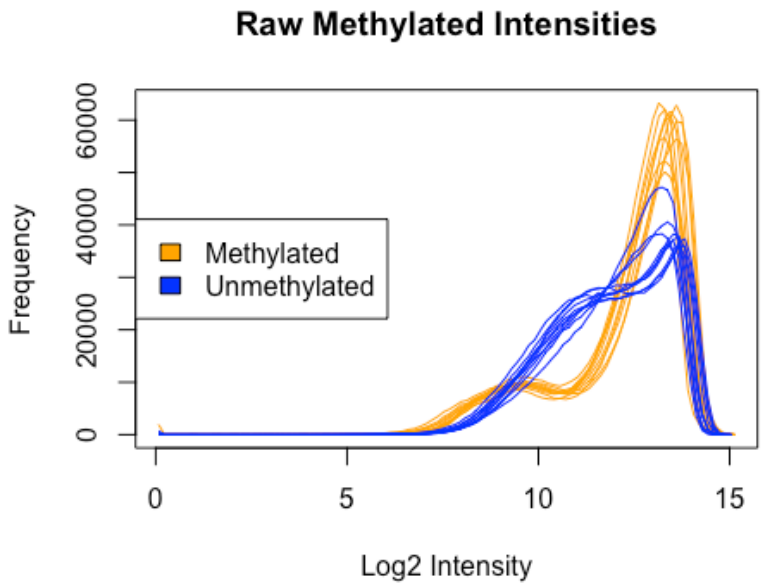
The raw data plots highlight the variability and technical biases present before preprocessing. The Red & Green Intensities and Meth & Unmeth Intensities show significant differences between samples, indicating the need for normalization to reduce technical noise. Similarly, the Sum of Meth & Unmeth Intensities and Combined Meth & Unmeth Intensities reveal inconsistent distributions across samples, reflecting batch effects and other biases. The Beta-values and M-values distributions are uneven and lack clear separation, further emphasizing the importance of preprocessing to make the data suitable for downstream methylation analysis.

```r
# Plot red and green intensities together (use RGChannelSet)
multifreqpoly(
  mat = log2(getGreen(fibromethyl_RG) + 1),
```

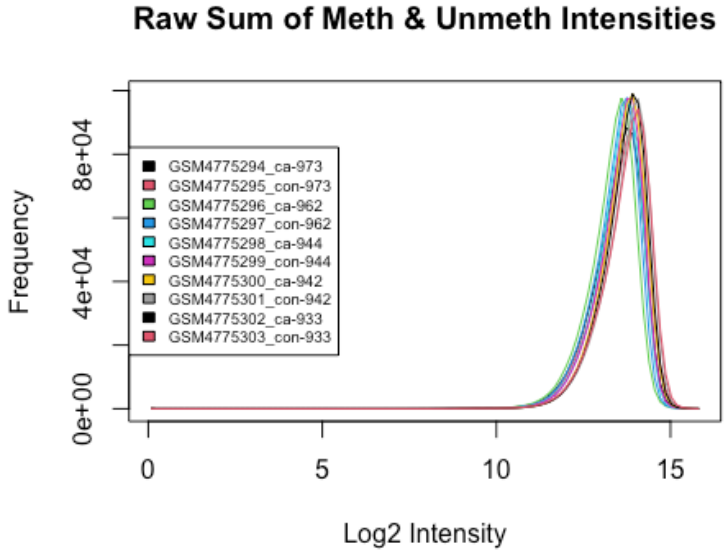
```
  col = "green",
  main = "Raw Green Intensities",
  xlab = "Log2 Intensity",
  legend = NULL # remove legend
)
multifreqpoly(
  mat = log2(getRed(fibromethyl_RG) + 1),
  col = "red",
  append = TRUE, # combine 2 plots
  legend = NULL # no legend here
)
legend("left", legend = c("Green", "Red"), fill = c("green", "red")) # add legend
manually
```



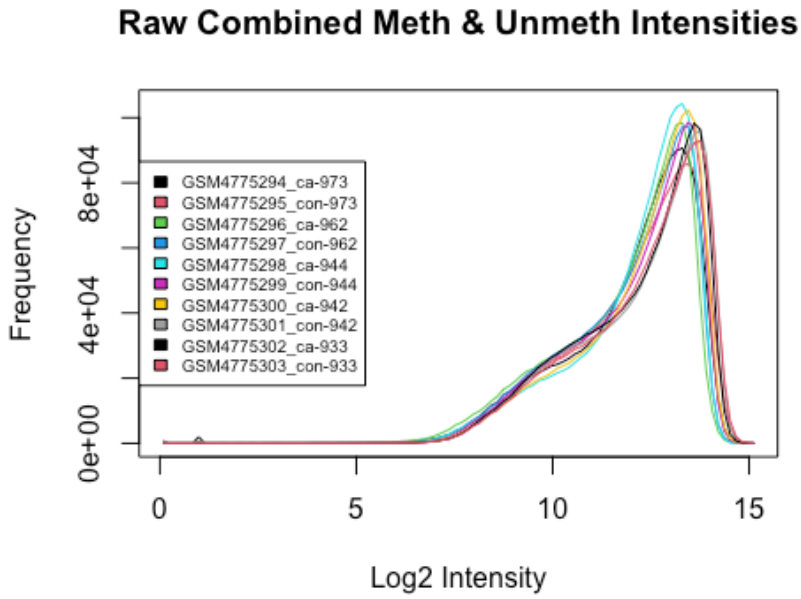

```
# Plot methylated and unmethylated intensities together (use MethylSet)
multifreqpoly(
  mat = log2(getMeth(fibromethyl_raw) + 1),
  col = "orange",
  main = "Raw Methylated Intensities",
  xlab = "Log2 Intensity",
  legend = NULL
)
multifreqpoly(
  mat = log2(getUnmeth(fibromethyl_raw) + 1),
  col = "blue",
  append = TRUE,
  legend = NULL # no legend here
)
legend("left", legend = c("Methylated", "Unmethylated"), fill = c("orange", "blue"))
# add legend manually
```



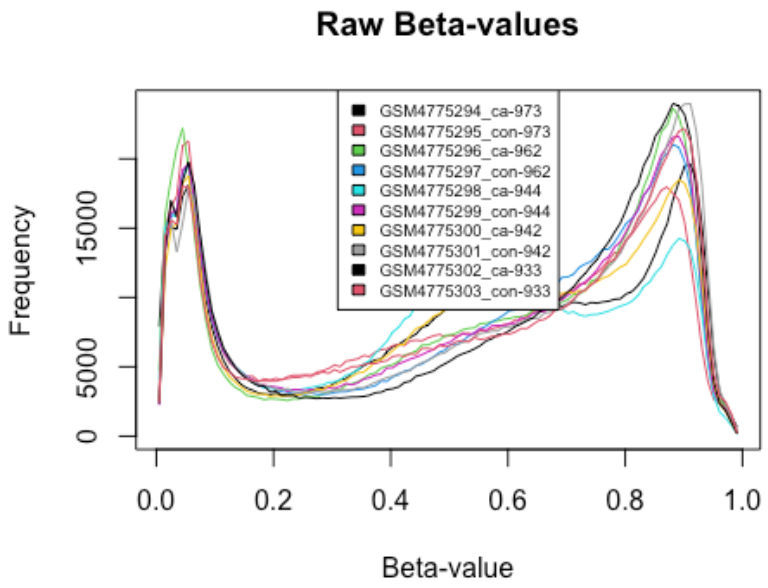

```r
# Plot summed methylated and unmethylated intensities
multifreqpoly(
  mat = log2(getMeth(fibromethyl_raw) + getUnmeth(fibromethyl_raw) + 1),
  main = "Raw Sum of Meth & Unmeth Intensities",
  xlab = "Log2 Intensity",
  legend = list(
    x = "left",
    legend = colnames(fibromethyl_raw), fill = 1:ncol(fibromethyl_raw),
    cex = 0.6
  )
)
```



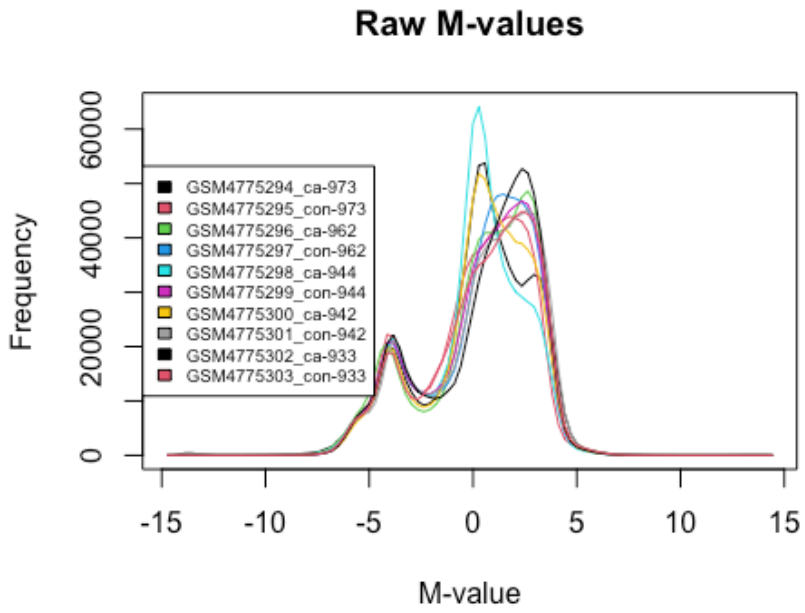
```r
# Plot combined methylated and unmethylated intensities
multifreqpoly(
  mat = log2(rbind(getMeth(fibromethyl_raw) + 1, getUnmeth(fibromethyl_raw)) + 1),
  main = "Raw Combined Meth & Unmeth Intensities",
  xlab = "Log2 Intensity",
  legend = list(
    x = "left",
    legend = colnames(fibromethyl_raw),
    fill = 1:ncol(fibromethyl_raw),
    cex = 0.6
  )
)
```




```r
# Plot raw Beta-values
multifreqpoly(
  mat = getB(fibromethyl_raw),
  main = "Raw Beta-values",
  xlab = "Beta-value",
  legend = list(
    x = "top",
    legend = colnames(fibromethyl_raw),
    fill = 1:ncol(fibromethyl_raw),
    cex = 0.6
  )
)
```




```r
# Plot raw M-values
multifreqpoly(
  mat = getM(fibromethyl_raw),
  main = "Raw M-values",
  xlab = "M-value",
  legend = list(
    x = "left",
    legend = colnames(fibromethyl_raw),
    fill = 1:ncol(fibromethyl_raw),
    cex = 0.6
  )
)
```



4. Filtering and Preprocessing

4.1. Quality control

- qc_CpG - A total of 5141 CpG sites were classified as low-quality, and majority of CpG sites have minimal low-quality data, supporting the dataset's overall reliability.
- freqpolygon_beta_beforeQC & frepolygon_beta_afterGC - The graph exhibits two peaks, one near 0 (unmethylated CpGs) and another near 1 (methylated CpGs). This bimodal pattern is typical for methylation data and indicates a clear separation of methylated and unmethylated sites. Minimal changes founded to the raw distribution. This may indicate that the data was already consistent and reliable.

```
# Quality control function
fibromethyl_QC <- QCinfo(fibromethyl_RG)

## 0  samples with percentage of low quality CpG value greater than  0.05  or
bisulfite intensity less than  11092.61
## 5141  CpGs with percentage of low quality value greater than  0.05
## Ploting qc_sample.jpg ...

## Done
## Ploting qc_CpG.jpg ...

## Done
## Identifying ourlier samples based on beta or total intensity values...
```

```
## After excluding low quality samples and CpGs
## 0  samples are outliers based on averaged total intensity value
## 0  samples are outliers in beta value distribution
## 0  outlier samples were added into badsample list
## WARNING: Sample size may be too small to correctly identify outlier samples!
## RECOMMAND: set outlier=FALSE or double check total intensity and beta value
##      distribution plots to confirm
## Ploting freqpolygon_beta_beforeQC.jpg ...

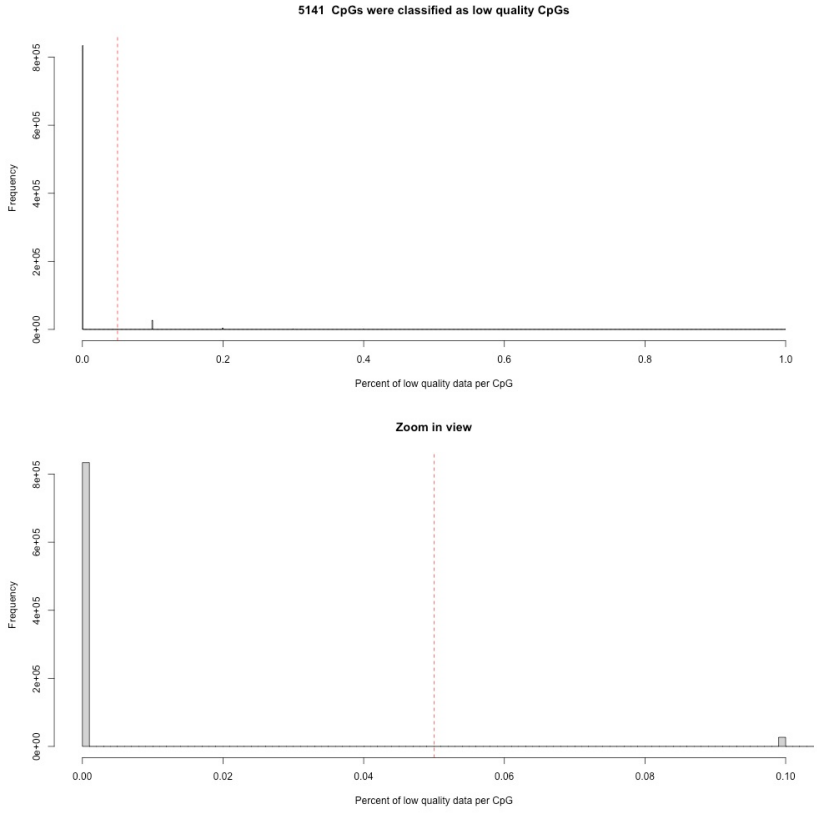
## Done
## Ploting freqpolygon_beta_afterQC.jpg ...

## Done

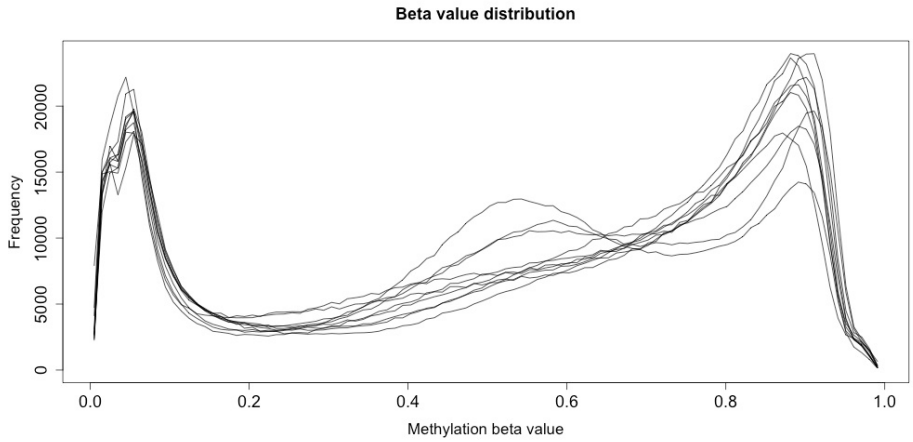
names(fibromethyl_QC)

## [1] "detP"          "nbead"         "bisul"         "badsample"
## [5] "badCpG"        "outlier_sample"

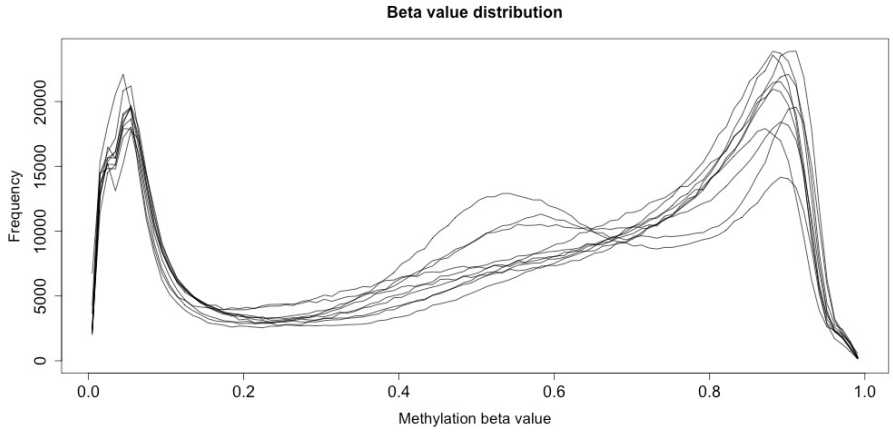
knitr::include_graphics("qc_CpG.jpg")
```




```
knitr::include_graphics("freqpolygon_beta_beforeQC.jpg")
```




```
knitr::include_graphics("freqpolygon_beta_afterQC.jpg")
```




```
length(fibromethyl_QC$badsample) #  no bad quality samples
```

```
## [1] 0
```

```
length(fibromethyl_QC$outlier_sample) #no outlier samples
```

```
## [1] 0
```

```
length(fibromethyl_QC$badCpG)
```

```
## [1] 5141
```

4.2. Checkout bad quality probes

```
# # bad quality probes
# NA probes
NA_CpGs <- is.na(rowSums(getB(fibromethyl_raw)))
sum(NA_CpGs)
```

```
## [1] 0
```

```r
NA_CpGs <- names(NA_CpGs)[NA_CpGs]

NA_CpGs # no NA probes

## character(0)
```

4.3 preprocessing with "ENmix" package

```r
fibromethyl_BC <- preprocessENmix(
  fibromethyl_RG ,
  bgParaEst = "oob",
  dyeCorr = "RELIC",
  QCinfo = fibromethyl_QC ,
  exQCsample = TRUE , # Exclude samples that don't meet quality control criteria (set
to TRUE to exclude)
  exQCcpg = TRUE , # Exclude CpGs that don't meet quality control criteria (set to
TRUE to exclude)
  exSample = NULL , # List of additional samples to exclude (set to NULL if none)
  exCpG = NA_CpGs  # List of CpGs to exclude (NA_CpGs should be excluded)
)

## 5141  CpGs were excluded before ENmix correction
## Analysis is running, please wait...!

fibromethyl_BC

## class: MethylSet
## dim: 860718 10
## metadata(0):
## assays(2): Meth Unmeth
## rownames(860718): cg18478105 cg09835024 ... cg10633746 cg12623625
## rowData names(0):
## colnames(10): GSM4775294_ca-973 GSM4775295_con-973 ...
##   GSM4775302_ca-933 GSM4775303_con-933
## colData names(3): SampleID PatientID TissueType
## Annotation
##   array: IlluminaHumanMethylationEPIC
##   annotation: ilm10b4.hg19
## Preprocessing
##   Method: NA
##   minfi version: NA
##   Manifest version: NA
```

4.4. Normalization with 'norm.quantile' function

```r
# 'fibromethyl_N' is the new object containing the normalized data.
# The quantile2 method adjusts the data to a common distribution across samples,
making cross-sample comparisons easier.
fibromethyl_N <- norm.quantile(fibromethyl_BC , method = "quantile2")

## Analysis is running, please wait...!

fibromethyl_N
```

```
## class: MethylSet
## dim: 860718 10
## metadata(0):
## assays(2): Meth Unmeth
## rownames(860718): cg18478105 cg09835024 ... cg10633746 cg12623625
## rowData names(0):
## colnames(10): GSM4775294_ca-973 GSM4775295_con-973 ...
##   GSM4775302_ca-933 GSM4775303_con-933
## colData names(3): SampleID PatientID TissueType
## Annotation
##   array: IlluminaHumanMethylationEPIC
##   annotation: ilm10b4.hg19
## Preprocessing
##   Method: Backgroud_corr: ENmix,oob
##   minfi version: dyeBiasCorrection: RELIC
##   Manifest version: Normalization: quantile2
```

5. Quality Control on preprocessed data

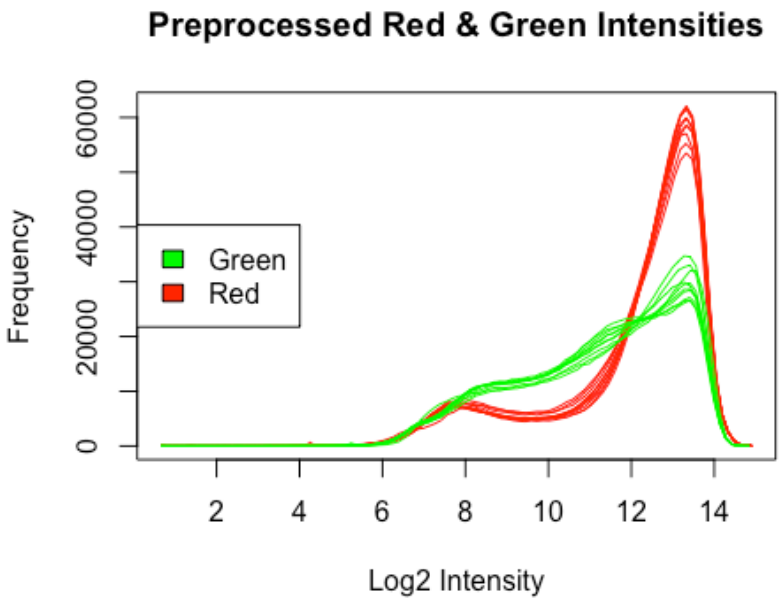
- The Red & Green Intensities now overlap, indicating balanced signal intensities, and the Meth & Unmeth Intensities show clear separation, reflecting proper adjustment. The Sum of Meth & Unmeth Intensities and Combined Meth & Unmeth Intensities have consistent distributions across samples, reducing technical variability. Finally, the Beta-values display a bimodal distribution, while the M-values are more clearly defined, ensuring that the data is ready for robust methylation analysis and comparison between conditions.

```r
# Which probes are green and red and TII
color <- getAnnotation(fibromethyl_N)$Color
green <- color == "Grn"
red <- color == "Red"
t2 <- color == ""

# Plot red and green intensities together
multifreqpoly(
  mat = log2(rbind(
    getMeth(fibromethyl_N)[(red | t2), ] + 1,
    getUnmeth(fibromethyl_N)[red, ] + 1 )),
  col = "red",
  main = "Preprocessed Red & Green Intensities",
  xlab = "Log2 Intensity",
  legend = NULL # remove legend
)

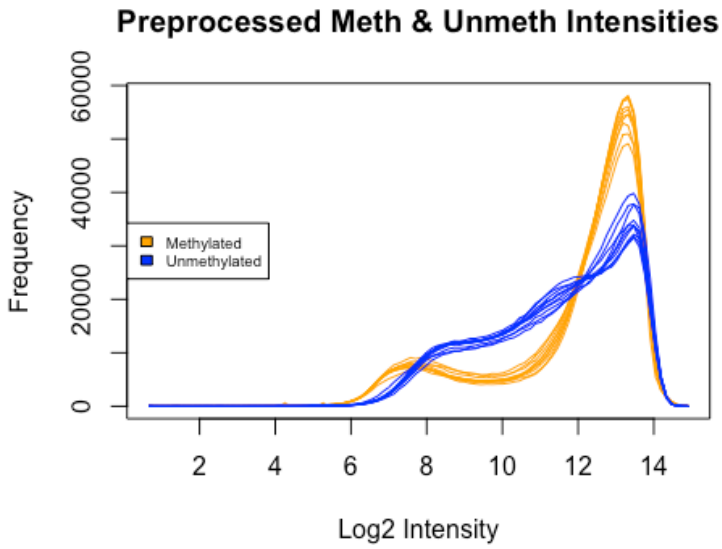
multifreqpoly(
  mat = log2(rbind(
    getMeth(fibromethyl_N)[green, ] + 1,
```

```
    getUnmeth(fibromethyl_N)[(green | t2), ] + 1
  )),
  col = "green",
  append = TRUE, # use append to combine 2 plots
  legend = list(
    x = "left", # change legend position
    legend = c("Green", "Red"), # change legend values
    fill = c("green", "red") # change legend color
  )
)
```

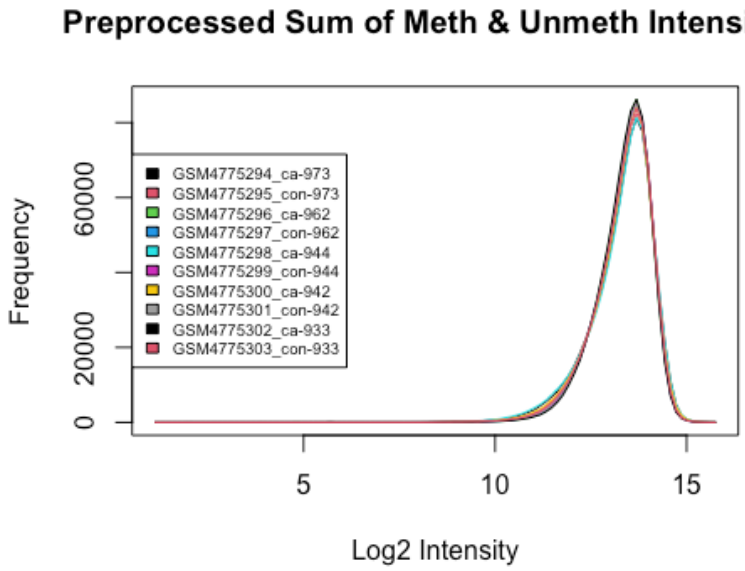



```
# Plot methylated and unmethylated intensities together (use MethylSet)
multifreqpoly(
  mat = log2(getMeth(fibromethyl_N) + 1),
  col = "orange",
  main = "Preprocessed Meth & Unmeth Intensities",
  xlab = "Log2 Intensity",
  legend = NULL
)

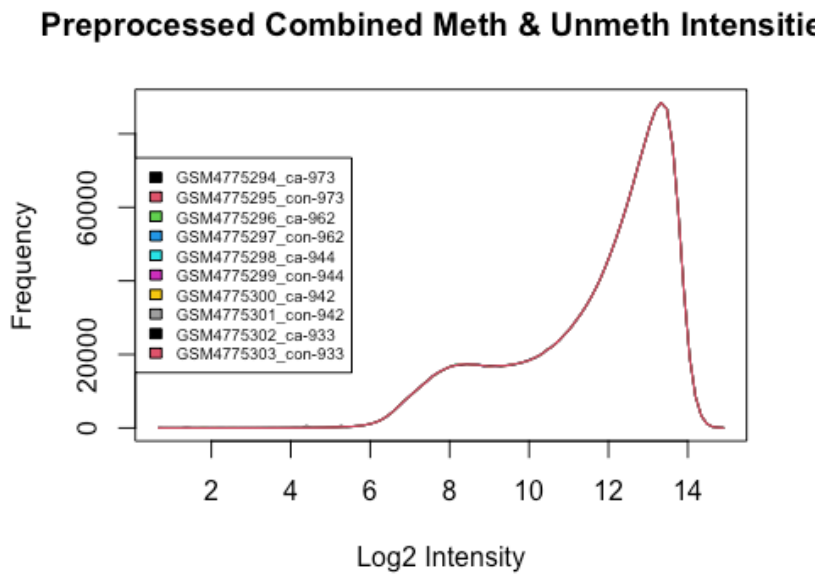
multifreqpoly(
  mat = log2(getUnmeth(fibromethyl_N) + 1),
  col = "blue",
  append = TRUE ,
  legend = list(
    x = "left",
    legend = c("Methylated", "Unmethylated"),
    fill = c("orange", "blue"),
    cex = 0.6
  )
)
```



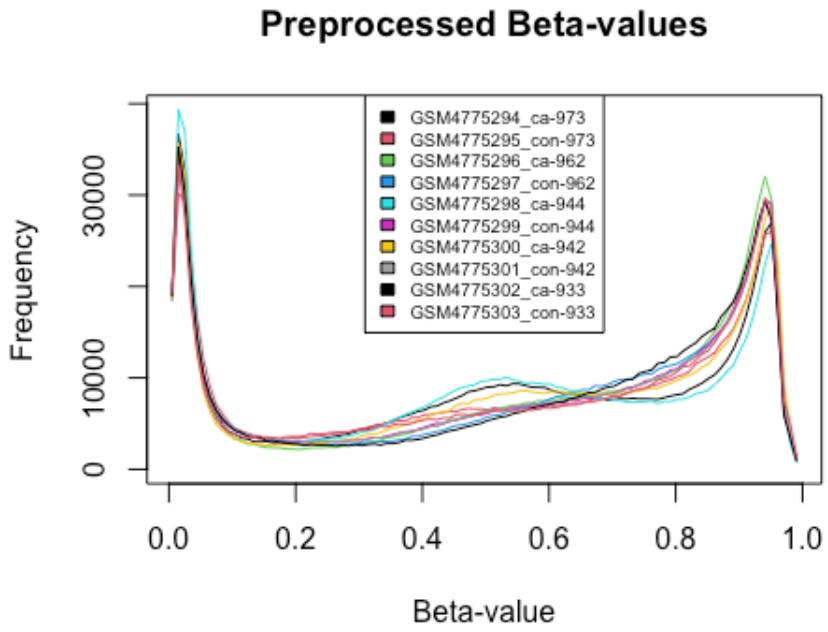

```r
# Plot summed meth and unmeth intensities
multifreqpoly(
  mat = log2(getMeth(fibromethyl_N) + getUnmeth(fibromethyl_N) + 1),
  main = "Preprocessed Sum of Meth & Unmeth Intensities",
  xlab = "Log2 Intensity",
  legend = list(
    x = "left",
    legend = colnames(fibromethyl_N),
    fill = 1:ncol(fibromethyl_N),
    cex = 0.6
  )
)
```



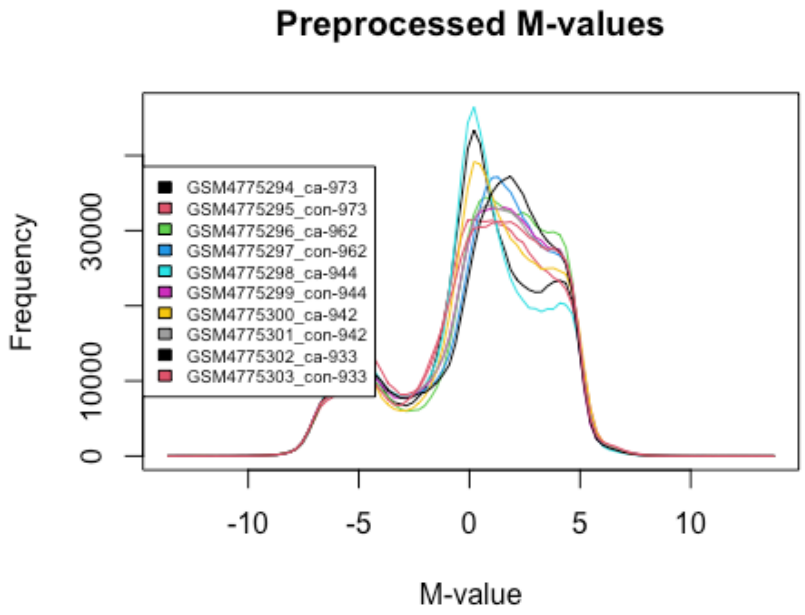
```r
# Plot combined meth and unmeth intensities
multifreqpoly(
  mat = log2(rbind(getMeth(fibromethyl_N) + 1, getUnmeth(fibromethyl_N)
  ) + 1),
  main = "Preprocessed Combined Meth & Unmeth Intensities",
  xlab = "Log2 Intensity",
  legend = list(
    x = "left",
    legend = colnames(fibromethyl_N),
    fill = 1:ncol(fibromethyl_N),
    cex = 0.6
  )
)
```




```r
# Plot beta-values
multifreqpoly(
  mat = getB(fibromethyl_N),
  main = "Preprocessed Beta-values",
  xlab = "Beta-value",
  legend = list(
    x = "top",
    legend = colnames(fibromethyl_N),
    fill = 1:ncol(fibromethyl_N),
    cex = 0.6
  )
)
```




```r
# Plot M-values
multifreqpoly(
  mat = getM(fibromethyl_N),
  main = "Preprocessed M-values",
  xlab = "M-value",
  legend = list(
    x = "left",
    legend = colnames(fibromethyl_N),
    fill = 1:ncol(fibromethyl_N),
    cex = 0.6
  )
)
```



6. Differetial Methylaton analysis

6.1. Define the experimental design

- The block variable is set to the PatientID column, which groups samples by patient. This blocking factor accounts for the paired structure of the data, where each patient may contribute samples for both tissue types. By using table, the balance of samples between tissue types for each patient is checked.
- Design matrix is created with model.matrix to specify the relationship between the response variable (e.g., methylation) and the predictor (TissueType). This matrix includes dummy variables for tissue types, allowing the model to evaluate the effects of "ESCC" compared to "NORMAL".

```
# 1. Get col_data information (patient data and tissue type)
col_data <- pData(fibromethyl_N)

# 2. Extract the tissue type information from the sample metadata and define it as a
factor with levels "ESCC" (tumor) and "NORMAL" (healthy tissue)
col_data$TissueType <- factor(col_data$TissueType, levels = c("NORMAL", "ESCC"))

# 3. Use sub() function to remove the "ca-" and "con-" prefixes from PatientID
col_data$PatientID <- sub("^ca-|^con-", "", col_data$PatientID)
```

```r
# 4. Set up the 'block' for adjusting patient-specific effects in the model
# Here, PatientID is used as the blocking factor to control for patient-specific
effects (e.g., each patient could have samples from both tissue types)
block <- col_data$PatientID
table(col_data$PatientID, col_data$TissueType)  # Check how many samples per patient
and tissue type
```

```
##
##      NORMAL ESCC
##   933      1    1
##   942      1    1
##   944      1    1
##   962      1    1
##   973      1    1
```

```r
# 5. Create the design matrix based on TissueType
# The model matrix is created to specify the design of the study, where the outcome
variable is modeled as a function of tissue type
design <- model.matrix(~ TissueType, data = col_data)

# 6. Check the design matrix to verify its structure
# The design matrix should show how each tissue type is coded (e.g., dummy variable
for "ESCC" versus "NORMAL")
print(design)
```

```
##                   (Intercept) TissueTypeESCC
## GSM4775294_ca-973           1             1
## GSM4775295_con-973          1             0
## GSM4775296_ca-962           1             1
## GSM4775297_con-962          1             0
## GSM4775298_ca-944           1             1
## GSM4775299_con-944          1             0
## GSM4775300_ca-942           1             1
## GSM4775301_con-942          1             0
## GSM4775302_ca-933           1             1
## GSM4775303_con-933          1             0
## attr(,"assign")
## [1] 0 1
## attr(,"contrasts")
## attr(,"contrasts")$TissueType
## [1] "contr.treatment"
```

6.2. Estimate the correlation between samples from the same patient (patient-level correlation)

- Estimated correlation is around 0.06. It suggests a weak correlation between samples from the same patient. Weak correlation (0.06) implies that most of the variability in the methylation data is likely due to differences in tissue type

("NORMAL" vs "ESCC") or other factors, rather than being heavily influenced by patient-specific characteristics.

```
# This is important when analyzing data from multiple samples per patient
#block = patient=specific block factor, getM  provides methylation data matrix
cor_fit <- duplicateCorrelation(getM(fibromethyl_N), design, block = block)

# A result around 0.06 suggests weak correlation, indicating that the samples from
the same patient do not exhibit strong similarity.
print(cor_fit$consensus.correlation)

## [1] 0.06511738
```

6.3. Fit a linear model to methylation M-values

-Fit a linear model to the methylation M-values using the lmFit function from the limma package. The design matrix specifies the comparison of interest (tissue type), the block accounts for patient-specific effects, and the correlation parameter incorporates the estimated correlation between paired samples from the same patient. This ensures that tissue-type differences are modeled while controlling for residual dependencies within patients.

```
# Step 2: Fit a linear model to methylation M-values
fit <- lmFit(getM(fibromethyl_N), design, block = block, correlation =
cor_fit$consensus.correlation)
```

6.4. Define contrasts for differential methylation

-A contrast matrix is created using makeContrasts to specify the comparison of interest—in this case, "ESCC vs NORMAL." This step sets up the framework for testing differential methylation between the two tissue types.

```
# Create a contrast matrix to compare ESCC vs NORMAL
contrast_matrix <- makeContrasts(ESCC_vs_NORMAL = TissueTypeESCC, levels = design)

## Warning in makeContrasts(ESCC_vs_NORMAL = TissueTypeESCC, levels = design):
## Renaming (Intercept) to Intercept
```

Empirical Bayes smoothing with eBayes is then performed to stabilize the variance estimates, improving statistical power and reliability in detecting differential methylation signals.

```
# Apply contrasts to the model using the contrast matrix and store in 'fit2'.
fit2 <- contrasts.fit(fit, contrast_matrix)
```

```r
#Perform empirical Bayes smoothing to improve statistical power
fit2 <- eBayes(fit2)
```

6.5. Extract differentially methylated CpG sites

-The topTable function generates a table of differentially methylated CpG probes, sorted by significance. Adjusted p-values are calculated using the Benjamini-Hochberg (BH) method to control the false discovery rate (FDR). The resulting table provides information on the most significant CpG sites, including fold changes, p-values, and adjusted p-values.

```r
# Generate a table of differentially methylated probes
# Adjust p-values using the Benjamini-Hochberg method (FDR control)
# Form result table
LIMMAout <- topTable(fit2, adjust = "BH", number = Inf)

# print result
head(LIMMAout)
```

```
##              logFC  AveExpr        t      P.Value    adj.P.Val        B
## cg23500559 5.231598 4.168630 25.35673 2.393338e-11 1.919478e-05 13.20075
## cg06770731 4.743592 3.199862 22.97741 7.176152e-11 1.919478e-05 12.68568
## cg22811668 5.368269 3.128400 22.63223 8.491750e-11 1.919478e-05 12.60171
## cg06492111 4.953505 3.924389 21.82208 1.273125e-10 1.919478e-05 12.39419
## cg01082422 4.741744 4.025537 21.80189 1.286276e-10 1.919478e-05 12.38882
## cg04808027 5.154883 3.576289 21.72451 1.338054e-10 1.919478e-05 12.36817
```

6.6. Count the number of genes with adjusted p-value < 0.05

-Out of all tested CpG sites, 35,994 (about 4.18%) are significantly differentially methylated at an adjusted p-value threshold of 0.05.

```r
sum(LIMMAout$adj.P.Val < 0.05)
```

```
## [1] 35994
```

```r
sum(LIMMAout$adj.P.Val<0.05)/length(LIMMAout$adj.P.Val)
```

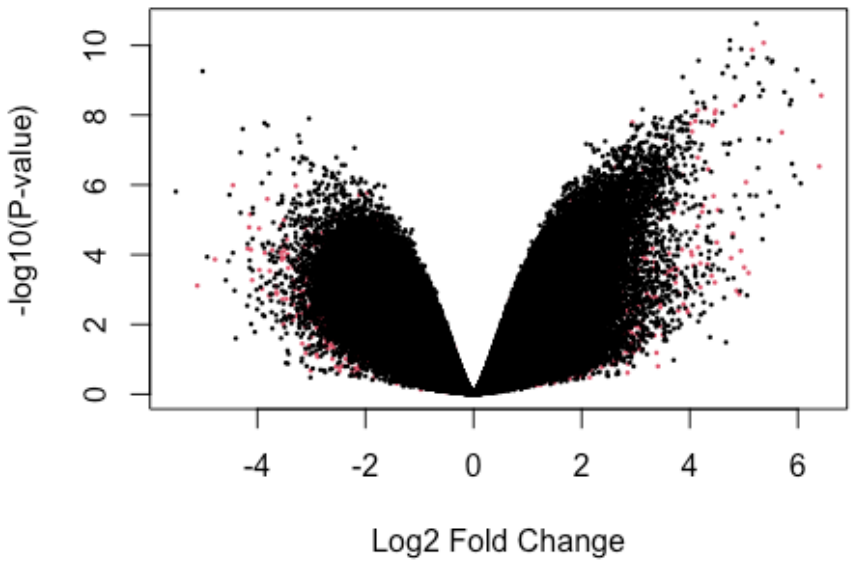
```
## [1] 0.04181857
```

6.7. Volcano plot

-Significant sites (adjusted p-value < 0.05) are highlighted in color (e.g., red), indicating differential methylation with either higher (positive log2 fold change) or lower (negative log2 fold change) methylation in "ESCC" compared to "NORMAL".

The larger the -log10(p-value), the stronger the statistical significance.The roughly symmetrical distribution around the y-axis suggests that both hypermethylation (positive log2 fold change) and hypomethylation (negative log2 fold change) are present in "ESCC" compared to "NORMAL."

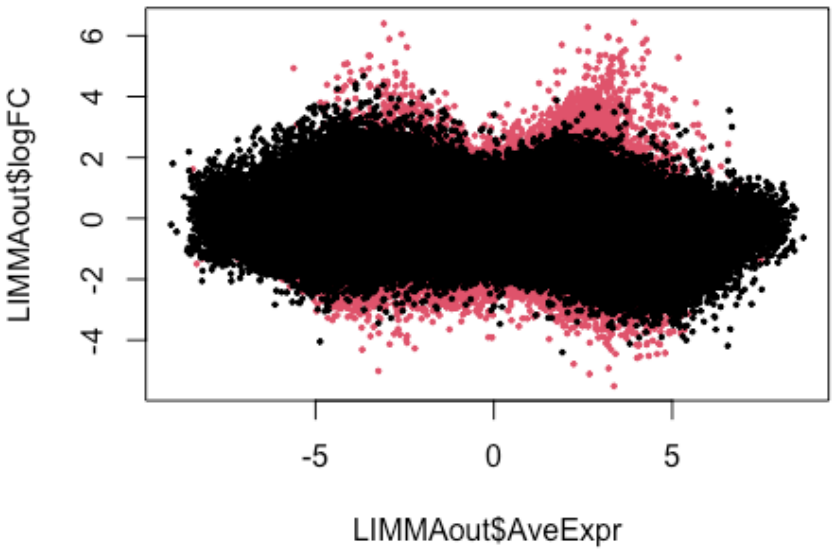
```
volcanoplot(fit2, style="p-value", col=as.factor(LIMMAout$adj.P.Val < 0.05))
```



6.8. MA plot

-The majority of points cluster around a log fold change of 0, indicating no substantial differences in methylation for many CpG sites. The colored points (red) represent CpG sites with significant differential methylation, as determined by their adjusted p-value. These are spread across both positive and negative logFC values, indicating both hypermethylation and hypomethylation in "ESCC" compared to "NORMAL.

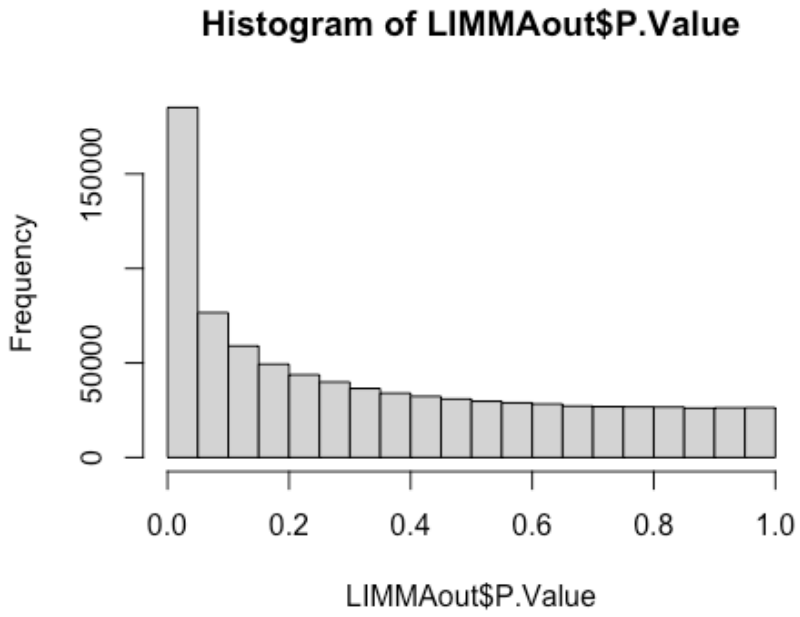
```
plot(LIMMAout$AveExpr, LIMMAout$logFC, col=as.factor(LIMMAout$adj.P.Val<0.05),
pch=20, cex=0.5)
```



6.9. p-value histogram

- The bar on the very left indicates a large number of CpG sites with very low p-values (close to 0). This suggests that many CpG sites show strong evidence of differential methylation between tissue types ("ESCC" vs "NORMAL"). Beyond approximately 0.1, the p-values are distributed more uniformly, which is expected under the null hypothesis.

```
hist(x = LIMMAout$P.Value)
```



7. Check Methylation for top results

7.1. Check M-values for top results

```
#Check M-values for top results
getM(fibromethyl_N)[rownames(head(LIMMAout)), ]
```

```
##           GSM4775294_ca-973 GSM4775295_con-973 GSM4775296_ca-962
## cg23500559          7.042695          1.2607643          6.829152
## cg06770731          5.539882          0.8153324          5.781019
## cg22811668          5.948366          0.1074965          5.715503
## cg06492111          6.019603          1.2807072          6.479344
## cg01082422          6.396022          1.8120104          6.579354
## cg04808027          6.766562          0.9467753          6.075986
##           GSM4775297_con-962 GSM4775298_ca-944 GSM4775299_con-944
## cg23500559           2.006711          6.720035          1.3034339
## cg06770731           1.386503          5.269010          0.6053885
## cg22811668           1.059007          5.558811          0.4097697
## cg06492111           2.023288          6.497081          1.2022266
## cg01082422           1.584071          6.170070          1.8697109
## cg04808027           1.540002          5.976971          0.9094709
##           GSM4775300_ca-942 GSM4775301_con-942 GSM4775302_ca-933
## cg23500559          6.812625          1.4718434          6.517637
## cg06770731          5.660398          0.6044385          5.607982
## cg22811668          6.310795          0.3652889          5.529195
## cg06492111          6.691989          1.1144202          6.317692
## cg01082422          6.840083          1.7733337          5.996518
## cg04808027          5.750769          0.8070853          6.198364
##           GSM4775303_con-933
## cg23500559          1.7214003
```

```
## cg06770731         0.7286678
## cg22811668         0.2797632
## cg06492111         1.6175409
## cg01082422         1.2342000
## cg04808027         0.7909058

# Check beta-values for top results
getB(fibromethyl_N)[rownames(head(LIMMAout)), ]

##           GSM4775294_ca-973 GSM4775295_con-973 GSM4775296_ca-962
## cg23500559         0.9881152         0.7032138         0.9868919
## cg06770731         0.9743836         0.6353753         0.9772876
## cg22811668         0.9786155         0.5169109         0.9763719
## cg06492111         0.9789182         0.7048864         0.9827594
## cg01082422         0.9796719         0.7733197         0.9812806
## cg04808027         0.9862625         0.6558413         0.9803935
##           GSM4775297_con-962 GSM4775298_ca-944 GSM4775299_con-944
## cg23500559          0.7977593         0.9864938          0.7091791
## cg06770731          0.7204559         0.9699283          0.6011142
## cg22811668          0.6730617         0.9741997          0.5685523
## cg06492111          0.7981340         0.9833155          0.6937993
## cg01082422          0.7451105         0.9783843          0.7797380
## cg04808027          0.7411806         0.9791933          0.6501532
##           GSM4775300_ca-942 GSM4775301_con-942 GSM4775302_ca-933
## cg23500559         0.9867557         0.7324584         0.9849898
## cg06770731         0.9758034         0.6009955         0.9753112
## cg22811668         0.9823566         0.5610731         0.9741889
## cg06492111         0.9846396         0.6807849         0.9815741
## cg01082422         0.9837269         0.7678668         0.9765842
## cg04808027         0.9769617         0.6339980         0.9816623
##           GSM4775303_con-933
## cg23500559          0.7647443
## cg06770731          0.6212733
## cg22811668          0.5464472
## cg06492111          0.7502327
## cg01082422          0.6969803
## cg04808027          0.6314376

# Add the mean betas of the CAF and NAF group, and their absolute difference
LIMMAout$normalBeta <- rowMeans(getB(fibromethyl_N)[rownames(LIMMAout),
pData(fibromethyl_N)$TissueType == "NORMAL"])
LIMMAout$tumorBeta <- rowMeans(getB(fibromethyl_N)[rownames(LIMMAout),
pData(fibromethyl_N)$TissueType == "ESCC"])

deltaBeta <- getB(fibromethyl_N)[, pData(fibromethyl_N)$TissueType == "ESCC"] -
getB(fibromethyl_N)[, pData(fibromethyl_N)$TissueType == "NORMAL"]

# Calculate the mean absolute value of differences between patients by CpG site
LIMMAout$deltaBeta <- rowMeans(abs(deltaBeta))
```

7.2. Annotation

```
# Extract annotation data from the preprocessed methylation object
anno <- getAnnotation(fibromethyl_N)
```

```r
# Sort in the same order as limma output, remove filtered CpGs, add missing CpGs
anno <- anno[rownames(LIMMAout),]

# Update row names of the annotation data to match the LIMMA output
rownames(anno) <- rownames(LIMMAout)
# note that if any CpGs were present in the limma output but not in the annotation ,
# these have now been added as NA rows
# any CpGs that were present in the annotation but not in LIMMAout have been removed

colnames(anno)
```

```
##  [1] "chr"
##  [2] "pos"
##  [3] "strand"
##  [4] "Name"
##  [5] "AddressA"
##  [6] "AddressB"
##  [7] "ProbeSeqA"
##  [8] "ProbeSeqB"
##  [9] "Type"
## [10] "NextBase"
## [11] "Color"
## [12] "Probe_rs"
## [13] "Probe_maf"
## [14] "CpG_rs"
## [15] "CpG_maf"
## [16] "SBE_rs"
## [17] "SBE_maf"
## [18] "Islands_Name"
## [19] "Relation_to_Island"
## [20] "Forward_Sequence"
## [21] "SourceSeq"
## [22] "UCSC_RefGene_Name"
## [23] "UCSC_RefGene_Accession"
## [24] "UCSC_RefGene_Group"
## [25] "Phantom4_Enhancers"
## [26] "Phantom5_Enhancers"
## [27] "DMR"
## [28] "X450k_Enhancer"
## [29] "HMM_Island"
## [30] "Regulatory_Feature_Name"
## [31] "Regulatory_Feature_Group"
## [32] "GencodeBasicV12_NAME"
## [33] "GencodeBasicV12_Accession"
## [34] "GencodeBasicV12_Group"
## [35] "GencodeCompV12_NAME"
## [36] "GencodeCompV12_Accession"
## [37] "GencodeCompV12_Group"
## [38] "DNase_Hypersensitivity_NAME"
## [39] "DNase_Hypersensitivity_Evidence_Count"
## [40] "OpenChromatin_NAME"
## [41] "OpenChromatin_Evidence_Count"
## [42] "TFBS_NAME"
```

```
## [43] "TFBS_Evidence_Count"
## [44] "Methyl27_Loci"
## [45] "Methyl450_Loci"
## [46] "Random_Loci"
```

7.3. Add interesting annotation to limma output

```
LIMMAout$Chromosome <- anno[,"chr"]  # - Chromosome: Chromosomal location of each CpG site
LIMMAout$Position <- anno[,"pos"] # - Position: Genomic position of the CpG site within the chromosome
LIMMAout$Gene_ID <- anno[,"UCSC_RefGene_Name"] # - Gene_ID: Gene(s) associated with the CpG site based on RefGene annotations
LIMMAout$CpG_Feature <- anno[,"UCSC_RefGene_Group"] # - CpG_Feature: Specific genomic feature where the CpG site is located
LIMMAout$CpG_Island <- anno[,"Relation_to_Island"] # - CpG_Island: Relation of the CpG site to CpG islands

head(LIMMAout)
```

```
##              logFC  AveExpr       t     P.Value    adj.P.Val       B
## cg23500559 5.231598 4.168630 25.35673 2.393338e-11 1.919478e-05 13.20075
## cg06770731 4.743592 3.199862 22.97741 7.176152e-11 1.919478e-05 12.68568
## cg22811668 5.368269 3.128400 22.63223 8.491750e-11 1.919478e-05 12.60171
## cg06492111 4.953505 3.924389 21.82208 1.273125e-10 1.919478e-05 12.39419
## cg01082422 4.741744 4.025537 21.80189 1.286276e-10 1.919478e-05 12.38882
## cg04808027 5.154883 3.576289 21.72451 1.338054e-10 1.919478e-05 12.36817
##            normalBeta tumorBeta   deltaBeta Chromosome  Position      Gene_ID
## cg23500559  0.7414710 0.9866493 0.003626046     chr17  60744836         MRC2
## cg06770731  0.6358428 0.9745428 0.005058628     chr16  87863812       SLC7A5
## cg22811668  0.5732090 0.9771465 0.091810723     chr19   2046303   MKNK2;MKNK2
## cg06492111  0.7255674 0.9822414 0.050626221     chr16  87874392       SLC7A5
## cg01082422  0.7526031 0.9799295 0.005092396      chr5 179224751 MGAT4B;MGAT4B
## cg04808027  0.6625222 0.9808947 0.019776366     chr11  66102146         RIN1
##            CpG_Feature CpG_Island
## cg23500559        Body     Island
## cg06770731      3'UTR    OpenSea
## cg22811668   Body;Body     Island
## cg06492111        Body    OpenSea
## cg01082422 3'UTR;3'UTR     Island
## cg04808027        Body     Island
```

8. Gene-level analysis

```
## Summarize data on the gene level

# remove CpGs with no gene annotation
LIMMAout_gene <- LIMMAout[LIMMAout$Gene_ID != "" & !is.na( LIMMAout$Gene_ID), ]

# Save CpG rownames (otherwise these are removed in the next step)
LIMMAout_gene$CpG <- rownames(LIMMAout_gene)

# Separate CpGs with multiple genes/features (this takes a while to run )
```

```r
LIMMAout_gene <- LIMMAout_gene %>%
  separate_rows(Gene_ID , CpG_Feature , CpG_Island , sep = ";") %>%
  distinct() %>%
  group_by(CpG, Gene_ID) %>%
  mutate(CpG_Feature = paste(unique(CpG_Feature), collapse = ";"),
         CpG_Island = paste(unique(CpG_Island), collapse = ";")) %>%
  ungroup() %>%
  distinct() %>%
  as.data.frame()

# Select the most interesting columns(CpG, GeneID,Cpg_features, CPG_island, logFC,
adg.p.val, normalBeeta, tumorBeta,)
#CpG: The CpG site identifier.
#Gene_ID: The gene associated with the CpG site.
#CpG_Feature: The feature of the CpG (e.g., promoter region, exon).
#CpG_Island: The CpG island associated with the site.
#logFC: The log fold-change of the methylation level between conditions (e.g., tumor
vs. normal).
#adj.P.Val: The adjusted p-value for statistical significance.
#normalBeeta: The methylation beta value in the normal tissue.
#tumorBeta: The methylation beta value in the tumor tissue.

LIMMAout_gene <- LIMMAout_gene[, c(15, 12, 13, 14, 1, 5, 7, 8, 9)]
head(LIMMAout_gene)

##         CpG Gene_ID CpG_Feature CpG_Island    logFC    adj.P.Val normalBeta
## 1 cg23500559    MRC2       Body     Island 5.231598 1.919478e-05  0.7414710
## 2 cg06770731  SLC7A5      3'UTR    OpenSea 4.743592 1.919478e-05  0.6358428
## 3 cg22811668   MKNK2       Body     Island 5.368269 1.919478e-05  0.5732090
## 4 cg06492111  SLC7A5       Body    OpenSea 4.953505 1.919478e-05  0.7255674
## 5 cg01082422  MGAT4B      3'UTR     Island 4.741744 1.919478e-05  0.7526031
## 6 cg04808027    RIN1       Body     Island 5.154883 1.919478e-05  0.6625222
##   tumorBeta   deltaBeta
## 1 0.9866493 0.003626046
## 2 0.9745428 0.005058628
## 3 0.9771465 0.091810723
## 4 0.9822414 0.050626221
## 5 0.9799295 0.005092396
## 6 0.9808947 0.019776366
```

8.1. Count Differentially Methylated Genes

Filter CpG sites associated with genes, separate rows with multiple annotations, and group CpGs by genes to summarize features.

```r
# Count all significant defferential methylation CpGs with adjusted p-values < 0.05.
A higher value indicates more CpG sites showing methylation differences.
sum(LIMMAout$adj.P.Val < 0.05)

## [1] 35994

sum(LIMMAout$adj.P.Val<0.05)/length(LIMMAout$adj.P.Val)
```

```
## [1] 0.04181857

# Count unique differentially methylated genes
length(unique(LIMMAout_gene$Gene_ID))

## [1] 27311

# Count unique differentially methylated genes with significant adjusted p-values.
This value represents key targets for biological interpretation and is a main result
of the study.
length(unique(LIMMAout_gene$Gene_ID[LIMMAout_gene$adj.P.Val < 0.05]))

## [1] 10317
```

8.2. Look for multiple CpGs in the same gene undergoing similar methylation differences

Count significant CpGs (adjusted p-value < 0.05). Identify unique differentially methylated genes, with or without adjusted p-value filtering.

```
# Filter by adjusted p-value < 0.05 and |logFC| > 1
#  For each top gene, print information related to differentially methylated CpG
sites.
topgenes <- unique(LIMMAout_gene$Gene_ID)[1:10] # ten most significant genes
for (gene in topgenes) {
  LIMMAout_subset <-
    LIMMAout_gene[(LIMMAout_gene$Gene_ID == gene) & (LIMMAout_gene$adj.P.Val < 0.05)
&
                  (abs(LIMMAout_gene$logFC) > 1), ]
  print(LIMMAout_subset)
}

##             CpG Gene_ID CpG_Feature CpG_Island    logFC    adj.P.Val
## 1     cg23500559    MRC2        Body     Island 5.231598 1.919478e-05
## 165   cg26168577    MRC2        Body    OpenSea 3.257395 6.162687e-04
## 246   cg02314166    MRC2        Body    OpenSea 3.766102 8.665471e-04
## 1308  cg09689317    MRC2        Body     Island 2.725378 3.391470e-03
## 1559  cg13636531    MRC2        Body    S_Shore 3.061417 3.984832e-03
## 2131  cg21768884    MRC2      3'UTR    OpenSea 2.939778 5.452166e-03
## 3048  cg19683251    MRC2        Body    S_Shelf 1.908320 7.518461e-03
## 3063  cg06873351    MRC2      3'UTR    OpenSea 1.916555 7.548388e-03
## 4815  cg13999321    MRC2        Body    OpenSea 1.574236 1.172052e-02
## 6285  cg06376951    MRC2        Body    N_Shelf 1.072474 1.498265e-02
## 14085 cg27320439    MRC2        Body    S_Shore 1.743332 2.933174e-02
## 18279 cg14497573    MRC2        Body    S_Shore 2.193979 3.623379e-02
## 21121 cg19831935    MRC2        Body    S_Shore 2.557798 4.059284e-02
## 21743 cg13108002    MRC2        Body    N_Shore 1.298865 4.161619e-02
## 25487 cg11192417    MRC2        Body    S_Shore 2.278540 4.761702e-02
## 26222 cg01522412    MRC2        Body    N_Shore 1.065033 4.876372e-02
##       normalBeta tumorBeta    deltaBeta
## 1      0.7414710 0.9866493 0.0036260464
## 165    0.6889926 0.9497156 0.0019112256
```

```
## 246    0.7460100 0.9705855 0.0872313346
## 1308   0.7376160 0.9389893 0.1589075023
## 1559   0.7126731 0.9462014 0.0433822361
## 2131   0.6970057 0.9339538 0.0013304755
## 3048   0.7608639 0.9178367 0.0274312598
## 3063   0.5295088 0.8006418 0.0690671492
## 4815   0.8310593 0.9335023 0.1296821789
## 6285   0.7673429 0.8707354 0.0116166321
## 14085  0.4631714 0.7284125 0.0255891726
## 18279  0.5884273 0.8438233 0.0014501855
## 21121  0.6871139 0.8985885 0.0226049688
## 21743  0.5786470 0.7636686 0.0486937747
## 25487  0.5759453 0.8365111 0.0008519207
## 26222  0.7512718 0.8594512 0.0687247143
##             CpG Gene_ID  CpG_Feature CpG_Island    logFC    adj.P.Val
## 2     cg06770731  SLC7A5        3'UTR    OpenSea 4.743592 1.919478e-05
## 4     cg06492111  SLC7A5         Body    OpenSea 4.953505 1.919478e-05
## 35    cg06071246  SLC7A5        3'UTR    OpenSea 3.123341 1.632642e-04
## 37    cg03486383  SLC7A5        3'UTR    OpenSea 3.929510 1.801198e-04
## 48    cg27135163  SLC7A5        3'UTR    OpenSea 3.470018 2.251219e-04
## 60    cg10034481  SLC7A5        3'UTR    OpenSea 4.004042 2.460341e-04
## 73    cg04264781  SLC7A5         Body    OpenSea 3.385898 2.817190e-04
## 96    cg26619943  SLC7A5         Body    OpenSea 3.483341 3.623082e-04
## 106   cg03879320  SLC7A5         Body    N_Shelf 2.868143 4.309852e-04
## 120   cg20757869  SLC7A5 ExonBnd;Body    OpenSea 3.306592 5.010289e-04
## 134   cg00653312  SLC7A5        3'UTR    OpenSea 3.051812 5.239395e-04
## 152   cg05393733  SLC7A5         Body    OpenSea 2.783449 5.680618e-04
## 201   cg10169763  SLC7A5         Body    OpenSea 3.187618 6.629683e-04
## 233   cg10099957  SLC7A5         Body    OpenSea 3.080374 7.744473e-04
## 263   cg04481596  SLC7A5        3'UTR    OpenSea 2.623416 9.289323e-04
## 313   cg00052925  SLC7A5         Body    OpenSea 2.937030 1.055272e-03
## 319   cg12365660  SLC7A5         Body    OpenSea 2.813832 1.076581e-03
## 412   cg03408354  SLC7A5         Body    OpenSea 2.322280 1.335943e-03
## 482   cg04802238  SLC7A5         Body    OpenSea 2.580464 1.454497e-03
## 586   cg16239834  SLC7A5         Body    OpenSea 1.998070 1.707792e-03
## 673   cg04171052  SLC7A5         Body    OpenSea 1.855024 1.907008e-03
## 698   cg26637881  SLC7A5         Body    OpenSea 2.569461 1.926752e-03
## 778   cg08094280  SLC7A5        3'UTR    OpenSea 1.791085 2.087982e-03
## 822   cg09618385  SLC7A5        3'UTR    OpenSea 1.730656 2.241591e-03
## 888   cg07558761  SLC7A5         Body    OpenSea 2.120085 2.381934e-03
## 889   cg13452738  SLC7A5         Body    OpenSea 3.107219 2.387582e-03
## 891   cg07021906  SLC7A5         Body    OpenSea 3.407113 2.394219e-03
## 911   cg02614661  SLC7A5         Body    OpenSea 2.532339 2.465868e-03
## 924   cg11297271  SLC7A5         Body    S_Shelf 2.108707 2.508493e-03
## 938   cg00069417  SLC7A5         Body    OpenSea 1.975439 2.522915e-03
## 1155  cg03553613  SLC7A5         Body    OpenSea 1.654515 2.969159e-03
## 1635  cg03801429  SLC7A5         Body     Island 1.507772 4.195696e-03
## 1666  cg01856752  SLC7A5         Body    OpenSea 2.128956 4.317189e-03
## 1867  cg02203067  SLC7A5         Body    OpenSea 1.965063 4.808036e-03
## 2217  cg01829163  SLC7A5         Body    OpenSea 2.211276 5.578469e-03
## 2251  cg05911082  SLC7A5         Body    OpenSea 1.845601 5.676192e-03
## 2301  cg06665333  SLC7A5         Body    OpenSea 2.486212 5.777188e-03
## 2725  cg08860287  SLC7A5         Body    OpenSea 1.505841 6.725022e-03
## 3362  cg05534653  SLC7A5         Body    N_Shelf 1.259166 8.341154e-03
```

```
## 8420  cg09285525  SLC7A5        Body    OpenSea 1.008527 1.941873e-02
## 8970  cg02454636  SLC7A5        Body    OpenSea 1.024530 2.040715e-02
## 13780 cg27467238  SLC7A5        Body    OpenSea 1.631707 2.887849e-02
## 14623 cg27555036  SLC7A5        Body    S_Shelf 2.589381 3.014473e-02
## 17551 cg27560818  SLC7A5        Body    OpenSea 1.881559 3.500927e-02
## 17841 cg08798345  SLC7A5        Body    OpenSea 1.525676 3.553382e-02
##       normalBeta tumorBeta   deltaBeta
## 2      0.6358428 0.9745428 0.005058628
## 4      0.7255674 0.9822414 0.050626221
## 35     0.5339823 0.9036311 0.002705924
## 37     0.6099555 0.9538025 0.098710142
## 48     0.6468668 0.9449307 0.036721215
## 60     0.6439844 0.9605991 0.022419743
## 73     0.8004700 0.9713585 0.054393915
## 96     0.5496089 0.9257820 0.007319308
## 106    0.5771645 0.8998299 0.031137831
## 120    0.6536694 0.9432375 0.083100517
## 134    0.6342321 0.9297037 0.058048600
## 152    0.6144654 0.9115580 0.124072664
## 201    0.6107036 0.9268692 0.005251928
## 233    0.6408401 0.9322154 0.013703580
## 263    0.6041801 0.8964762 0.103803885
## 313    0.5660373 0.9006377 0.002157308
## 319    0.6270383 0.9152810 0.158106909
## 412    0.5390938 0.8483088 0.053360507
## 482    0.7169388 0.9310916 0.110843886
## 586    0.5404808 0.8217807 0.078050030
## 673    0.6027043 0.8422565 0.098984235
## 698    0.6256917 0.9014636 0.005486620
## 778    0.6276577 0.8501573 0.021986761
## 822    0.5263061 0.7836572 0.003657920
## 888    0.5650907 0.8426766 0.104689152
## 889    0.5932710 0.9133274 0.004419630
## 891    0.6283181 0.9356214 0.061072289
## 911    0.4929465 0.8385133 0.009799716
## 924    0.4176892 0.7498636 0.077549109
## 938    0.6205203 0.8596165 0.229048478
## 1155   0.5862209 0.8119139 0.002084310
## 1635   0.5987398 0.8050927 0.008684184
## 1666   0.4851299 0.7964912 0.140306808
## 1867   0.5688260 0.8288447 0.010326125
## 2217   0.5534564 0.8409096 0.006888911
## 2251   0.5609942 0.8150819 0.224831000
## 2301   0.5207144 0.8437887 0.007795880
## 2725   0.5063465 0.7393515 0.141481522
## 3362   0.5616822 0.7507552 0.002552011
## 8420   0.5709845 0.7262993 0.215016450
## 8970   0.5480922 0.7097378 0.003905447
## 13780  0.4828548 0.7280913 0.034552925
## 14623  0.6676823 0.9024627 0.004575732
## 17551  0.6668649 0.8680180 0.092589329
## 17841  0.5530455 0.7685325 0.002659272
##             CpG Gene_ID CpG_Feature CpG_Island    logFC    adj.P.Val
## 3     cg22811668   MKNK2        Body     Island 5.368269 1.919478e-05
```

```
## 8     cg11634930   MKNK2       Body     Island 5.439430 2.369801e-05
## 27    cg03246954   MKNK2       Body     Island 4.949019 1.106738e-04
## 179   cg13220129   MKNK2       Body     Island 3.431136 6.574630e-04
## 3493  cg11319132   MKNK2       Body    S_Shore 2.704558 8.672947e-03
## 7173  cg26723446   MKNK2       Body    S_Shore 2.484140 1.691058e-02
## 8211  cg14292923   MKNK2       Body    S_Shore 2.154280 1.900195e-02
## 8309  cg06521145   MKNK2       Body    N_Shore 1.391089 1.919791e-02
## 9358  cg17585440   MKNK2    TSS1500    S_Shore 1.083446 2.101144e-02
## 10459 cg23694389   MKNK2       Body    N_Shore 1.183851 2.268639e-02
## 12745 cg21964793   MKNK2  Body;3'UTR     Island 1.091853 2.700765e-02
## 13202 cg18179735   MKNK2  Body;3'UTR     Island 1.145525 2.794902e-02
## 14452 cg08324490   MKNK2       Body    S_Shore 1.329501 2.988280e-02
##       normalBeta tumorBeta   deltaBeta
## 3      0.5732090 0.9771465 0.091810723
## 8      0.6381371 0.9801963 0.033934223
## 27     0.7515918 0.9841822 0.135793373
## 179    0.5499751 0.9223921 0.058170709
## 3493   0.8123409 0.9569958 0.008104267
## 7173   0.7678519 0.9419549 0.154385690
## 8211   0.8157285 0.9470003 0.127451066
## 8309   0.2988329 0.5239696 0.015369385
## 9358   0.3045550 0.4801226 0.003494906
## 10459  0.6251154 0.7883215 0.146683173
## 12745  0.6529415 0.7969112 0.016375891
## 13202  0.5392880 0.7160531 0.210538546
## 14452  0.7695081 0.8809253 0.019296394
##             CpG Gene_ID  CpG_Feature CpG_Island    logFC    adj.P.Val
## 5     cg01082422  MGAT4B       3'UTR     Island 4.741744 1.919478e-05
## 255   cg18484008  MGAT4B       3'UTR     Island 2.666980 9.076237e-04
## 4439  cg04942710  MGAT4B Body;TSS1500    S_Shore 1.337412 1.081444e-02
## 5616  cg23011886  MGAT4B  Body;TSS200    S_Shore 1.633629 1.352736e-02
## 8851  cg18133875  MGAT4B        Body     Island 1.937836 2.024997e-02
## 17952 cg06189266  MGAT4B        Body    N_Shore 1.397244 3.572069e-02
##       normalBeta tumorBeta   deltaBeta
## 5      0.7526031 0.9799295 0.005092396
## 255    0.6998690 0.9305111 0.056780204
## 4439   0.5175925 0.7276033 0.066283391
## 5616   0.6762016 0.8566758 0.140156397
## 8851   0.7042965 0.8877478 0.047917750
## 17952  0.6871855 0.8428847 0.004626366
##             CpG Gene_ID CpG_Feature CpG_Island    logFC    adj.P.Val
## 6     cg04808027    RIN1       Body     Island 5.154883 1.919478e-05
## 16    cg12912663    RIN1       Body     Island 4.609287 3.425004e-05
## 17    cg20749584    RIN1       Body     Island 3.871577 3.897635e-05
## 251   cg22777174    RIN1       Body    N_Shore 2.501113 8.894055e-04
## 371   cg12068791    RIN1       Body    S_Shore 2.350167 1.190420e-03
## 388   cg07543230    RIN1       Body    N_Shore 2.669071 1.230719e-03
## 552   cg07948432    RIN1       Body     Island 2.227603 1.619385e-03
## 1701  cg17274976    RIN1       Body    N_Shore 1.883427 4.399004e-03
## 1720  cg19554304    RIN1       Body    N_Shore 1.529751 4.468002e-03
## 23815 cg22719871    RIN1    TSS1500    S_Shelf 1.009564 4.513553e-02
##       normalBeta tumorBeta    deltaBeta
## 6      0.6625222 0.9808947 0.0197763660
## 16     0.5824681 0.9651364 0.1511727292
```

```
## 17     0.6220875 0.9542991 0.0387827699
## 251    0.6379298 0.9041587 0.0011364240
## 371    0.6270422 0.8908467 0.0185997317
## 388    0.6118150 0.9028406 0.0179960889
## 552    0.6733963 0.9017720 0.1878850879
## 1701   0.6023587 0.8429745 0.0005563855
## 1720   0.4899490 0.7328928 0.0577624611
## 23815  0.4903674 0.6562629 0.0700039820
##             CpG Gene_ID CpG_Feature CpG_Island    logFC    adj.P.Val
## 7     cg03598185  SCAMP4       Body    S_Shore 5.165343 2.369801e-05
## 148   cg01009414  SCAMP4       Body    S_Shore 3.081909 5.335930e-04
## 6842  cg08617069  SCAMP4       Body     Island 1.451703 1.625480e-02
## 16049 cg20243900  SCAMP4      5'UTR     Island 2.710389 3.254099e-02
## 23888 cg09759578  SCAMP4      5'UTR     Island 1.386509 4.528708e-02
##       normalBeta tumorBeta   deltaBeta
## 7      0.5604802 0.9727679 0.081412072
## 148    0.5019399 0.8890338 0.003336986
## 6842   0.8199474 0.9232984 0.039425172
## 16049  0.5115110 0.8336811 0.116785528
## 23888  0.8502727 0.9346349 0.066649973
##             CpG Gene_ID CpG_Feature CpG_Island    logFC    adj.P.Val
## 9     cg23634417  TMEM79       Body    OpenSea 4.162054 2.369801e-05
## 4538  cg15793285  TMEM79       Body    OpenSea 1.717668 1.103738e-02
## 23645 cg07234388  TMEM79  Body;3'UTR    S_Shore 1.017528 4.478596e-02
##       normalBeta tumorBeta   deltaBeta
## 9      0.7613842 0.9760728 0.060076021
## 4538   0.6929781 0.8782497 0.006954987
## 23645  0.3836959 0.5551006 0.104975123
##             CpG Gene_ID   CpG_Feature CpG_Island    logFC    adj.P.Val
## 10    cg07551314   PAQR7         Body    S_Shelf 5.529181 2.369801e-05
## 181   cg00465927   PAQR7         Body    S_Shelf 2.914898 6.574630e-04
## 1494  cg18128212   PAQR7      TSS1500    N_Shelf 2.684117 3.884025e-03
## 1916  cg19906087   PAQR7        5'UTR    OpenSea 2.313283 4.913303e-03
## 2288  cg01566199   PAQR7 1stExon;5'UTR    OpenSea 2.359152 5.746393e-03
## 19842 cg26875775   PAQR7      TSS1500    N_Shelf 2.310848 3.867677e-02
##       normalBeta tumorBeta   deltaBeta
## 10     0.6452556 0.9783074 0.003732244
## 181    0.6726108 0.9338941 0.069777710
## 1494   0.7166658 0.9344953 0.010731820
## 1916   0.6089295 0.8819786 0.001767613
## 2288   0.6759467 0.9060543 0.170217351
## 19842  0.6040530 0.8505582 0.006532866
##            CpG Gene_ID CpG_Feature CpG_Island    logFC    adj.P.Val normalBeta
## 11   cg09102257    EGR1       Body     Island 5.512564 2.410676e-05  0.4288241
## 229  cg12766556    EGR1       Body    S_Shore 4.285075 7.737816e-04  0.6540784
## 365  cg07336840    EGR1       Body     Island 3.523693 1.183424e-03  0.6029026
## 1264 cg23029363    EGR1       Body     Island 4.520912 3.267318e-03  0.6295428
## 7689 cg01107476    EGR1      3'UTR    S_Shore 1.754191 1.782142e-02  0.7579333
##      tumorBeta   deltaBeta
## 11   0.9463687 0.013712186
## 229  0.9632734 0.003931916
## 365  0.9221677 0.005125584
## 1264 0.9616977 0.003121432
## 7689 0.9027306 0.150525301
```

```
##             CpG Gene_ID CpG_Feature CpG_Island    logFC    adj.P.Val
## 12    cg13224075  SLC7A1       Body    OpenSea 5.059598 2.480818e-05
## 14    cg10752421  SLC7A1       Body    OpenSea 5.980626 3.077974e-05
## 63    cg09750646  SLC7A1      3'UTR     Island 3.119785 2.516104e-04
## 90    cg04783530  SLC7A1      5'UTR    OpenSea 3.368510 3.621356e-04
## 97    cg03284642  SLC7A1      5'UTR    OpenSea 3.983699 3.623082e-04
## 556   cg05126855  SLC7A1      5'UTR    OpenSea 2.846302 1.624056e-03
## 784   cg01213557  SLC7A1      5'UTR    OpenSea 2.273359 2.091680e-03
## 789   cg15003482  SLC7A1      5'UTR    OpenSea 3.020990 2.091680e-03
## 866   cg04769514  SLC7A1       Body     Island 2.762276 2.347621e-03
## 1121  cg03536202  SLC7A1      5'UTR    OpenSea 3.029042 2.868157e-03
## 1303  cg09214535  SLC7A1       Body    S_Shelf 1.610974 3.375324e-03
## 1521  cg05941208  SLC7A1      5'UTR    OpenSea 2.408558 3.924981e-03
## 1807  cg26117398  SLC7A1      5'UTR    OpenSea 1.922492 4.710376e-03
## 1994  cg10778996  SLC7A1       Body    OpenSea 2.725658 5.125159e-03
## 2236  cg12014634  SLC7A1      3'UTR    N_Shore 1.759301 5.645097e-03
## 3514  cg12823441  SLC7A1       Body    OpenSea 1.943335 8.710623e-03
## 3570  cg08495784  SLC7A1      5'UTR    OpenSea 2.170366 8.815224e-03
## 3829  cg10905736  SLC7A1       Body    OpenSea 1.460197 9.395026e-03
## 4288  cg02381064  SLC7A1       Body     Island 1.950493 1.043177e-02
## 4435  cg11069855  SLC7A1      5'UTR    OpenSea 1.631293 1.080875e-02
## 5087  cg09887627  SLC7A1       Body    S_Shore 2.338560 1.239649e-02
## 8285  cg21690645  SLC7A1       Body    OpenSea 3.102462 1.912770e-02
## 9983  cg04688292  SLC7A1      5'UTR    OpenSea 1.467072 2.196128e-02
## 11145 cg03904840  SLC7A1       Body    OpenSea 1.289758 2.400492e-02
## 18734 cg17798847  SLC7A1       Body    OpenSea 2.160145 3.697051e-02
## 21392 cg11760099  SLC7A1       Body    OpenSea 1.566168 4.105913e-02
## 21933 cg02497486  SLC7A1      3'UTR    N_Shelf 1.522965 4.189754e-02
## 22229 cg25493901  SLC7A1       Body     Island 1.315183 4.244247e-02
##       normalBeta tumorBeta   deltaBeta
## 12     0.6710920 0.9790536 0.002598548
## 14     0.5314048 0.9768909 0.076213209
## 63     0.5735870 0.9163209 0.053093014
## 90     0.7026508 0.9542910 0.123818246
## 97     0.6306627 0.9532739 0.034838576
## 556    0.7609771 0.9518156 0.009015770
## 784    0.7086357 0.9160705 0.060158971
## 789    0.6895719 0.9389187 0.076090393
## 866    0.5822609 0.8962410 0.003474171
## 1121   0.6767714 0.9318097 0.057247181
## 1303   0.7819040 0.9128274 0.003724563
## 1521   0.5799041 0.8688864 0.159230301
## 1807   0.5456679 0.8135757 0.155078291
## 1994   0.7413099 0.9430031 0.005299450
## 2236   0.7719866 0.9170863 0.038809083
## 3514   0.8229133 0.9400921 0.148880049
## 3570   0.5362271 0.8251402 0.092006487
## 3829   0.5554890 0.7673703 0.279102106
## 4288   0.5304848 0.8036527 0.090656394
## 4435   0.6559003 0.8463273 0.119837321
## 5087   0.7015831 0.9096264 0.004098289
## 8285   0.6693293 0.9436582 0.056368255
## 9983   0.4601259 0.6954984 0.042700462
## 11145  0.7315039 0.8620954 0.337168058
```

```
## 18734  0.6888090 0.9115451 0.221464550
## 21392  0.7152229 0.8638731 0.003307736
## 21933  0.6162338 0.8081864 0.048883060
## 22229  0.5268736 0.7274738 0.006600670
```

8.3. Look for multiple CpGs in "promoter" regions undergoing similar methylation differences

Examine significant CpGs (adjusted p-value < 0.05, |logFC| > 1) for the top 10 genes. Print detailed information for CpGs undergoing similar methylation changes within the same gene.

```
LIMMAout_prom <- LIMMAout_gene[grepl("TSS|1stExon", LIMMAout_gene$CpG_Feature),]

# Extract top 10 genes based on promoter-region CpGs
topgenes_prom <- unique(LIMMAout_prom$Gene_ID)[1:10]
for (gene in topgenes_prom) {
  LIMMAout_subset <- LIMMAout_gene[(LIMMAout_gene$Gene_ID == gene) &
                                   (LIMMAout_gene$adj.P.Val < 0.05) &
                                   (abs(LIMMAout_gene$logFC) > 1), ]
  print(LIMMAout_subset)
}
```

```
##             CpG Gene_ID         CpG_Feature CpG_Island     logFC    adj.P.Val
## 15    cg26559315   LYNX1 5'UTR;TSS200;1stExon     Island -5.012281 3.155983e-05
## 633   cg20541178   LYNX1 5'UTR;TSS200;1stExon     Island -2.395904 1.811496e-03
## 5537  cg19021188   LYNX1      TSS200;TSS1500    S_Shore  1.360272 1.341528e-02
## 6260  cg23180489   LYNX1      TSS200;TSS1500    S_Shore  1.618603 1.493045e-02
## 6984  cg22687807   LYNX1 5'UTR;TSS1500;TSS200     Island -2.030340 1.649052e-02
## 7174  cg00399059   LYNX1      TSS200;TSS1500    S_Shore  1.652115 1.691058e-02
## 12023 cg22861548   LYNX1      TSS200;TSS1500    S_Shore  1.579000 2.561314e-02
## 19483 cg14078481   LYNX1 5'UTR;TSS1500;TSS200     Island -1.676142 3.813766e-02
##       normalBeta  tumorBeta   deltaBeta
## 15    0.37686674 0.01848182 0.040404604
## 633   0.44460486 0.13566773 0.003206597
## 5537  0.78166715 0.89752181 0.068522702
## 6260  0.74868462 0.89739808 0.170720628
## 6984  0.05351832 0.01203907 0.216520943
## 7174  0.67733056 0.86226808 0.004585998
## 12023 0.72447265 0.88083060 0.001404304
## 19483 0.05759548 0.01656680 0.033141800
##             CpG Gene_ID CpG_Feature CpG_Island    logFC    adj.P.Val
## 20    cg09186408  MIR33B     TSS1500    S_Shore 5.354809 7.880409e-05
## 1020  cg04805065  MIR33B     TSS1500    S_Shore 2.393231 2.703555e-03
## 1340  cg24161106  MIR33B        Body    S_Shore 4.357819 3.483005e-03
## 9685  cg09494646  MIR33B      TSS200    S_Shore 2.015692 2.151063e-02
## 22688 cg11393407  MIR33B        Body    S_Shore 1.108190 4.315823e-02
##       normalBeta tumorBeta   deltaBeta
## 20     0.5454586 0.9710167 0.100213600
## 1020   0.5667297 0.8662002 0.104546943
## 1340   0.6646099 0.9620120 0.013174287
```

```
## 9685   0.8300868 0.9448818 0.002585406
## 22688  0.8814366 0.9382320 0.004355187
##            CpG  Gene_ID CpG_Feature CpG_Island     logFC    adj.P.Val
## 45   cg11256132 C1orf100      TSS200    OpenSea  3.358188 0.0002227901
## 6540 cg15797341 C1orf100        Body    OpenSea -2.578105 0.0155165613
##      normalBeta tumorBeta  deltaBeta
## 45    0.5802006 0.9269854 0.05428335
## 6540  0.9119444 0.6598192 0.00274622
##             CpG Gene_ID CpG_Feature CpG_Island    logFC    adj.P.Val
## 55    cg07733567 MIR1180      TSS200     Island 4.337818 0.0002439811
## 280   cg24553547 MIR1180      TSS200     Island 2.997281 0.0009693345
## 693   cg19157647 MIR1180      TSS200     Island 1.942658 0.0019141035
## 3243  cg24547137 MIR1180      TSS200     Island 1.377482 0.0079487114
## 6064  cg04864152 MIR1180      TSS200     Island 1.771365 0.0144967207
## 11974 cg05896714 MIR1180        Body     Island 1.929805 0.0255339080
##       normalBeta tumorBeta   deltaBeta
## 55     0.6514701 0.9660486 0.014685207
## 280    0.6024511 0.9135871 0.002722963
## 693    0.6568314 0.8757438 0.203405546
## 3243   0.6511554 0.8264463 0.025546168
## 6064   0.6775928 0.8694585 0.065338122
## 11974  0.5038459 0.7755602 0.114102685
##             CpG Gene_ID   CpG_Feature CpG_Island     logFC    adj.P.Val
## 58    cg27650212 SLC13A3 5'UTR;TSS1500    S_Shore  3.661991 0.0002439811
## 13259 cg04145275 SLC13A3        5'UTR    OpenSea -1.726671 0.0280448372
## 14893 cg23364614 SLC13A3      TSS1500    OpenSea  1.736897 0.0304963128
## 21754 cg12127137 SLC13A3         Body    OpenSea -1.071173 0.0416334411
## 22753 cg10341355 SLC13A3         Body    OpenSea -1.615833 0.0432628539
##       normalBeta tumorBeta   deltaBeta
## 58     0.7815419 0.9704460 0.003023098
## 13259  0.8947866 0.7255618 0.069759318
## 14893  0.7659731 0.9069149 0.073124531
## 21754  0.7274920 0.5623433 0.125337463
## 22753  0.8116987 0.5883045 0.033916714
##             CpG Gene_ID   CpG_Feature CpG_Island    logFC    adj.P.Val
## 62    cg02376018    BCL6 5'UTR;TSS1500    N_Shore 3.398516 0.0002489914
## 209   cg18146226    BCL6         Body    OpenSea 3.553807 0.0006685821
## 1180  cg09181580    BCL6         Body    OpenSea 2.847295 0.0030688374
## 2747  cg09643398    BCL6  5'UTR;TSS200    N_Shore 1.669873 0.0067370998
## 2773  cg10819062    BCL6 5'UTR;TSS1500    N_Shelf 1.648828 0.0068286619
## 9759  cg06070445    BCL6 5'UTR;TSS1500    N_Shore 1.842631 0.0215860310
## 10475 cg21739584    BCL6  5'UTR;TSS200    N_Shelf 1.478951 0.0227403136
## 15511 cg05663031    BCL6 5'UTR;TSS1500    N_Shore 1.238357 0.0316336305
## 18725 cg13552858    BCL6 5'UTR;TSS1500    N_Shore 1.008211 0.0369672271
##       normalBeta tumorBeta   deltaBeta
## 62     0.6424487 0.9433554 0.011414882
## 209    0.6183102 0.9391522 0.169839102
## 1180   0.6008968 0.9068196 0.005851157
## 2747   0.5723708 0.8048910 0.004220105
## 2773   0.7997421 0.9189906 0.095316152
## 9759   0.4586686 0.7371679 0.162721322
## 10475  0.6145756 0.8067279 0.046715022
## 15511  0.4731128 0.6737073 0.005300192
## 18725  0.5086524 0.6717047 0.001808258
```

```
##             CpG Gene_ID CpG_Feature CpG_Island    logFC    adj.P.Val
## 78    cg09229423 MIR24-1     TSS1500    N_Shore 3.959660 0.0002880614
## 308   cg02268620 MIR24-1     TSS1500    N_Shore 3.345215 0.0010299029
## 968   cg09951047 MIR24-1        Body     Island 3.506372 0.0026118733
## 1057  cg07975356 MIR24-1     TSS1500    N_Shore 3.248058 0.0027713333
## 1198  cg26417468 MIR24-1     TSS1500    N_Shore 2.857575 0.0031124717
## 1370  cg00532885 MIR24-1      TSS200     Island 3.484964 0.0035303111
## 1818  cg02324722 MIR24-1      TSS200     Island 2.755305 0.0047225718
## 10666 cg20355301 MIR24-1      TSS200    N_Shore 1.630566 0.0230722452
##       normalBeta tumorBeta   deltaBeta
## 78     0.5932930 0.9458353 0.114794551
## 308    0.6633520 0.9369806 0.168417176
## 968    0.5548181 0.9190847 0.046058132
## 1057   0.5178688 0.8956363 0.004747814
## 1198   0.5930421 0.9018136 0.108022136
## 1370   0.4613562 0.8837701 0.026417716
## 1818   0.5845020 0.8909160 0.044459909
## 10666  0.4114772 0.6725705 0.025244474
##             CpG Gene_ID        CpG_Feature CpG_Island    logFC    adj.P.Val
## 89    cg00898447   BCAR1 TSS1500;TSS200;Body    S_Shore 3.682239 0.0003621356
## 1274  cg00319595   BCAR1 TSS1500;TSS200;Body    S_Shore 2.334216 0.0033071796
## 3092  cg06528803   BCAR1       Body;TSS1500    S_Shelf 2.321867 0.0076234008
## 5720  cg12185172   BCAR1       Body;TSS1500    S_Shore 1.708461 0.0137501237
## 6491  cg09066997   BCAR1 TSS1500;TSS200;Body    S_Shore 1.725173 0.0154782608
## 6754  cg22837325   BCAR1       TSS1500;Body    N_Shore 1.281472 0.0160425032
## 7613  cg02853019   BCAR1       Body;TSS1500    S_Shelf 1.182774 0.0176809933
## 11961 cg10361675   BCAR1 Body;TSS1500;TSS200    S_Shore 1.445599 0.0255022136
## 12124 cg04060571   BCAR1              Body    N_Shore 1.190876 0.0257196535
## 13065 cg03451204   BCAR1       Body;TSS1500    N_Shore 1.145685 0.0277247604
## 16678 cg05389731   BCAR1  Body;5'UTR;TSS1500    S_Shore 1.135534 0.0337865225
## 20877 cg15317982   BCAR1         5'UTR;Body    OpenSea 1.147529 0.0401975696
##       normalBeta tumorBeta   deltaBeta
## 89     0.7209481 0.9645709 0.162774953
## 1274   0.4975648 0.8241779 0.181097535
## 3092   0.6479085 0.8902404 0.004008397
## 5720   0.6902989 0.8703511 0.089319274
## 6491   0.8118111 0.9273383 0.153117512
## 6754   0.5088628 0.7112834 0.057681779
## 7613   0.5677076 0.7453717 0.001715570
## 11961  0.6063604 0.8006450 0.197311110
## 12124  0.2933008 0.4853162 0.089114977
## 13065  0.6091091 0.7712727 0.003474752
## 16678  0.4061332 0.5965163 0.121544297
## 20877  0.5652003 0.7368936 0.005537411
##             CpG Gene_ID CpG_Feature CpG_Island    logFC    adj.P.Val
## 91    cg08611602   BRMS1     TSS1500     Island 2.994397 0.0003623082
## 163   cg14370595   BRMS1     TSS1500     Island 3.851575 0.0006157255
## 23814 cg22719871   BRMS1  3'UTR;Body    S_Shelf 1.009564 0.0451355268
## 25446 cg23242341   BRMS1     TSS1500    N_Shore 1.060360 0.0475813391
##       normalBeta tumorBeta  deltaBeta
## 91     0.6791918 0.9388910 0.09267614
## 163    0.5396734 0.9326559 0.06494666
## 23814  0.4903674 0.6562629 0.07000398
## 25446  0.6039996 0.7563414 0.08178723
```

```
##             CpG Gene_ID CpG_Feature CpG_Island    logFC    adj.P.Val
## 92    cg08611602  B3GNT1     1stExon     Island 2.994397 0.0003623082
## 164   cg14370595  B3GNT1     1stExon     Island 3.851575 0.0006157255
## 25447 cg23242341  B3GNT1        Body    N_Shore 1.060360 0.0475813391
##       normalBeta tumorBeta  deltaBeta
## 92     0.6791918 0.9388910 0.09267614
## 164    0.5396734 0.9326559 0.06494666
## 25447  0.6039996 0.7563414 0.08178723
```

9. Gene set analysis

```
# filter significant probes
sign_probes <- LIMMAout[LIMMAout$adj.P.Val < 0.01, ]

# laod annotation
library(IlluminaHumanMethylationEPICanno.ilm10b4.hg19)
hm850 <- getAnnotation(IlluminaHumanMethylationEPICanno.ilm10b4.hg19)

probenames <- rownames(sign_probes)
probes <- hm850[probenames, ]

# trnasform to Granges
library(GenomicRanges)
probes_gr <- GRanges(
  seqnames = probes$chr,
  ranges = IRanges(start = probes$pos, end = probes$pos),
  strand = "*"
)

ProbeToGene <- getNearestGene(probes_gr)

head(ProbeToGene)

##   queryHits subjectHits distance nearestGeneSymbol
## 1         1       24916        0             MRC2
## 2         2       21987        0           SLC7A5
## 3         3       10072        0            MKNK2
## 4         4       21987        0           SLC7A5
## 5         5        3589        0           MGAT4B
## 6         6       24621        0             RIN1

# filterout probes with distnce <500
ProbeToGene <- ProbeToGene[ProbeToGene$distance < 500, ]
table(ProbeToGene$distance < 500)

##
## TRUE
## 3763

EntrezIDs <- mapIds(org.Hs.eg.db, ProbeToGene$nearestGeneSymbol, 'ENTREZID',
'SYMBOL')

## 'select()' returned 1:1 mapping between keys and columns
```

```
EntrezIDs <- EntrezIDs[!(duplicated(EntrezIDs) | is.na(EntrezIDs))]
goanaOut <- goana(de=EntrezIDs, species="Hs", trend=T)
goanaOut <- goanaOut[order(goanaOut$P.DE, decreasing=FALSE),]
goanaOut$FDR.DE <- p.adjust(goanaOut$P.DE, method="BH")
topGOcpg <- topGO(goanaOut, ontology="BP", number=50)
topGOcpg
```

```
##                                                                    Term
## GO:0007275                                multicellular organism development
## GO:0048731                                             system development
## GO:0016043                                cellular component organization
## GO:0048856                                anatomical structure development
## GO:0009653                                anatomical structure morphogenesis
## GO:0007399                                       nervous system development
## GO:0071840                     cellular component organization or biogenesis
## GO:0032502                                         developmental process
## GO:0022008                                                neurogenesis
## GO:0030182                                       neuron differentiation
## GO:0048699                                        generation of neurons
## GO:0030154                                         cell differentiation
## GO:0048869                                cellular developmental process
## GO:0034330                                    cell junction organization
## GO:0030030                                  cell projection organization
## GO:0051128                     regulation of cellular component organization
## GO:0023051                                       regulation of signaling
## GO:0010646                                regulation of cell communication
## GO:0120036                plasma membrane bounded cell projection organization
## GO:0048518                          positive regulation of biological process
## GO:0048666                                             neuron development
## GO:0048522                          positive regulation of cellular process
## GO:0048468                                           cell development
## GO:0000902                                          cell morphogenesis
## GO:0031175                               neuron projection development
## GO:0051179                                                 localization
## GO:0007155                                               cell adhesion
## GO:0009966                               regulation of signal transduction
## GO:0048513                                       animal organ development
## GO:0048583                               regulation of response to stimulus
## GO:0032501                                 multicellular organismal process
## GO:0050794                                   regulation of cellular process
## GO:0050789                                regulation of biological process
## GO:0050793                              regulation of developmental process
## GO:0048858                                   cell projection morphogenesis
## GO:0031344                          regulation of cell projection organization
## GO:0120039                plasma membrane bounded cell projection morphogenesis
## GO:0007154                                             cell communication
## GO:0023052                                                   signaling
## GO:0032989                           cellular anatomical entity morphogenesis
## GO:0035556                                 intracellular signal transduction
## GO:0009719                                   response to endogenous stimulus
## GO:0048812                                   neuron projection morphogenesis
## GO:0065007                                            biological regulation
## GO:0120035 regulation of plasma membrane bounded cell projection organization
## GO:0009887                                        animal organ morphogenesis
```

```
## GO:0051234                                    establishment of localization
## GO:0071495                          cellular response to endogenous stimulus
## GO:0006810                                                       transport
## GO:0009888                                                tissue development
##            Ont     N   DE         P.DE       FDR.DE
## GO:0007275  BP  4764  743 7.651162e-43 8.630893e-39
## GO:0048731  BP  4072  641 7.161575e-37 4.039308e-33
## GO:0016043  BP  6581  926 1.395453e-36 6.296562e-33
## GO:0048856  BP  6016  862 3.796255e-36 1.427455e-32
## GO:0009653  BP  2757  474 2.737031e-35 7.111302e-32
## GO:0007399  BP  2539  446 2.836830e-35 7.111302e-32
## GO:0071840  BP  6791  938 8.297934e-34 1.872097e-30
## GO:0032502  BP  6593  913 5.976749e-33 1.225831e-29
## GO:0022008  BP  1741  330 9.584236e-32 1.663307e-28
## GO:0030182  BP  1426  282 4.108803e-30 6.179913e-27
## GO:0048699  BP  1508  293 6.054228e-30 8.536839e-27
## GO:0030154  BP  4448  650 1.987092e-27 2.490599e-24
## GO:0048869  BP  4450  650 2.264693e-27 2.689144e-24
## GO:0034330  BP   766  177 8.429546e-27 9.508949e-24
## GO:0030030  BP  1611  296 5.026238e-26 5.154407e-23
## GO:0051128  BP  2459  405 1.017517e-25 9.980957e-23
## GO:0023051  BP  3554  538 1.393356e-25 1.309812e-22
## GO:0010646  BP  3559  538 1.971577e-25 1.779230e-22
## GO:0120036  BP  1569  286 1.446015e-24 1.208280e-21
## GO:0048518  BP  6411  856 2.121442e-24 1.709352e-21
## GO:0048666  BP  1157  228 4.441081e-24 3.455008e-21
## GO:0048522  BP  5861  794 5.424893e-24 4.079701e-21
## GO:0048468  BP  2838  445 8.204932e-24 5.971338e-21
## GO:0000902  BP   973  200 1.977714e-23 1.394350e-20
## GO:0031175  BP  1010  205 2.675934e-23 1.775640e-20
## GO:0051179  BP  5335  732 3.564134e-23 2.297441e-20
## GO:0007155  BP  1533  276 7.254304e-23 4.546232e-20
## GO:0009966  BP  3131  470 3.353744e-21 2.044968e-18
## GO:0048513  BP  3052  460 4.609371e-21 2.736632e-18
## GO:0048583  BP  4116  582 1.650292e-20 9.546725e-18
## GO:0032501  BP  7918  996 2.632020e-20 1.484525e-17
## GO:0050794  BP 11713 1376 1.225284e-19 6.428752e-17
## GO:0050789  BP 12273 1429 2.217177e-19 1.136858e-16
## GO:0050793  BP  2555  393 2.530001e-19 1.268430e-16
## GO:0048858  BP   668  144 5.373247e-19 2.579273e-16
## GO:0031344  BP   676  145 6.466494e-19 3.039387e-16
## GO:0120039  BP   663  143 6.809660e-19 3.135362e-16
## GO:0007154  BP  6687  857 1.057010e-18 4.675924e-16
## GO:0023052  BP  6595  846 1.710969e-18 7.283242e-16
## GO:0032989  BP   786  160 1.856127e-18 7.754828e-16
## GO:0035556  BP  2721  409 2.409243e-18 9.882714e-16
## GO:0009719  BP  1758  290 2.824535e-18 1.137934e-15
## GO:0048812  BP   647  139 3.085671e-18 1.221330e-15
## GO:0065007  BP 12701 1464 3.213383e-18 1.249951e-15
## GO:0120035  BP   659  140 6.437457e-18 2.420591e-15
## GO:0009887  BP  1025  192 7.426225e-18 2.746608e-15
## GO:0051234  BP  4698  633 1.816732e-17 6.404263e-15
## GO:0071495  BP  1505  253 5.423713e-17 1.882529e-14
```

```
## GO:0006810  BP  4385  595 6.632608e-17 2.233407e-14
## GO:0009888  BP  2032  319 7.358249e-17 2.441315e-14
```

```
# print result
topGOcpg_sorted <- topGOcpg[order(topGOcpg$N), ]
head(topGOcpg_sorted)
```

```
##                                                                      Term
## GO:0048812                                    neuron projection morphogenesis
## GO:0120035 regulation of plasma membrane bounded cell projection organization
## GO:0120039             plasma membrane bounded cell projection morphogenesis
## GO:0048858                                  cell projection morphogenesis
## GO:0031344                       regulation of cell projection organization
## GO:0034330                                     cell junction organization
##           Ont   N  DE        P.DE      FDR.DE
## GO:0048812  BP 647 139 3.085671e-18 1.221330e-15
## GO:0120035  BP 659 140 6.437457e-18 2.420591e-15
## GO:0120039  BP 663 143 6.809660e-19 3.135362e-16
## GO:0048858  BP 668 144 5.373247e-19 2.579273e-16
## GO:0031344  BP 676 145 6.466494e-19 3.039387e-16
## GO:0034330  BP 766 177 8.429546e-27 9.508949e-24
```

```
library(doParallel)
registerDoParallel(cores = 4)
fibromethyl_gr <- preprocessFunnorm(fibromethyl_RG)
```

```
## [preprocessFunnorm] Background and dye bias correction with noob
```

```
## [preprocessFunnorm] Mapping to genome
```

```
## [preprocessFunnorm] Quantile extraction
```

```
## [preprocessFunnorm] Normalization
```

```
dmrs <- bumphunter(fibromethyl_gr, design = design, coef=2, cutoff = 0.25, B=500,
maxGap=500, verbose=TRUE, nullMethod="bootstrap")
```

```
## [bumphunterEngine] Parallelizing using 4 workers/cores (backend: doParallelMC,
version: 1.0.17).
```

```
## [bumphunterEngine] Computing coefficients.
```

```
## [bumphunterEngine] Performing 500 bootstraps.
```

```
## Loading required package: rngtools
```

```
## [bumphunterEngine] Computing marginal bootstrap p-values.
```

```
## [bumphunterEngine] cutoff: 0.25
```

```
## [bumphunterEngine] Finding regions.
```

```
## [bumphunterEngine] Found 6885 bumps.
```

```
## [bumphunterEngine] Computing regions for each bootstrap.
```

```
## [bumphunterEngine] Estimating p-values and FWER.

head(names(dmrs))

## [1] "table"          "coef"           "fitted"         "pvaluesMarginal"
## [5] "null"           "algorithm"
```

10. Identify differentially methylated regions (DMRs): bumphunter

```
sign_bumps <- dmrs$table[dmrs$table$fwerArea<0.10,]
#For the significant bumps, construct a GRanges object which can then be used to get
the nearest gene symbol.
dmrs_GR <- makeGRangesFromDataFrame(sign_bumps[,1:3])
bumpsToGenes <- getNearestGene(dmrs_GR)
head(bumpsToGenes)

##      queryHits subjectHits distance nearestGeneSymbol
## 2152         1       3126        0          SEPTIN9
## 2440         2      12122        0           ZNF790
## 1177         3      20447        0           TXNRD1

#Filter out bumps that are far away (>=500bp) from their nearest gene, their impact
is probably not as high on the gene as closely differentially methylated regions.
table(bumpsToGenes$distance < 500)

##
## TRUE
##    3

bumpsToGenes <- bumpsToGenes[bumpsToGenes$distance < 500,]
#Goana uses Entrez gene identifiers, we need to convert our gene symbols to entrez
ids. For this purpose we use the org.Hs.eg.db package:
EntrezIDs_bmp <- mapIds(org.Hs.eg.db, bumpsToGenes$nearestGeneSymbol, 'ENTREZID',
'SYMBOL')

## 'select()' returned 1:1 mapping between keys and columns

### 'select()' returned 1:1 mapping between keys and columns
#subset for non duplicated and mapped genes
EntrezIDs_bmp <- EntrezIDs_bmp[!(duplicated(EntrezIDs_bmp) | is.na(EntrezIDs_bmp))]

#Overrepresentation analysis with goana:
goanaOut_bmp <- goana(de=EntrezIDs_bmp, species="Hs", trend=T)

#FDR multiple testing adjustment:
goanaOut_bmp <- goanaOut_bmp[order(goanaOut_bmp$P.DE, decreasing=FALSE),]
goanaOut_bmp$FDR.DE <- p.adjust(goanaOut_bmp$P.DE, method="BH")
topGObumps <- topGO(goanaOut_bmp, ontology="BP", number=50)
head(topGObumps)

##                                                 Term Ont  N DE
## GO:1902857 positive regulation of non-motile cilium assembly  BP  9  1
## GO:1902855         regulation of non-motile cilium assembly  BP 12  1
## GO:0045724           positive regulation of cilium assembly  BP 29  1
## GO:0045454                          cell redox homeostasis  BP 41  1
```

```
## GO:1905515                      non-motile cilium assembly  BP 66  1
## GO:1902017                    regulation of cilium assembly  BP 74  1
##                  P.DE FDR.DE
## GO:1902857 0.001300139     1
## GO:1902855 0.001733269     1
## GO:0045724 0.004185303     1
## GO:0045454 0.005913730     1
## GO:1905515 0.009508196     1
## GO:1902017 0.010656592     1

#Order topGObumps on number of genes in the GO term, this will show more "specific"
GO terms (less genes in the term means a term lower in the hierarchy).
topGObumps[order(topGObumps$N),]

##                                                                      Term
## GO:1902857                      positive regulation of non-motile cilium assembly
## GO:1902855                              regulation of non-motile cilium assembly
## GO:0045724                                positive regulation of cilium assembly
## GO:0045454                                             cell redox homeostasis
## GO:1905515                                          non-motile cilium assembly
## GO:1902017                                        regulation of cilium assembly
## GO:0001707                                                 mesoderm formation
## GO:0048332                                               mesoderm morphogenesis
## GO:1902117                                 positive regulation of organelle assembly
## GO:0098869                                        cellular oxidant detoxification
## GO:0120034 positive regulation of plasma membrane bounded cell projection assembly
## GO:0061640                                   cytoskeleton-dependent cytokinesis
## GO:1990748                                             cellular detoxification
## GO:0097237                                cellular response to toxic substance
## GO:0001704                                    formation of primary germ layer
## GO:0007498                                          mesoderm development
## GO:0098754                                               detoxification
## GO:0000910                                                  cytokinesis
## GO:0007369                                                  gastrulation
## GO:0120032         regulation of plasma membrane bounded cell projection assembly
## GO:0060491                                regulation of cell projection assembly
## GO:1902115                                     regulation of organelle assembly
## GO:0009636                                         response to toxic substance
## GO:0031346                   positive regulation of cell projection organization
## GO:0060271                                                  cilium assembly
## GO:0044782                                               cilium organization
## GO:0010638                         positive regulation of organelle organization
## GO:0044089                  positive regulation of cellular component biogenesis
## GO:0120031                     plasma membrane bounded cell projection assembly
## GO:0030031                                          cell projection assembly
## GO:0048729                                             tissue morphogenesis
## GO:0048598                                         embryonic morphogenesis
## GO:0051301                                                   cell division
## GO:0120035     regulation of plasma membrane bounded cell projection organization
## GO:0031344                           regulation of cell projection organization
## GO:0019725                                               cellular homeostasis
## GO:0044087                          regulation of cellular component biogenesis
## GO:0070925                                            organelle assembly
## GO:0051130                 positive regulation of cellular component organization
```

```
## GO:0009790                                                    embryo development
## GO:0033043                                  regulation of organelle organization
## GO:0048646              anatomical structure formation involved in morphogenesis
## GO:0022402                                                    cell cycle process
## GO:0120036                 plasma membrane bounded cell projection organization
## GO:0030030                                       cell projection organization
## GO:0042592                                              homeostatic process
## GO:0007049                                                      cell cycle
## GO:0050794                                    regulation of cellular process
## GO:0050789                                  regulation of biological process
## GO:0065007                                            biological regulation
##            Ont     N DE       P.DE FDR.DE
## GO:1902857  BP     9  1 0.001300139      1
## GO:1902855  BP    12  1 0.001733269      1
## GO:0045724  BP    29  1 0.004185303      1
## GO:0045454  BP    41  1 0.005913730      1
## GO:1905515  BP    66  1 0.009508196      1
## GO:1902017  BP    74  1 0.010656592      1
## GO:0001707  BP    76  1 0.010943552      1
## GO:0048332  BP    78  1 0.011230457      1
## GO:1902117  BP    84  1 0.012090838      1
## GO:0098869  BP    99  1 0.014239608      1
## GO:0120034  BP   111  1 0.015956379      1
## GO:0061640  BP   117  1 0.016814016      1
## GO:1990748  BP   119  1 0.017099785      1
## GO:0097237  BP   129  1 0.018527796      1
## GO:0001704  BP   131  1 0.018813232      1
## GO:0007498  BP   140  1 0.020097010      1
## GO:0098754  BP   158  1 0.022661205      1
## GO:0000910  BP   188  1 0.026924916      1
## GO:0007369  BP   200  1 0.028626923      1
## GO:0120032  BP   204  1 0.029193817      1
## GO:0060491  BP   206  1 0.029477181      1
## GO:1902115  BP   223  1 0.031883552      1
## GO:0009636  BP   260  1 0.037107191      1
## GO:0031346  BP   360  1 0.051131041      1
## GO:0060271  BP   385  1 0.054615598      1
## GO:0044782  BP   412  1 0.058369326      1
## GO:0010638  BP   506  1 0.071360328      1
## GO:0044089  BP   507  1 0.071497883      1
## GO:0120031  BP   583  1 0.081912430      1
## GO:0030031  BP   596  1 0.083686029      1
## GO:0048729  BP   616  1 0.086410179      1
## GO:0048598  BP   618  1 0.086682297      1
## GO:0051301  BP   643  1 0.090079210      1
## GO:0120035  BP   659  1 0.092248808      1
## GO:0031344  BP   676  1 0.094550225      1
## GO:0019725  BP   824  1 0.114421960      1
## GO:0044087  BP   989  1 0.136231064      1
## GO:0070925  BP  1002  1 0.137933975      1
## GO:0051130  BP  1122  1 0.153547587      1
## GO:0009790  BP  1128  1 0.154323279      1
## GO:0033043  BP  1183  1 0.161411719      1
## GO:0048646  BP  1271  1 0.172670648      1
```

```
## GO:0022402  BP  1294  1 0.175596618     1
## GO:0120036  BP  1569  1 0.210048359     1
## GO:0030030  BP  1611  1 0.215224044     1
## GO:0042592  BP  1724  1 0.229036759     1
## GO:0007049  BP  1816  1 0.240162062     1
## GO:0050794  BP 11713  3 0.179612545     1
## GO:0050789  BP 12273  3 0.206628163     1
## GO:0065007  BP 12701  3 0.229010101     1
```

```
#Look at the overlap between GO terms for the CpG based analysis & the region based analysis.
intersect(topGObumps$Term, topGOcpg$Term)
```

```
## [1] "regulation of plasma membrane bounded cell projection organization"
## [2] "regulation of cell projection organization"
## [3] "regulation of cellular process"
## [4] "regulation of biological process"
## [5] "plasma membrane bounded cell projection organization"
## [6] "cell projection organization"
## [7] "biological regulation"
```

```
length(intersect(topGObumps$Term, topGOcpg$Term))/50
```

```
## [1] 0.14
```

```
which(topGObumps$Term %in% intersect(topGObumps$Term, topGOcpg$Term))
```

```
## [1] 34 35 44 45 46 47 48
```

```
which(topGOcpg$Term %in% intersect(topGObumps$Term, topGOcpg$Term))
```

```
## [1] 15 19 32 33 36 44 45
```

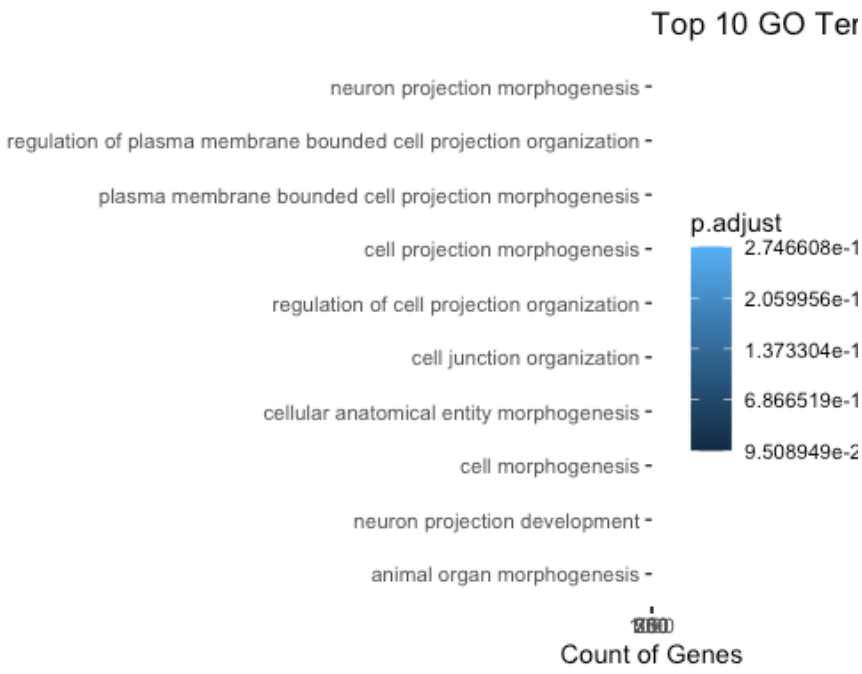
11. Create a bar plot for the top 10 GO terms

- Top GO Terms: The plot shows the most significant biological processes, with "neuron projection morphogenesis" having the most genes (~1000) involved.
- Gene Count: Longer bars indicate more genes associated with each term. For example, "neuron projection morphogenesis" is the most gene-rich process.
- Adjusted p-value (Color): The color gradient indicates statistical significance, with blue (low p-values) representing more significant terms like "neuron projection morphogenesis" and "animal organ morphogenesis". These terms are strongly associated with the CpG changes observed in your analysis.

```
# Sort the GO results by the number of genes (N)
topGOcpg_sorted <- topGOcpg[order(topGOcpg$N), ]

# Create a bar plot for the top 10 GO terms
ggplot(topGOcpg_sorted[1:10, ], aes(x = reorder(Term, -N), y = N, fill = FDR.DE)) +
```

```
  geom_bar(stat = "identity", width = 0.8) +  # Create bars with adjusted width
  coord_flip() +  # Flip coordinates for better readability of GO terms
  labs(title = "Top 10 GO Terms for CpG Analysis",  # Title of the plot
       x = "GO Term",                              # Label for x-axis
       y = "Count of Genes",                       # Label for y-axis
       fill = "p.adjust"
  )
```



12. Writing data for comparison of results

```
write.table(EntrezIDs, sep= "\t", file="EntrezIDs_CpG_results.txt")
write.table(EntrezIDs_bmp, sep= "\t", file="EntrezIDs_bmp_results.txt")
CpG_GSA_res <- topGO(goanaOut, ontology="BP", number=100)
bmp_GSA_res <- topGO(goanaOut_bmp, ontology="BP", number=100)
write.table(CpG_GSA_res, sep= "\t", file="CpG_GSA_results.txt")
write.table(bmp_GSA_res, sep= "\t", file="bmp_GSA_results.txt")
```

Appendix B:

RNA-Seq Analysis of Esophageal Squamous Cell Carcinoma (ESCC) and Paired Normal Tissue

Chaeyoung Kim

2024-12-13

This R script identifies differentially expressed genes between esophageal squamous cell carcinoma (ESCC) tissue and paired normal tissue from six individual patients. MultiQC analysis was performed using an HPC environment, and this script processes pre-aligned gene count data published on GEO. The analysis begins with general data exploration, followed by preprocessing, normalization, and statistical analysis using the "edgeR" package.

General info

- Species: Homo Sapiens
- Sequencing device: Illumina HiSeq 2500
- Single- or paired-end sequencing: Paired-end sequencing
- Samples: This dataset includes six ESCC tumor and six adjacent normal tissue samples obtained from six individuals
- Muli Qc performed with HPC. Trimming and mapping has not been done due to the issue regarding the access into HPC. Instead, statistical analysis performed with raw count data downloaded from NCBI (https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE194116).

load required packages

```
##Load required package
library(edgeR)
library(limma)
library(AnnotationDbi)
library(org.Hs.eg.db)
library(GO.db)
library(clusterProfiler)
```

Setup working directory

```
setwd("/Users/kimchaeyoung/Desktop/Applied_HT_analysis/Project")
```

Load dataset

```
# Read in text file
rnadatafull <- read.table("GSE194116_Raw_gene_counts_matrix.txt", header = T, sep =
"\t", quote = "")

#Have a look at the data
head(rnadatafull)
```

```
##       ensembl_id gene_name Normal.1.count Normal.2.count Normal.3.count
## 1 ENSG00000000003    TSPAN6          2043          2635          1566
## 2 ENSG00000000005      TNMD             3             8             3
## 3 ENSG00000000419      DPM1           339           320           304
## 4 ENSG00000000457     SCYL3           717           689           567
## 5 ENSG00000000460  C1orf112           206           229           172
## 6 ENSG00000000938       FGR           178           702           636
##   Normal.4.count Normal.5.count Normal.6.count ECA.1.count ECA.2.count
## 1           1150           1396           2053        1321         787
## 2              6              5              3           0           0
## 3            296            410            427         887         528
## 4            747            725            787         422         594
## 5            231            234            331         483         796
## 6            551            230            304         276         144
##   ECA.3.count ECA.4.count ECA.5.count ECA.6.count
## 1         417         999        1202         646
## 2           0           0           1           0
## 3         873         488         530         778
## 4         710         550         530         441
## 5         690         622         625         564
## 6         241         262         254         706
```

```
# save ensemble IDs and gene names in separate variable
geneIDs <- rnadatafull[,1:2]

head(geneIDs)
```

```
##       ensembl_id gene_name
## 1 ENSG00000000003    TSPAN6
## 2 ENSG00000000005      TNMD
## 3 ENSG00000000419      DPM1
```

```
## 4 ENSG00000000457     SCYL3
## 5 ENSG00000000460  C1orf112
## 6 ENSG00000000938       FGR
```

1. Preprocess data

```
## Separate numerical data (count data) from annotation
rownames(rnadatafull) <- 1:nrow(rnadatafull)
rnadata <- rnadatafull[, 3:14]

## Make annotation for design later on
individuals <- factor(c("patient1", "patient2", "patient3", "patient4", "patient5",
"patient6","patient1", "patient2", "patient3", "patient4", "patient5", "patient6"))

cvst <- factor(c("Normal", "Normal", "Normal", "Normal", "Normal",
"Normal","ESCC","ESCC","ESCC","ESCC","ESCC","ESCC"))
```

1.1. Further preprocessing

```
# Create DGEList object: 'rnadata' contains the count data, and 'rnadatafull[, 1:2]'
contains gene information
y <- DGEList(rnadata, genes = rnadatafull[, 1:2])

# Optional filter
# 'lib.size' refers to the library size of each sample. We calculate CPM (Counts Per
Million) values,and apply a filtering threshold (cutoff) to remove genes with low
counts.
cutoff <- 3 / (mean(y$samples$lib.size) / 1000000)

#Keep genes with CPM greater than the cutoff in at least 6 samples:
keep <- rowSums(cpm(y) > cutoff) >= 6
#This provides information on how many genes passed the filtering criteria
summary(keep)

##    Mode   FALSE    TRUE
## logical   15849   24315

# Update DGEList object to include only the filtered genes
y <- y[keep, , keep.lib.sizes = FALSE]
```

1.2. Normalization

```
y <- calcNormFactors(y)

# Have a look at the library sizes and normalization factors
y$samples

##               group lib.size norm.factors
## Normal.1.count     1 22294014    0.9438172
## Normal.2.count     1 21970627    0.9856696
## Normal.3.count     1 21787999    0.9169301
## Normal.4.count     1 19353923    1.0573374
## Normal.5.count     1 20592391    1.0111336
```

```
## Normal.6.count     1 18508014    1.0568192
## ECA.1.count        1 20507630    1.0195277
## ECA.2.count        1 17650965    1.0776411
## ECA.3.count        1 17247682    1.0271229
## ECA.4.count        1 19620147    0.9853978
## ECA.5.count        1 22725078    0.9369623
## ECA.6.count        1 19252943    0.9958464
```

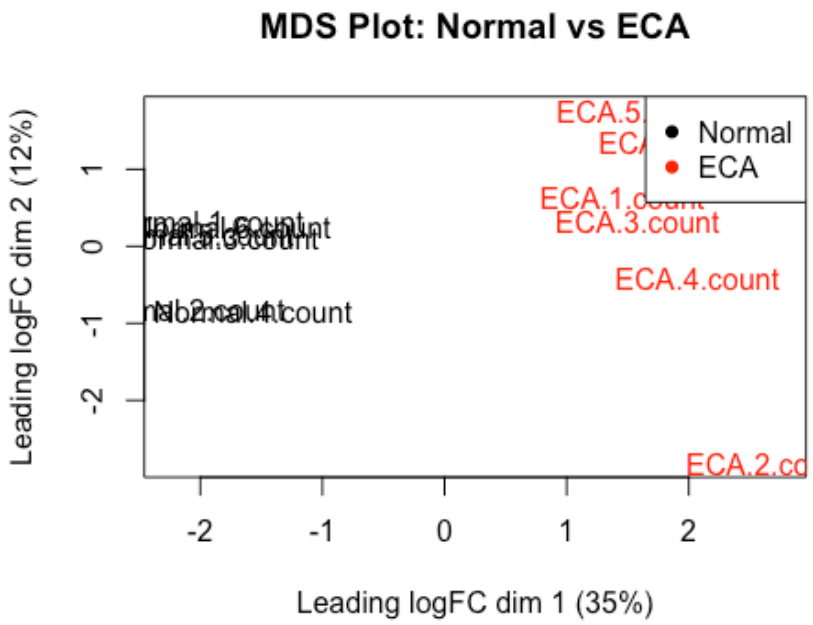
1.3. MDS plot

- Visualize Sample Similarity -The MDS plot clearly distinguishes between the Normal and ECA groups, effectively highlighting significant biological differences between the two conditions. The variability within the ECA group suggests heterogeneity, offering potential insights through further analysis.

```
# Define colors for Normal (black) and ECA (red)
colors <- ifelse(grepl("^Normal", rownames(y$samples)), "black", "red")

# Plot MDS with specified colors
plotMDS(y, col=colors, main = "MDS Plot: Normal vs ECA")

# Add legend
legend("topright", legend=c("Normal", "ECA"), col=c("black", "red"), pch=16)
```



2. EdgeR differential expression analysis - Paired analysis

- as my dataset has been obtained from paired design experiment, perform paired analysis

2.1. Create design matrix

In paired data analysis, I need to account for within-individual variation to correctly identify the effects of the variable of interest (cvst in this case). The design matrix (model.matrix) is created to include both the condition (cvst) and the individual effect (individuals) as factors. This ensures that any differences detected between conditions are not confounded by differences inherent to the individuals.

```
# Create a design matrix: The 'cvst' variable represents the condition (e.g., Normal vs ECA), and 'individuals' represents the paired samples (e.g., patient1, patient2, etc.).

# The design matrix specifies how the data is modeled in terms of the experimental design.
design <- model.matrix(~ cvst + individuals)
design
```

```
##    (Intercept) cvstNormal individualspatient2 individualspatient3
## 1            1         1                   0                   0
## 2            1         1                   1                   0
## 3            1         1                   0                   1
## 4            1         1                   0                   0
## 5            1         1                   0                   0
## 6            1         1                   0                   0
## 7            1         0                   0                   0
## 8            1         0                   1                   0
## 9            1         0                   0                   1
## 10           1         0                   0                   0
## 11           1         0                   0                   0
## 12           1         0                   0                   0
##    individualspatient4 individualspatient5 individualspatient6
## 1                    0                   0                   0
## 2                    0                   0                   0
## 3                    0                   0                   0
## 4                    1                   0                   0
## 5                    0                   1                   0
## 6                    0                   0                   1
## 7                    0                   0                   0
## 8                    0                   0                   0
## 9                    0                   0                   0
## 10                   1                   0                   0
## 11                   0                   1                   0
## 12                   0                   0                   1
```

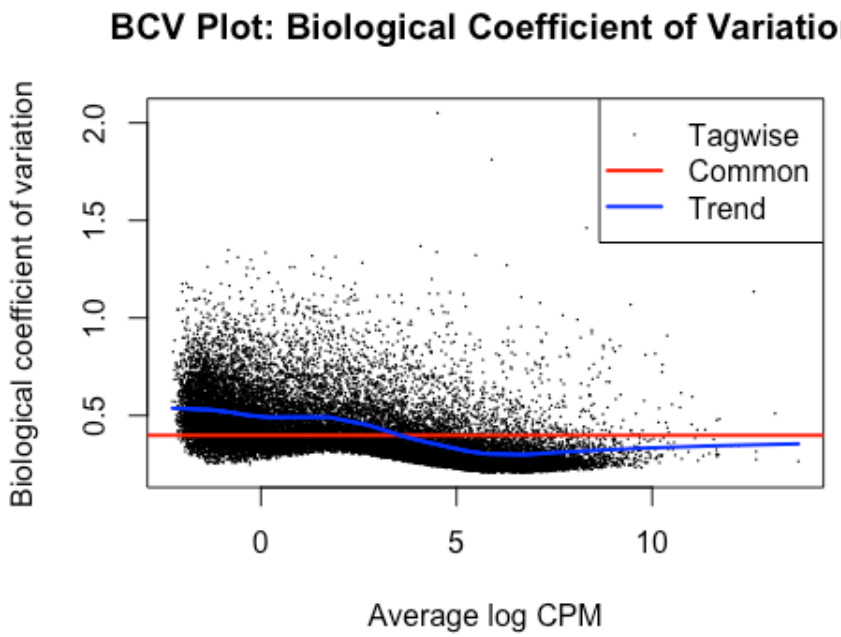
```
## attr(,"assign")
## [1] 0 1 2 2 2 2 2
## attr(,"contrasts")
## attr(,"contrasts")$cvst
## [1] "contr.treatment"
##
## attr(,"contrasts")$individuals
## [1] "contr.treatment"

rownames(design) <- colnames(y)
y <- estimateDisp(y, design)
```

2.2 Plot BCV (biological coefficient of variation) plot

- shows the relationship between the biological variation of gene expression
- The graph shows high variability in lowly expressed genes, with variability decreasing and stabilizing as expression levels increase. The blue trend line effectively captures the decreasing variability pattern, and most genes are close to the red common BCV line, indicating that dispersion estimates are appropriately calculated. Highly variable genes may require further analysis.

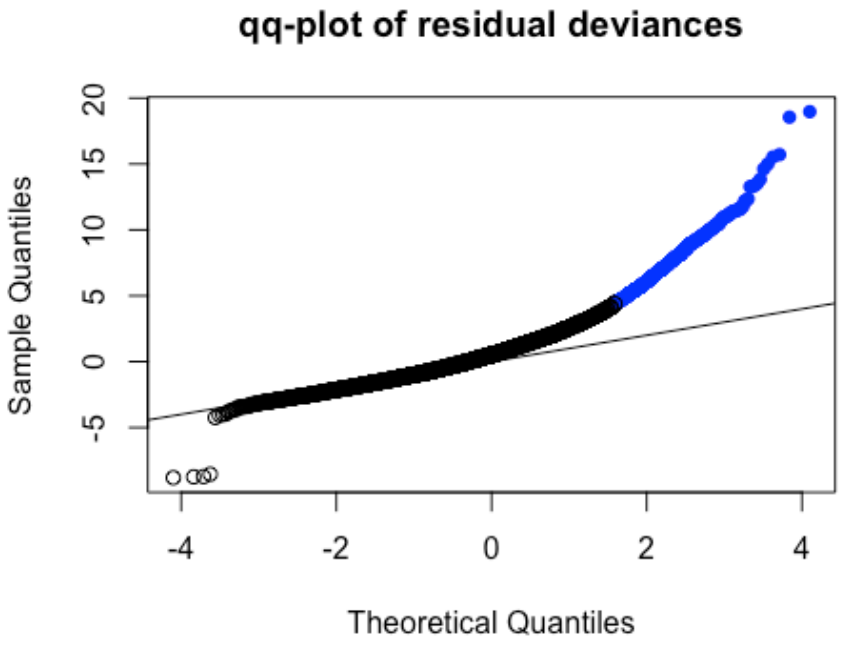
```
plotBCV(y, main = "BCV Plot: Biological Coefficient of Variation")
```



2.3. Fitting model and making a qq-plot to evaluate goodness of fit.

-The central points follow the theoretical distribution well, indicating the model fits most of the data appropriately. However, deviations at the tails suggest the model may not handle extreme values effectively.

```
# glmQLFit, glmLRT 수행
fit <- glmQLFit(y, design)
gof(fit,plot=TRUE)
```



2.4. glmQSFTest

- Evaluate the significance of coefficients (in this case, the second coefficient).The argument `coef = 2` specifies that the second coefficient in the model. The null hypothesis tested here is that there is no difference in expression between the groups being compared (e.g., Normal vs. ESCC).

```
# glmQLFTest() performs a Likelihood Ratio Test (LRT) based on the fitted model
qlf <- glmQLFTest(fit, coef = 2)
summary(qlf)

##                     Length Class           Mode
## coefficients         170205 -none-          numeric
## fitted.values        291780 -none-          numeric
```

```
## deviance              24315 -none-          numeric
## iter                  24315 -none-          numeric
## failed                24315 -none-          logical
## method                    1 -none-          character
## unshrunk.coefficients 170205 -none-         numeric
## df.residual           24315 -none-          numeric
## design                   84 -none-          numeric
## offset               291780 CompressedMatrix numeric
## dispersion                1 -none-          numeric
## prior.count               1 -none-          numeric
## leverage             291780 -none-          numeric
## unit.deviance.adj    291780 -none-          numeric
## unit.df.adj          291780 -none-          numeric
## df.residual.adj       24315 -none-          numeric
## deviance.adj          24315 -none-          numeric
## average.ql.dispersion     1 -none-          numeric
## df.prior                  1 -none-          numeric
## var.post              24315 -none-          numeric
## var.prior             24315 -none-          numeric
## samples                   3 data.frame      list
## genes                     2 data.frame      list
## AveLogCPM             24315 -none-          numeric
## table                     4 data.frame      list
## comparison                1 -none-          character
## df.test               24315 -none-          numeric
## df.total              24315 -none-          numeric
```

```
resP <- topTags(qlf, n = nrow(qlf$table))
# Save the full result of the differential expression analysis to a text file
write.table(resP$table , file = "resP_edgeR_ESCC.txt", col.names = T, row.names = F,
sep = "\t", quote = F)

rownames(resP$table) <- y$genes$ensembl_id
# Filter the result to only include genes with a p-value less than 0.05
(statistically significant genes)
resP_edgeR <- resP$table[resP$table[, ncol(resP$table)] < 0.05, ]

# Display the number of rows (genes) in the filtered result
num_rows <- nrow(resP_edgeR)
cat("Number of rows in resP_edgeR:", num_rows, "\n")
```

```
## Number of rows in resP_edgeR: 7305
```

```
dim(resP_edgeR)
```

```
## [1] 7305    7
```

```
down_regulated <- sum(resP_edgeR$logFC < 0)
down_regulated
```

```
## [1] 3525
```

```
up_regulated <- sum(resP_edgeR$logFC > 0)
up_regulated
```

```
## [1] 3780
```

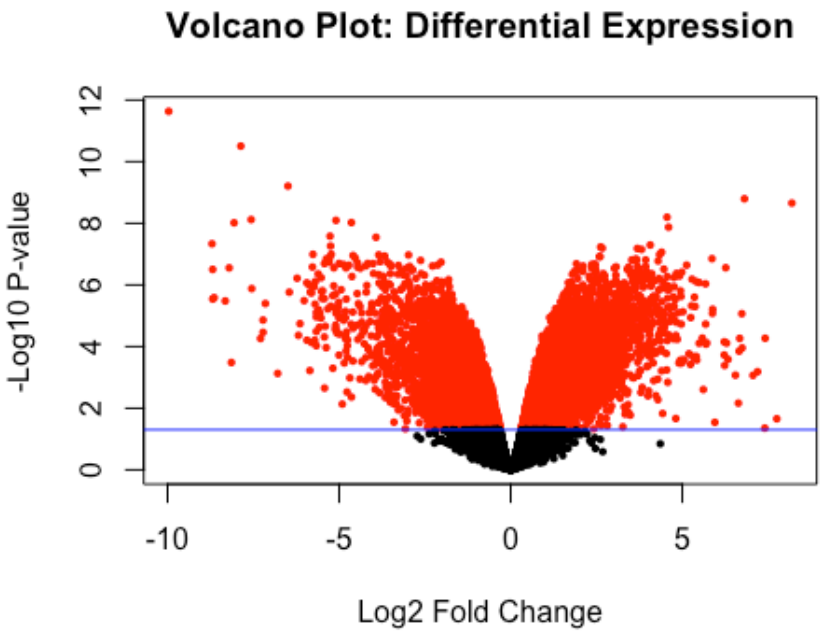
2.5. Create Volvano plot

- visualize the results of differential expression analysis
- Genes on the left and right are downregulated and upregulated, respectively. Black points represent genes with no significant change.

```r
# -log10 transform of Pvalue
resP$table$negLogP <- -log10(resP$table$PValue)

with(resP$table, plot(logFC, negLogP,
                      pch = 16, cex = 0.6,
                      col = ifelse(PValue < 0.05, "red", "black"),
                      main = "Volcano Plot: Differential Expression",
                      xlab = "Log2 Fold Change",
                      ylab = "-Log10 P-value"))

abline(h = -log10(0.05), col = "blue")
```



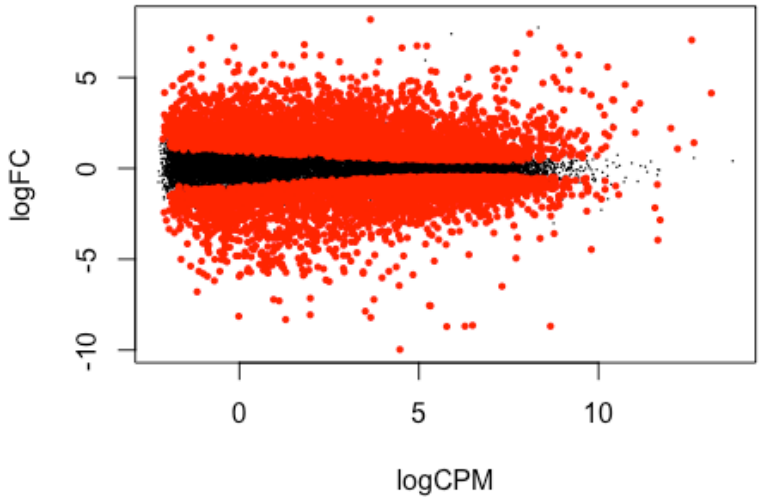
2.6. Creating MA plot

-Visualize the relationship between the mean expression level (average expression) and the log fold change (difference in expression) for each gene between two conditions. -The red points, which are spread out, indicate genes with significant

differences in expression between the tumor and normal tissue samples. These are the genes that are most likely biologically relevant for further investigation.

```
with(resP$table , plot(logCPM , logFC , pch = 16, cex = 0.2, main = "MA-plot of log
fold changes between expression values in tumor and normal tissue."))
with(resP$table, points(logCPM[FDR < 0.05], logFC[FDR < 0.05], pch =
                        16, col = "red", cex = 0.6))
```

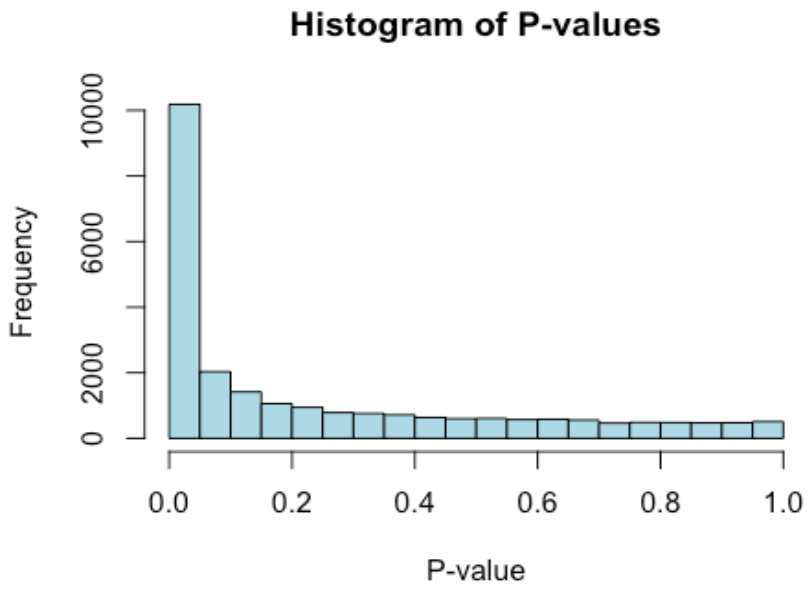
g fold changes between expression values in tumor a



2.7 Plot histogram of P-values

- visualize the distribution of p-values obtained from statistical tests, to evaluate the reliability of the results.
- The majority of p-values are very small, indicating strong evidence for the effects tested in most of the comparisons.

```
hist(resP$table$PValue, main = "Histogram of P-values", xlab = "P-value", col =
"lightblue")
```



2.8. Top10 differentially expressed genes

- Display the most significant differential expression results (top10) between normal and ESCC tissue in the RNA-seq analysis.

```
head(resP$table[, c(1, 2, 3, 6, 7)], 10)
```

```
##                     ensembl_id gene_name     logFC       PValue          FDR
## ENSG00000000003 ENSG00000170373      CST1 -9.966594 2.335211e-12 5.678066e-08
## ENSG00000000419 ENSG00000170454     KRT75 -7.865582 3.137313e-11 3.814188e-07
## ENSG00000000457 ENSG00000099953     MMP11 -6.491084 6.212125e-10 5.034927e-06
## ENSG00000000460 ENSG00000187848     P2RX2  6.810847 1.607231e-09 9.769953e-06
## ENSG00000000938 ENSG00000181092    ADIPOQ  8.193276 2.219090e-09 1.079144e-05
## ENSG00000000971 ENSG00000137766    UNC13C  4.555104 6.374277e-09 2.351332e-05
## ENSG00000001036 ENSG00000123500   COL10A1 -7.563470 7.600829e-09 2.351332e-05
## ENSG00000001084 ENSG00000173157  ADAMTS20 -5.091863 8.057611e-09 2.351332e-05
## ENSG00000001167 ENSG00000099812      MISP -4.641954 9.502751e-09 2.351332e-05
## ENSG00000001460 ENSG00000127928     GNGT1 -8.060504 9.670295e-09 2.351332e-05
```

3. Gene Set Analysis

```
# Display the first 10 rows of resP_edgeR
head(resP_edgeR, 10)
```

```
##                     ensembl_id gene_name     logFC   logCPM        F
## ENSG00000000003 ENSG00000170373      CST1 -9.966594 4.469789 492.1460
## ENSG00000000419 ENSG00000170454     KRT75 -7.865582 3.506998 299.4413
## ENSG00000000457 ENSG00000099953     MMP11 -6.491084 7.313593 560.6491
## ENSG00000000460 ENSG00000187848     P2RX2  6.810847 1.809978 177.2143
## ENSG00000000938 ENSG00000181092    ADIPOQ  8.193276 3.649818 266.4165
## ENSG00000000971 ENSG00000137766    UNC13C  4.555104 1.393889 211.5872
```

```
## ENSG00000001036 ENSG00000123500   COL10A1 -7.563470 5.320874 337.5784
## ENSG00000001084 ENSG00000173157  ADAMTS20 -5.091863 1.056572 169.7713
## ENSG00000001167 ENSG00000099812      MISP -4.641954 2.615540 276.4453
## ENSG00000001460 ENSG00000127928     GNGT1 -8.060504 1.972294 201.4226
##                       PValue         FDR
## ENSG00000000003 2.335211e-12 5.678066e-08
## ENSG00000000419 3.137313e-11 3.814188e-07
## ENSG00000000457 6.212125e-10 5.034927e-06
## ENSG00000000460 1.607231e-09 9.769953e-06
## ENSG00000000938 2.219090e-09 1.079144e-05
## ENSG00000000971 6.374277e-09 2.351332e-05
## ENSG00000001036 7.600829e-09 2.351332e-05
## ENSG00000001084 8.057611e-09 2.351332e-05
## ENSG00000001167 9.502751e-09 2.351332e-05
## ENSG00000001460 9.670295e-09 2.351332e-05

EntrezIDs <- mapIds(org.Hs.eg.db, rownames(resP_edgeR), 'ENTREZID', 'ENSEMBL')

## 'select()' returned 1:many mapping between keys and columns

sign_genes_entrez <- resP_edgeR[!(duplicated(EntrezIDs) | is.na(EntrezIDs)), ]
rownames(sign_genes_entrez) <- EntrezIDs[!(duplicated(EntrezIDs) | is.na(EntrezIDs))]

goanaOut <- goana(de=rownames(sign_genes_entrez), species="Hs", trend=T)
goanaOut <- goanaOut[order(goanaOut$P.DE, decreasing=FALSE),]
goanaOut$FDR.DE <- p.adjust(goanaOut$P.DE, method="BH")
topGORNA <- topGO(goanaOut, ontology="BP", number=50)

goanaOut_BP <- goanaOut[goanaOut$Ont == "BP",]
print(paste("Amount of significant GO Biological Process terms:",
as.character(sum(goanaOut_BP$FDR.DE < 0.05))))

## [1] "Amount of significant GO Biological Process terms: 2066"

topGORNA[order(topGORNA$N),]

##                                                          Term Ont     N
## GO:0046907                                intracellular transport  BP  1357
## GO:0015031                                    protein transport  BP  1415
## GO:0045184                establishment of protein localization  BP  1701
## GO:0033554                          cellular response to stress  BP  1847
## GO:0071705                          nitrogen compound transport  BP  1903
## GO:0051649                establishment of localization in cell  BP  1947
## GO:0071702                          organic substance transport  BP  2370
## GO:0051128          regulation of cellular component organization  BP  2459
## GO:0008104                               protein localization  BP  2505
## GO:0070727                 cellular macromolecule localization  BP  2516
## GO:0006796       phosphate-containing compound metabolic process  BP  2559
## GO:0006793                       phosphorus metabolic process  BP  2585
## GO:0009056                                catabolic process  BP  2668
## GO:0035556                  intracellular signal transduction  BP  2721
## GO:0036211                      protein modification process  BP  2958
## GO:0033036                        macromolecule localization  BP  3004
## GO:0051173 positive regulation of nitrogen compound metabolic process  BP  3083
```

```
## GO:0043412                              macromolecule modification  BP  3191
## GO:0044085                            cellular component biogenesis  BP  3368
## GO:0051641                                  cellular localization  BP  3384
## GO:0031325         positive regulation of cellular metabolic process  BP  3440
## GO:0010604    positive regulation of macromolecule metabolic process  BP  3506
## GO:0023051                                regulation of signaling  BP  3554
## GO:0010646                        regulation of cell communication  BP  3559
## GO:0006996                             organelle organization  BP  3623
## GO:0009893                 positive regulation of metabolic process  BP  3816
## GO:0006950                                 response to stress  BP  3984
## GO:0048731                                 system development  BP  4072
## GO:0048583                      regulation of response to stimulus  BP  4116
## GO:0006810                                          transport  BP  4385
## GO:0051234                          establishment of localization  BP  4698
## GO:0007275                       multicellular organism development  BP  4764
## GO:0019538                              protein metabolic process  BP  4915
## GO:0051179                                        localization  BP  5335
## GO:0048522                  positive regulation of cellular process  BP  5861
## GO:1901564                 organonitrogen compound metabolic process  BP  5992
## GO:0048856                         anatomical structure development  BP  6016
## GO:0048518                positive regulation of biological process  BP  6411
## GO:0016043                       cellular component organization  BP  6581
## GO:0032502                            developmental process  BP  6593
## GO:0071840            cellular component organization or biogenesis  BP  6791
## GO:0051716                          cellular response to stimulus  BP  7616
## GO:0050896                                 response to stimulus  BP  9228
## GO:0006807                    nitrogen compound metabolic process  BP 10169
## GO:0044238                              primary metabolic process  BP 10655
## GO:0071704                    organic substance metabolic process  BP 11820
## GO:0008152                                  metabolic process  BP 12343
## GO:0065007                               biological regulation  BP 12701
## GO:0009987                                   cellular process  BP 17647
## GO:0008150                                 biological_process  BP 18888
##             DE         P.DE       FDR.DE
## GO:0046907  726  1.805689e-48 7.023818e-46
## GO:0015031  741  5.490232e-45 1.769502e-42
## GO:0045184  857  1.775893e-43 5.488484e-41
## GO:0033554  913  4.027524e-42 1.180065e-39
## GO:0071705  962  5.990434e-50 2.502781e-47
## GO:0051649 1012  2.552662e-60 1.745170e-57
## GO:0071702 1124  1.083296e-41 3.133365e-39
## GO:0051128 1198  3.758547e-52 1.630704e-49
## GO:0008104 1249  8.118462e-62 6.105354e-59
## GO:0070727 1254  5.747424e-62 4.471297e-59
## GO:0006796 1188  8.048540e-39 2.087162e-36
## GO:0006793 1203  5.700192e-40 1.549422e-37
## GO:0009056 1246  1.796958e-42 5.334363e-40
## GO:0035556 1291  1.174837e-48 4.650088e-46
## GO:0036211 1381  1.551464e-47 5.833763e-45
## GO:0033036 1458  1.247079e-63 1.042050e-60
## GO:0051173 1388  1.647746e-37 4.040738e-35
## GO:0043412 1495  3.598045e-53 1.623510e-50
## GO:0044085 1516  2.144000e-41 6.122883e-39
## GO:0051641 1672  3.325261e-82 4.688826e-79
```

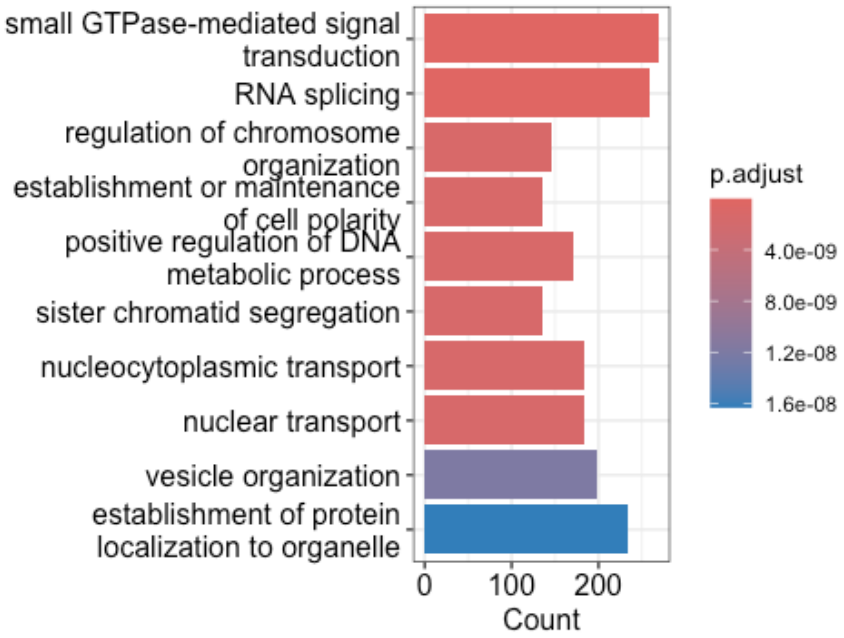
```
## GO:0031325 1532  8.929622e-39 2.289332e-36
## GO:0010604 1583  1.783354e-44 5.666794e-42
## GO:0023051 1612  6.779363e-47 2.353065e-44
## GO:0010646 1607  2.836963e-45 9.276047e-43
## GO:0006996 1696  2.854111e-61 2.012237e-58
## GO:0009893 1712  1.409038e-46 4.816561e-44
## GO:0006950 1749  1.449126e-40 4.036263e-38
## GO:0048731 1813  3.624565e-47 1.297997e-44
## GO:0048583 1830  3.364395e-47 1.224260e-44
## GO:0006810 1977  1.973898e-57 1.141875e-54
## GO:0051234 2108  3.960634e-60 2.553024e-57
## GO:0007275 2126  2.087326e-58 1.239267e-55
## GO:0019538 2204  2.354865e-63 1.897432e-60
## GO:0051179 2416  1.302743e-76 1.469560e-73
## GO:0048522 2575  9.664818e-67 8.721919e-64
## GO:1901564 2699  8.710776e-86 1.403742e-82
## GO:0048856 2587  2.623950e-56 1.345430e-53
## GO:0048518 2801  7.704438e-72 7.900901e-69
## GO:0016043 2953  6.361506e-95 1.104015e-91
## GO:0032502 2813  3.844905e-59 2.344457e-56
## GO:0071840 3055 1.260280e-101 2.369432e-98
## GO:0051716 3129  8.980741e-48 3.434144e-45
## GO:0050896 3659  4.849122e-40 1.334159e-37
## GO:0006807 4095  3.608089e-60 2.394179e-57
## GO:0044238 4270  1.375643e-61 1.001157e-58
## GO:0071704 4573  6.011754e-43 1.832854e-40
## GO:0008152 4775  1.747721e-47 6.463990e-45
## GO:0065007 4846  1.565525e-38 3.924422e-36
## GO:0009987 6561  1.545568e-72 1.660456e-69
## GO:0008150 6892  3.116839e-68 3.057348e-65

#### Gene Set Analysis
# GO analysis (Biological Process)

go_results <- enrichGO(gene = rownames(sign_genes_entrez),
                       OrgDb = org.Hs.eg.db,
                       ont = "BP",       # GO Biological Process (BP)
                       pAdjustMethod = "BH",
                       pvalueCutoff = 0.05,
                       qvalueCutoff = 0.05)

go_results_df <- as.data.frame(go_results)

#visualize only top10 Go results
barplot(go_results, showCategory = 10)
```




```
RNAseq_GSA_res <- topGO(goanaOut, ontology="BP", number=100)

# write GSA results into text file for further analysis
write.table(RNAseq_GSA_res, sep= "\t", file="RNAseq_GSA_results.txt")
```

4. save the results for futher analysis

```
edgeR_res <- resP_edgeR[order(resP_edgeR$FDR, decreasing=F),]
write.table(edgeR_res, sep= "\t", file="RNAseq_results.txt")

RNAseq_GSA_res <- topGO(goanaOut, ontology="BP", number=100)
write.table(RNAseq_GSA_res, sep= "\t", file="RNAseq_GSA_results.txt")
```

Appendix C:

Result comparison

Chaeyoung Kim

2024-12-19

This R markdown the comparison of results of two data analysis.

```
setwd("/Users/kimchaeyoung/Desktop/Applied_HT_analysis/Project/results/")
dir <- "/Users/kimchaeyoung/Desktop/Applied_HT_analysis/Project/results/"

library(org.Hs.eg.db)
```

```
## Loading required package: AnnotationDbi
```

```
## Loading required package: stats4
```

```
## Loading required package: BiocGenerics
```

```
##
## Attaching package: 'BiocGenerics'
```

```
## The following objects are masked from 'package:stats':
##
##     IQR, mad, sd, var, xtabs
```

```
## The following objects are masked from 'package:base':
##
##     anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##     colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##     get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
##     match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
##     Position, rank, rbind, Reduce, rownames, sapply, setdiff, table,
##     tapply, union, unique, unsplit, which.max, which.min
```

```
## Loading required package: Biobase
```

```
## Welcome to Bioconductor
##
##     Vignettes contain introductory material; view with
##     'browseVignettes()'. To cite Bioconductor, see
##     'citation("Biobase")', and for packages 'citation("pkgname")'.
```

```
## Loading required package: IRanges
```

```
## Loading required package: S4Vectors

##
## Attaching package: 'S4Vectors'

## The following object is masked from 'package:utils':
##
##     findMatches

## The following objects are masked from 'package:base':
##
##     expand.grid, I, unname

##

library(VennDiagram)

## Loading required package: grid

## Loading required package: futile.logger
```

Load in data

```
results_files <- list.files(pattern = ".*.txt")
for (i in 1:length(results_files)) { #loop through .txt files, assign them to their respective variables
assign(sub("\\..*","",results_files[i]), read.table(results_files[i])) }
```

1. Fitration of RNA_seq results

This section filters the RNA-seq results based on an FDR threshold of 0.05, sorts the data by row names, and then performs an analysis based on CpG methylation data. A Venn diagram is created to visualize the overlap between the significant RNA-seq results and the CpG analysis.

```
# RNAseq filtering
# Filtering RNAseq results with an FDR less than 0.05 and sorting the results by row names
RNAseq_filtered <- RNAseq_results[RNAseq_results$FDR < 0.05, ]
RNAseq_filtered <- RNAseq_filtered[order(rownames(RNAseq_filtered)), ]
ENS_RNA <- rownames(RNAseq_filtered[RNAseq_filtered$FDR<0.05,])
```

2.Comparison between RNA-seq analysis vs differentially methylated regions (DMRs)

- no overlappings

```
EntrezIDs_bmp <- read.table("EntrezIDs_bmp_results.txt", header = TRUE)

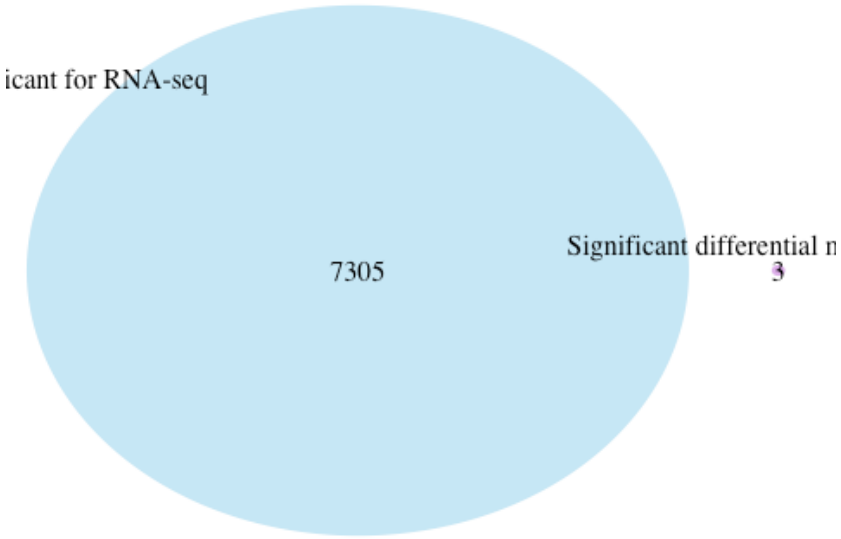
# Extracting the 'x' column as a vector
```

```
EntrezIDs_bmp_vec <- EntrezIDs_bmp$x
ENS_bmp <- mapIds(org.Hs.eg.db, as.character(EntrezIDs_bmp_vec), 'ENSEMBL',
'ENTREZID')

## 'select()' returned 1:many mapping between keys and columns

ENS_bmp_uniq <- ENS_bmp[!(duplicated(ENS_bmp) | is.na(ENS_bmp))]

# Create a Venn diagram to visualize the overlap between RNAseq results and DMRs
grid.newpage()
draw.pairwise.venn(
  area1 = length(rownames(RNAseq_filtered)),  # RNAseq filtered area
  area2 = length(ENS_bmp_uniq),               # CpG analysis area
  cross.area = length(intersect(rownames(RNAseq_filtered), ENS_bmp_uniq)),  #
Intersection area
  category = c("Significant for RNA-seq", "Significant differential methylation
(bumps) "),  # Category labels
  fill = c("skyblue", "mediumorchid"),
  lty = "blank"
)
```




```
## (polygon[GRID.polygon.1], polygon[GRID.polygon.2], polygon[GRID.polygon.3],
polygon[GRID.polygon.4], text[GRID.text.5], text[GRID.text.6], text[GRID.text.7],
text[GRID.text.8])
```

3. Comparison betwen RNA-seq analysis and CpG based analysis

```
EntrezIDs_CpG <- read.table("EntrezIDs_CpG_results.txt", header = TRUE)

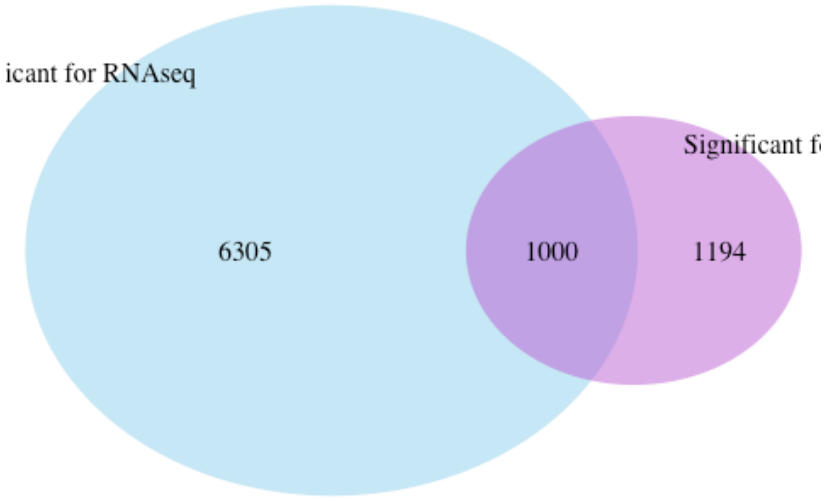
# Extracting the 'x' column as a vector
EntrezIDs_CpG_vec <- EntrezIDs_CpG$x
```

```r
ENS_CpG <- mapIds(org.Hs.eg.db, as.character(EntrezIDs_CpG_vec), 'ENSEMBL',
'ENTREZID')

## 'select()' returned 1:many mapping between keys and columns

ENS_CpG_uniq <- ENS_CpG[!(duplicated(ENS_CpG) | is.na(ENS_CpG))]

# Create a Venn diagram to visualize the overlap between RNAseq results and CpG
methylation results
grid.newpage()
draw.pairwise.venn(
  area1 = length(rownames(RNAseq_filtered)),  # RNAseq filtered area
  area2 = length(ENS_CpG_uniq),              # CpG analysis area
  cross.area = length(intersect(rownames(RNAseq_filtered), ENS_CpG_uniq)),  #
Intersection area
  category = c("Significant for RNAseq", "Significant for CpG"),  # Category labels
  fill = c("skyblue", "mediumorchid"),
  lty = "blank"
)
```




```
## (polygon[GRID.polygon.9], polygon[GRID.polygon.10], polygon[GRID.polygon.11],
polygon[GRID.polygon.12], text[GRID.text.13], text[GRID.text.14], text[GRID.text.15],
text[GRID.text.16], text[GRID.text.17])
```

4. Genes and Terms of interest

The diagram reveals that there are 26 GO terms that are shared by both RNA-seq and methylation analyses, which might represent biological processes or functions that are affected by both gene expression and DNA methylation

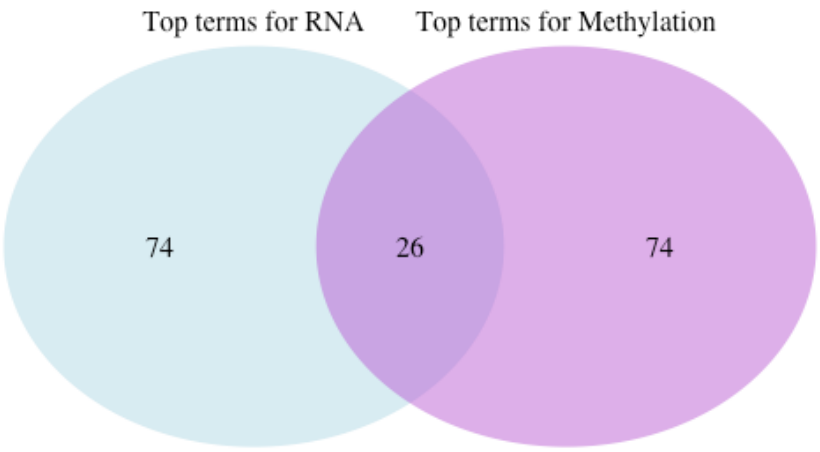
```r
print("Ensembl gene IDs that are significant:")
```

```
## [1] "Ensembl gene IDs that are significant:"
```

```r
# Extract GO terms from RNA-seq analysis results
Metterms <- bmp_GSA_results$Term  # Extract GO terms related to RNA-seq

# Extract GO terms from methylation analysis results
RNAterms <- RNAseq_GSA_results$Term  # Extract GO terms related to methylation
analysis

# Create a new graphics page for the Venn diagram
grid.newpage()
grid.newpage()
draw.pairwise.venn(length(RNAterms), length(Metterms),
length(intersect(RNAterms,Metterms)), category = c("Top terms for RNA", "Top terms
for Methylation"),
lty = rep("blank", 2), fill = c("light blue", "mediumorchid"), alpha = rep(0.5, 2),
cat.pos = c(0, 0), cat.dist = rep(0.025, 2))
```




```
## (polygon[GRID.polygon.18], polygon[GRID.polygon.19], polygon[GRID.polygon.20],
polygon[GRID.polygon.21], text[GRID.text.22], text[GRID.text.23], text[GRID.text.24],
text[GRID.text.25], text[GRID.text.26])
```

5. Ensembl gene IDs that are significant for the two analyses (RNAseq and CpG)

This section finds the common genes between RNA-seq and CpG methylation analysis.

```
# Find common genes between RNAseq results and CpG analysis
print("Ensembl gene IDs that are significant for the two analyses:")
```

```
## [1] "Ensembl gene IDs that are significant for the two analyses:"
```

```
common_genes <- intersect(ENS_RNA, ENS_CpG)  # Identify genes that are significant in both RNAseq and CpG analyses\
head(common_genes)
```

```
## [1] "ENSG00000001460" "ENSG00000001617" "ENSG00000001629" "ENSG00000002746"
## [5] "ENSG00000002822" "ENSG00000003147"
```

```
# Top GO terms overlapping for the two analyses
print("Top overlapping GO terms:")
```

```
## [1] "Top overlapping GO terms:"
```

```
common_terms <- intersect(RNAterms, Metterms)  # Identify common GO terms between RNAseq and methylation analysis
head(common_terms)
```

```
## [1] "cellular component organization or biogenesis"
## [2] "cellular component organization"
## [3] "cellular process"
## [4] "positive regulation of biological process"
## [5] "positive regulation of cellular process"
## [6] "organelle organization"
```

```
# Genes of interest (GOI) based on the common genes found in RNAseq and CpG
# Extract genes of interest from RNAseq results based on the common genes.

GOI <- RNAseq_results[which(rownames(RNAseq_results) %in% common_genes), ]
rownames(GOI) <- mapIds(org.Hs.eg.db, rownames(GOI), 'SYMBOL', 'ENSEMBL')  # Convert ENSEMBL IDs to SYMBOLs
```

```
## 'select()' returned 1:many mapping between keys and columns
```

```
# Terms of interest (POI) based on the common GO terms
# Extract terms of interest based on the common GO terms.
POI <- RNAseq_GSA_results[which(RNAseq_GSA_results$Term %in% common_terms), ]

# View top 10 rows of both GOI and POI
# Display the top 10 genes and GO terms of interest.
```

5.1. GOI

In this analysis, positive logFC values indicate up-regulated genes, meaning these genes have increased expression in the experimental condition compared to the control. Genes like CFLAR, KDM1A, and RBM6 are examples of up-regulated genes. On the other hand, negative logFC values indicate down-regulated genes, meaning these genes have decreased expression in the experimental condition. Examples of down-regulated genes include STPG1, SEMA3F, ANKIB1, HECW1, MAD1L1, and ICA1, which all show reduced expression. Therefore, genes with positive logFC values are up-regulated, while genes with negative logFC values are down-regulated in the experimental group.

```
head(GOI, n=10)
```

```
##            ensembl_id gene_name     logFC    logCPM        F       PValue
## STPG1  ENSG00000127928     GNGT1 -8.060504 1.9722937 201.4226 9.670295e-09
## SEMA3F ENSG00000060718   COL11A1 -8.703936 5.7733561 216.5736 4.633264e-08
## ANKIB1 ENSG00000228742 LINC02577 -5.249074 1.6900926 120.7926 5.485484e-08
## HECW1  ENSG00000095203  EPB41L4B -2.977733 4.0113937 188.9327 1.070544e-07
## MAD1L1 ENSG00000223485 LINC01615 -4.594110 0.1554300  86.6495 1.114826e-07
## ICA1   ENSG00000214336     FOXI3 -5.372894 0.6769829 129.1362 1.431481e-07
## CFLAR  ENSG00000116983    HPCAL4  3.527323 2.3264447 177.2282 1.553250e-07
## MTMR7  ENSG00000119547   ONECUT2 -3.808518 0.9149784 132.8048 1.764565e-07
## KDM1A  ENSG00000197256     KANK2  1.914136 7.4096116 163.7816 2.069153e-07
## RBM6   ENSG00000076555     ACACB  2.451443 5.8016488 163.0520 2.112539e-07
##                FDR
## STPG1  2.351332e-05
## SEMA3F 8.046987e-05
## ANKIB1 8.336221e-05
## HECW1  8.606043e-05
## MAD1L1 8.606043e-05
## ICA1   8.606043e-05
## CFLAR  8.606043e-05
## MTMR7  8.606043e-05
## KDM1A  8.606043e-05
## RBM6   8.606043e-05
```

5.2. POI

The terms with higher DE values indicate biological processes that are most impacted by the experimental conditions. For example, the “cellular process” term (GO:0009987) is associated with 6561 DE genes, meaning that a significant portion of the genes in this category show altered expression.

```r
head(POI, n=10)
```

```
##                                                 Term Ont     N   DE
## GO:0071840 cellular component organization or biogenesis  BP  6791 3055
## GO:0016043              cellular component organization  BP  6581 2953
## GO:0009987                          cellular process  BP 17647 6561
## GO:0048518    positive regulation of biological process  BP  6411 2801
## GO:0048522      positive regulation of cellular process  BP  5861 2575
## GO:0006996                     organelle organization  BP  3623 1696
## GO:0032502                      developmental process  BP  6593 2813
## GO:0007275           multicellular organism development  BP  4764 2126
## GO:0048856             anatomical structure development  BP  6016 2587
## GO:0051128 regulation of cellular component organization  BP  2459 1198
##                    P.DE      FDR.DE
## GO:0071840 1.260280e-101 2.369432e-98
## GO:0016043  6.361506e-95 1.104015e-91
## GO:0009987  1.545568e-72 1.660456e-69
## GO:0048518  7.704438e-72 7.900901e-69
## GO:0048522  9.664818e-67 8.721919e-64
## GO:0006996  2.854111e-61 2.012237e-58
## GO:0032502  3.844905e-59 2.344457e-56
## GO:0007275  2.087326e-58 1.239267e-55
## GO:0048856  2.623950e-56 1.345430e-53
## GO:0051128  3.758547e-52 1.630704e-49
```